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(54) **EXPANDABLE INTERBODY IMPLANT AND CORRESPONDING SURGICAL TOOL**

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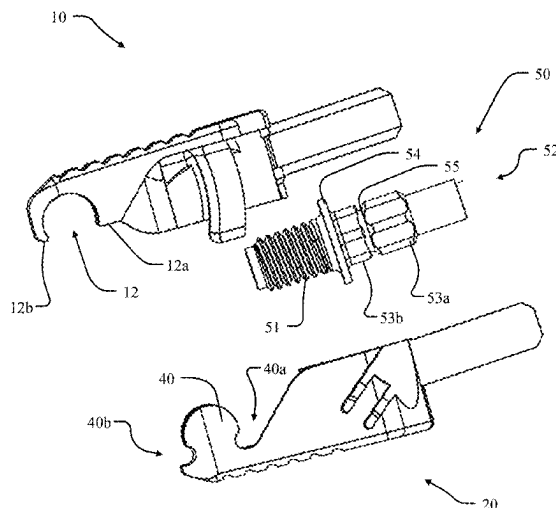
(58) **Field of Classification Search**

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(57) **ABSTRACT**

An interbody system including an implant and a tool for inserting and expanding the medical implant and locking the implant in place is disclosed. The medical implant may include an expandable body defined by a superior endplate and an inferior endplate that are hingedly coupled and may be expanded and lordosed. The superior and inferior endplate may include radially disposed and opposed surfaces that mate and/or directly contact each other when a locking screw is threaded through a screw aperture. The implant may include a threaded breakoff screw disposed in the threaded screw aperture and movable between a locked position and an unlocked position, for example. In the locked position, the threaded locking screw may urge the distal engagement surface of the first core into direct contact with the proximal engagement surface of the second core. When broken, the breakoff screw may comprise a recessed fracture surface.

17 Claims, 56 Drawing Sheets



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2016/0310291	A1	10/2016	Greenhalgh	2019/0133434	A1	5/2019	Lee et al.
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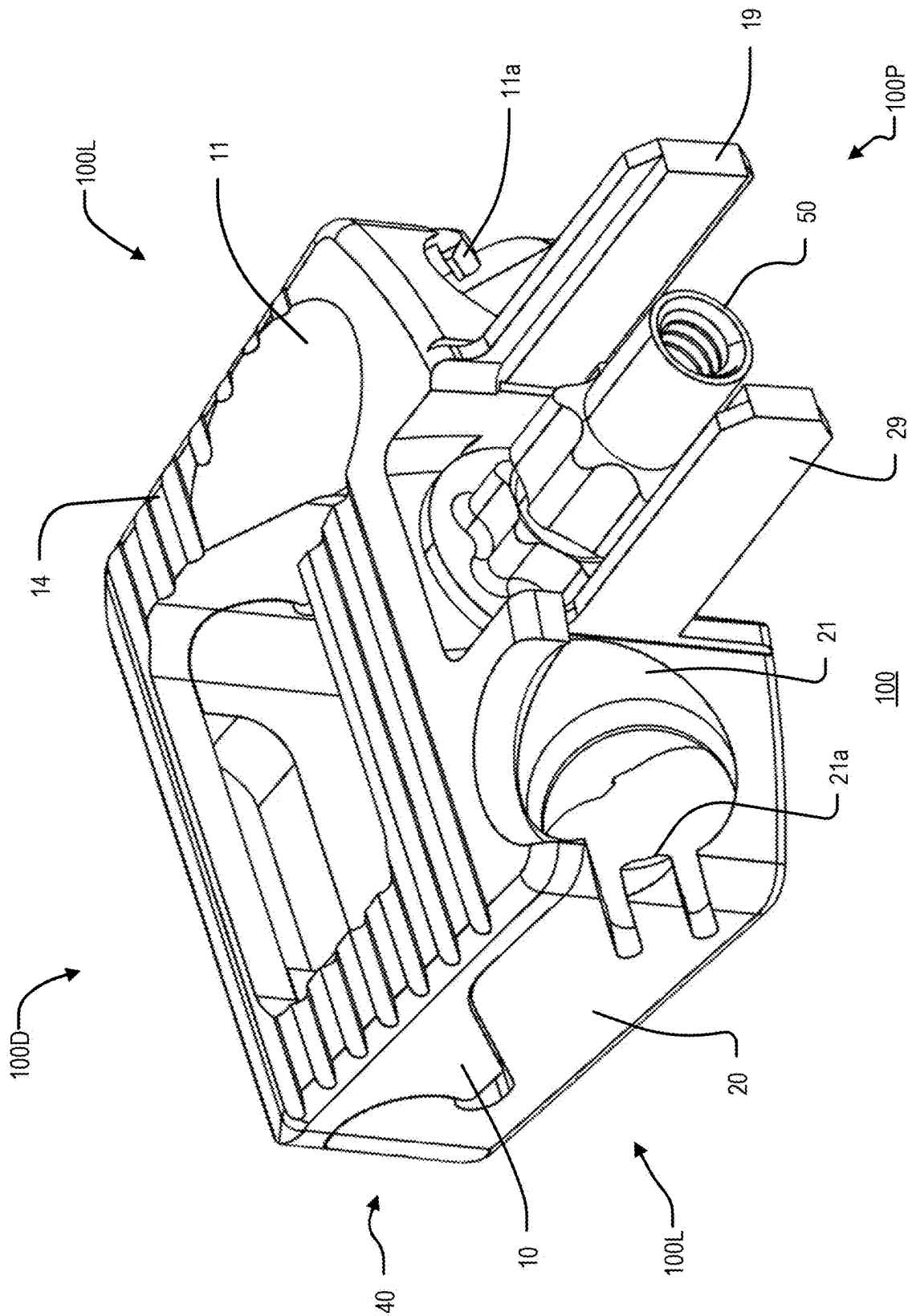
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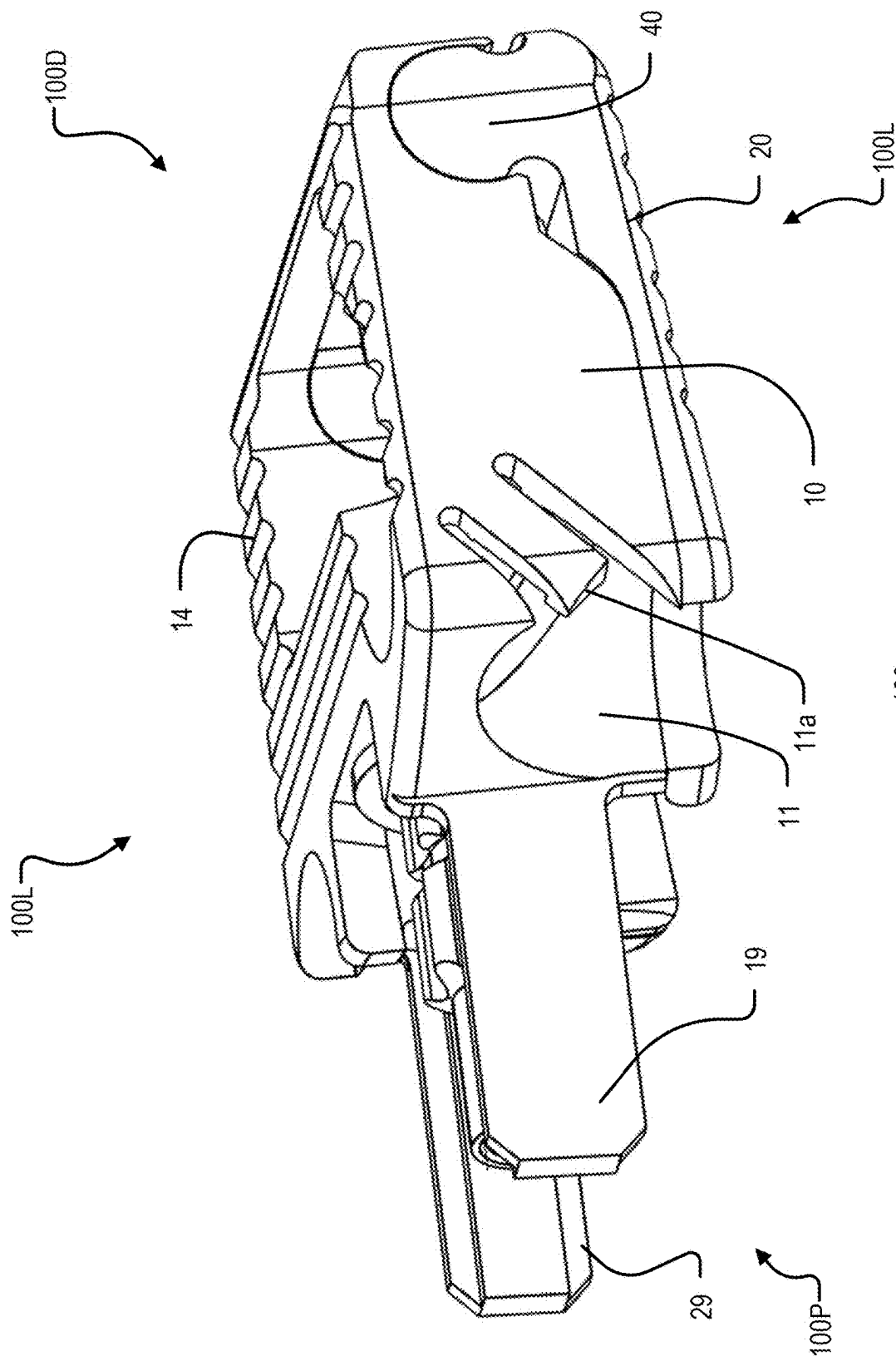
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* cited by examiner



100
FIG. 1



100
FIG. 2

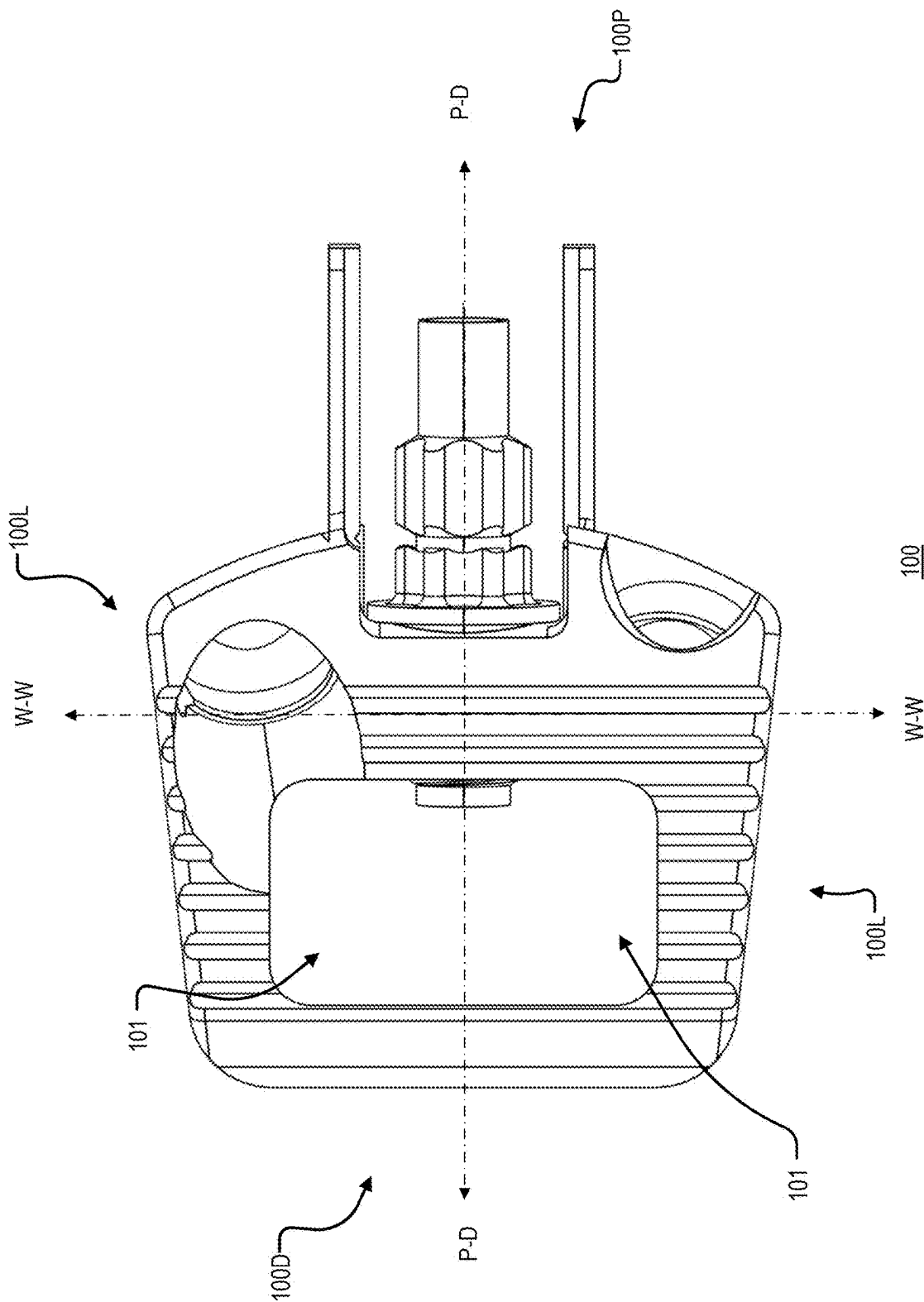
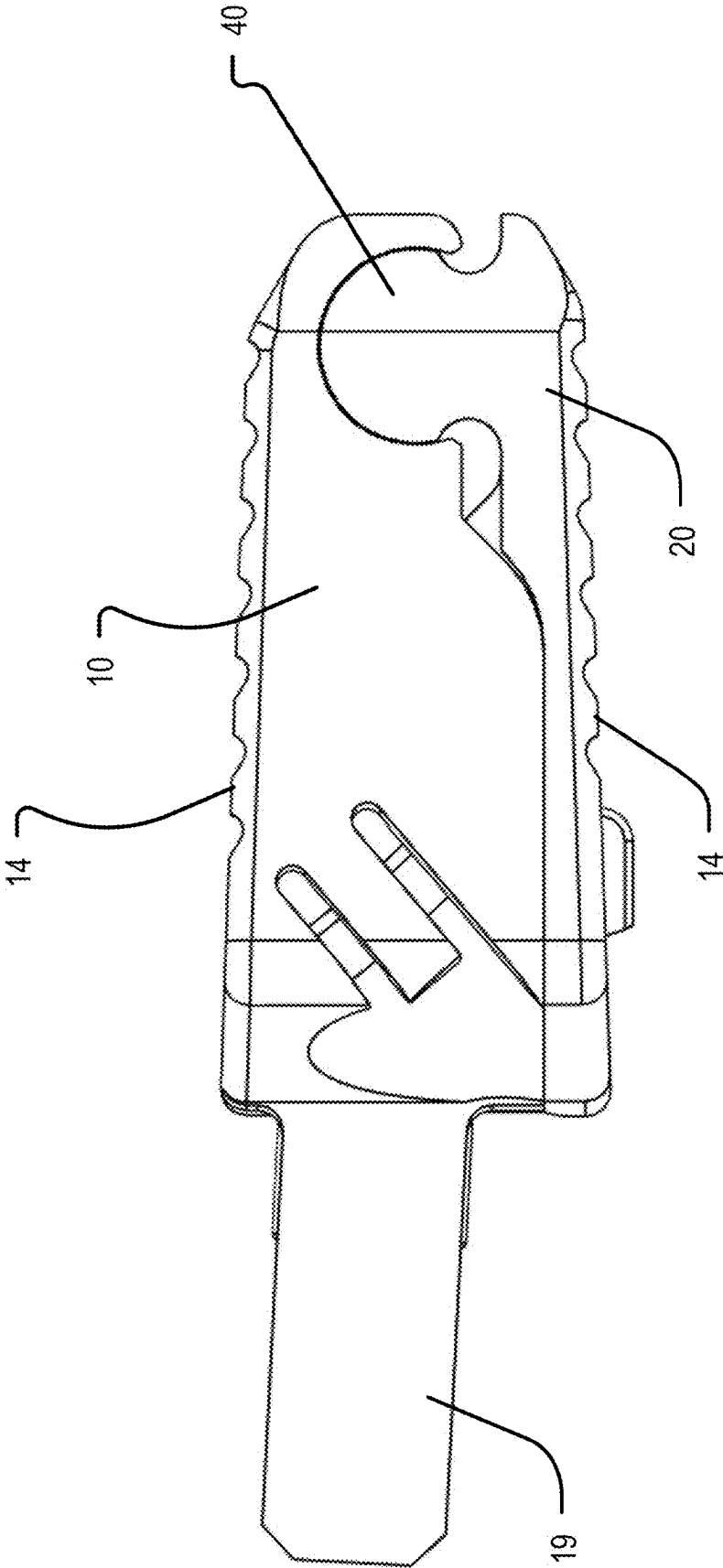
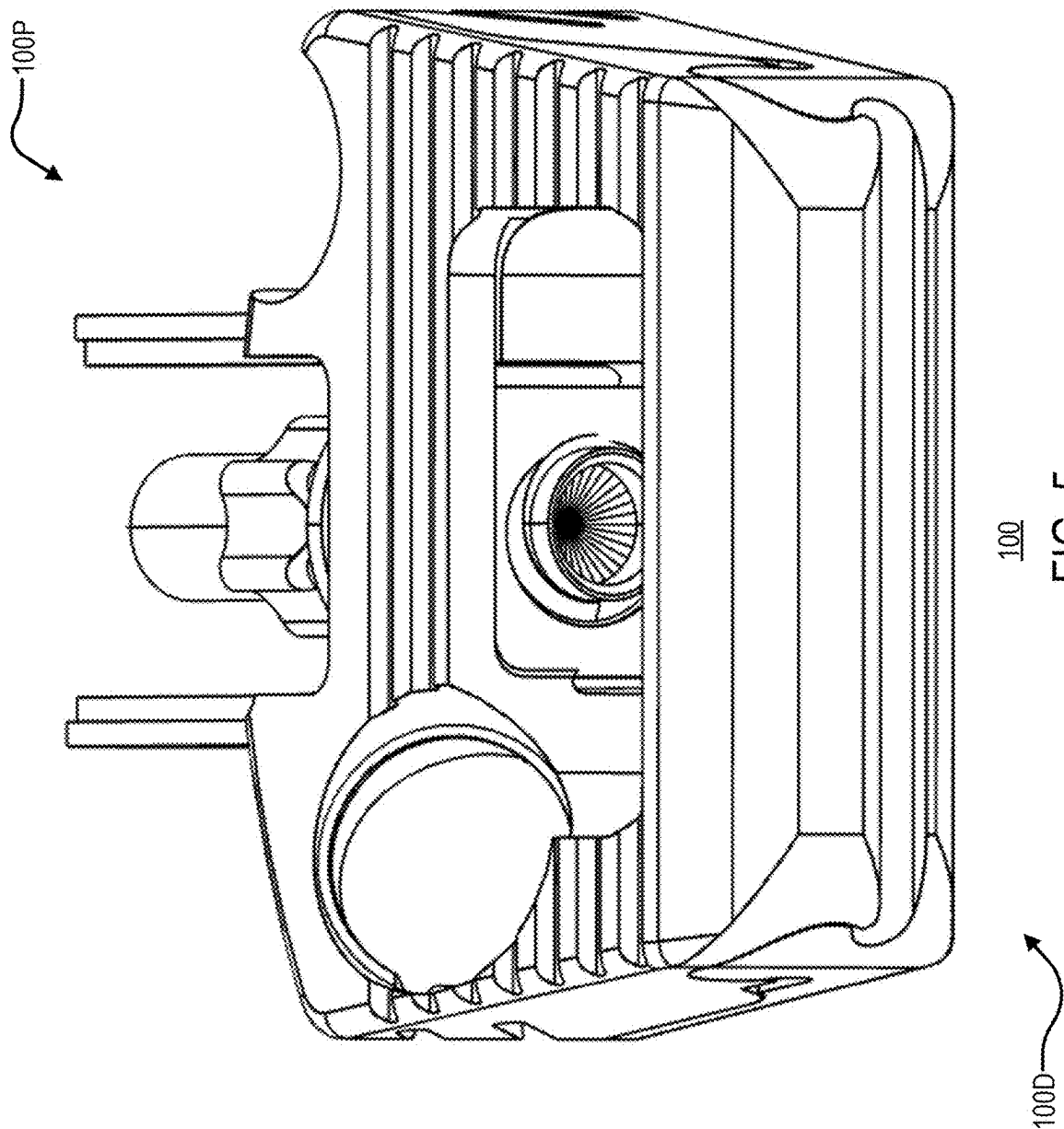


FIG. 3



100
FIG. 4



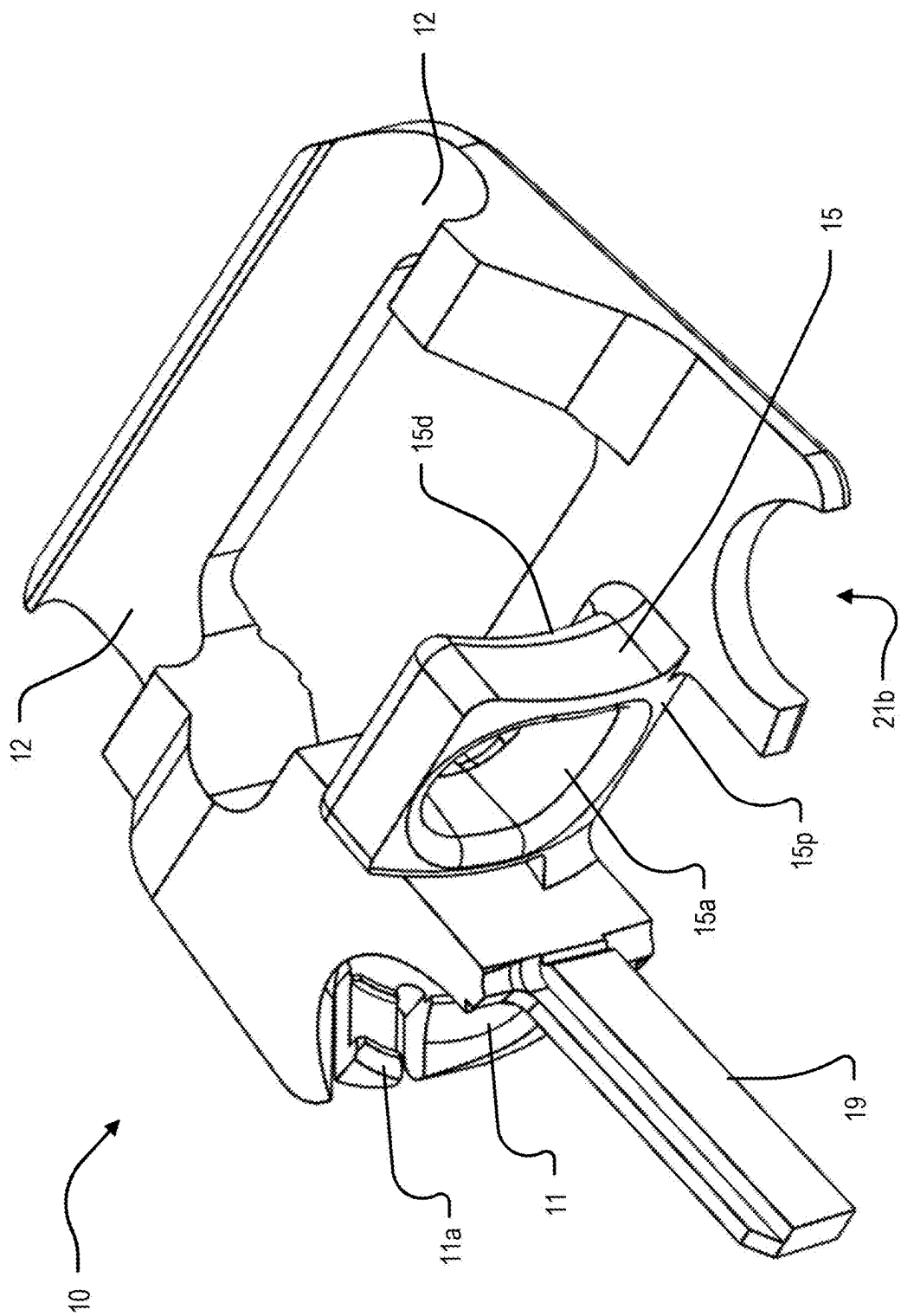


FIG. 6

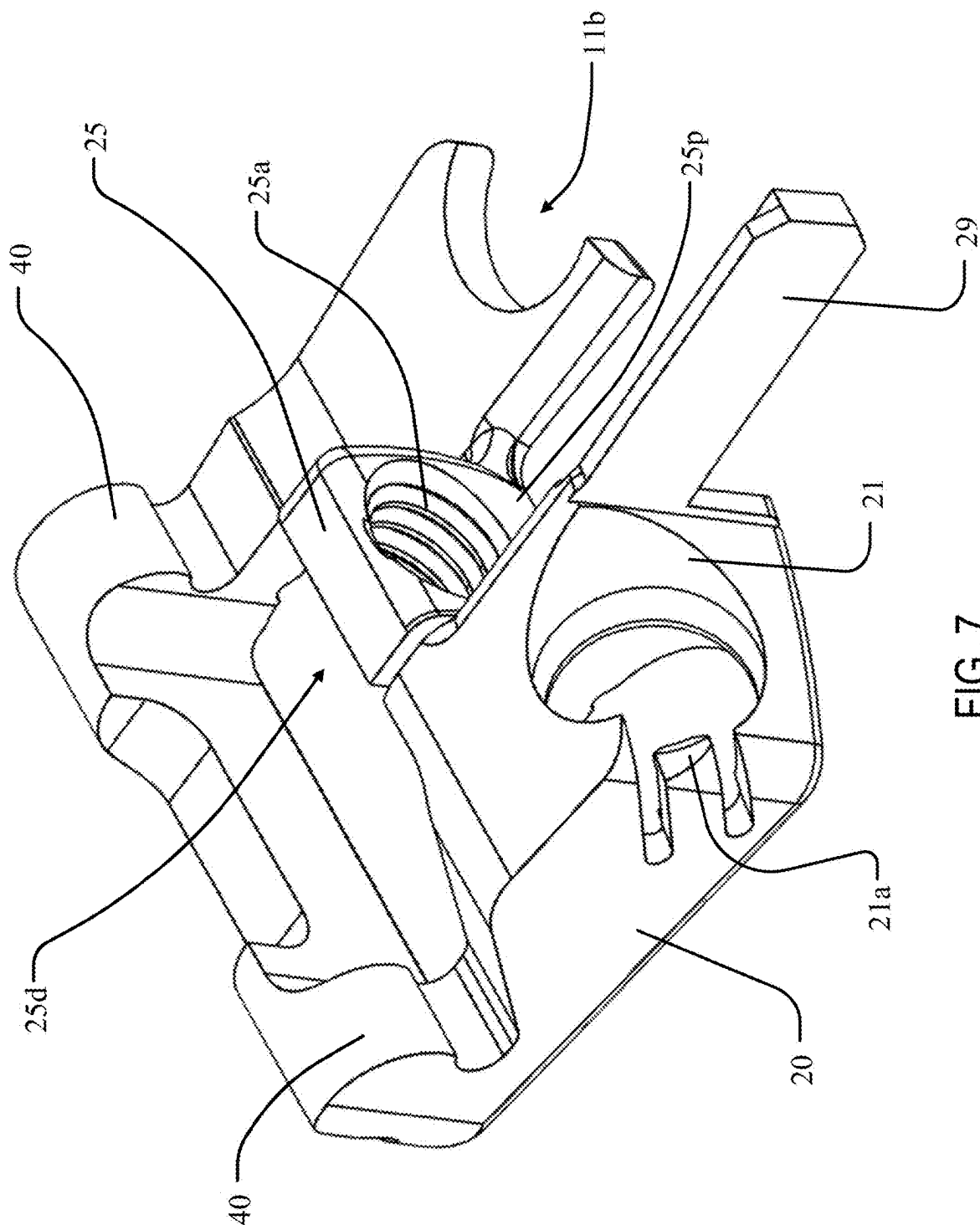
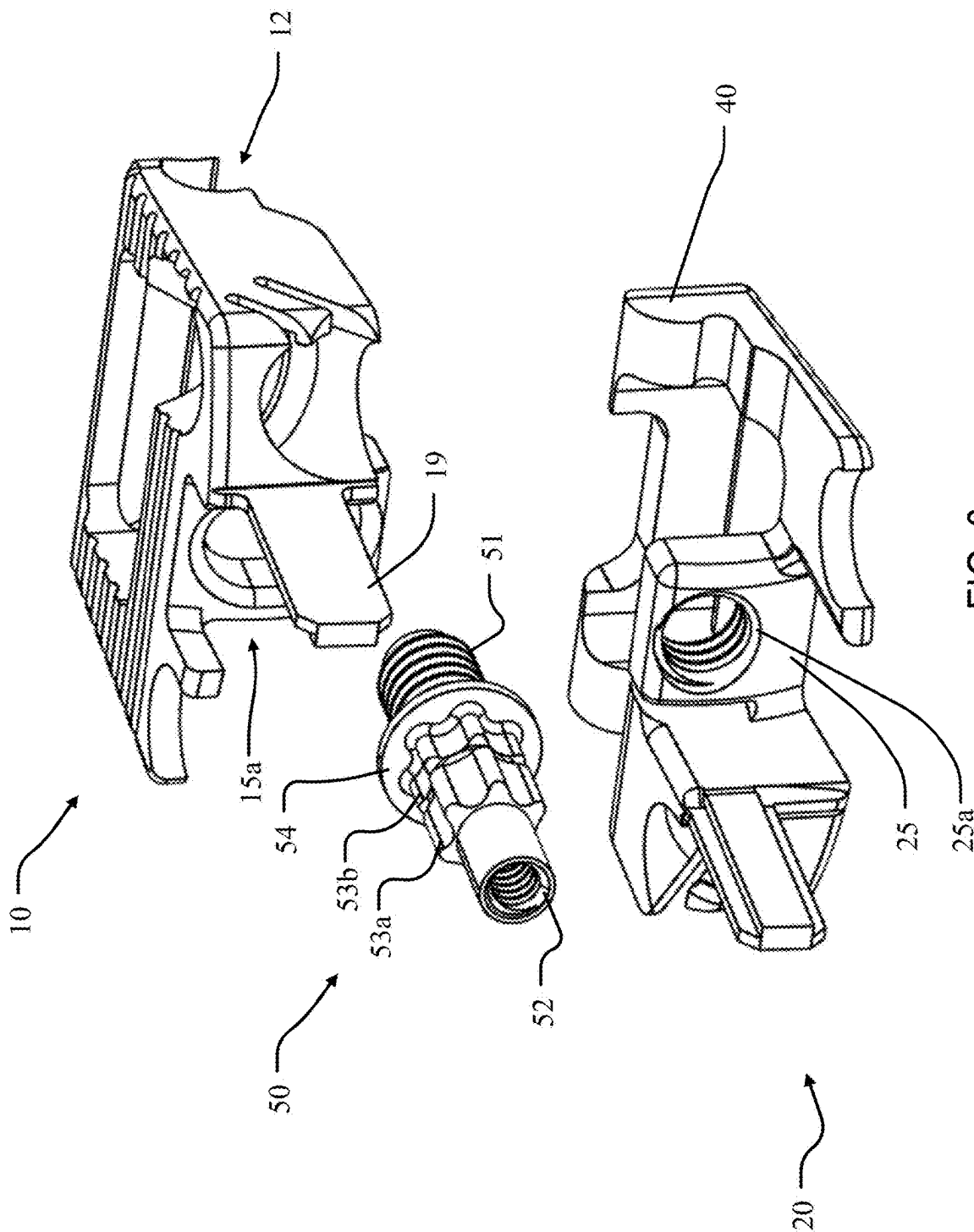
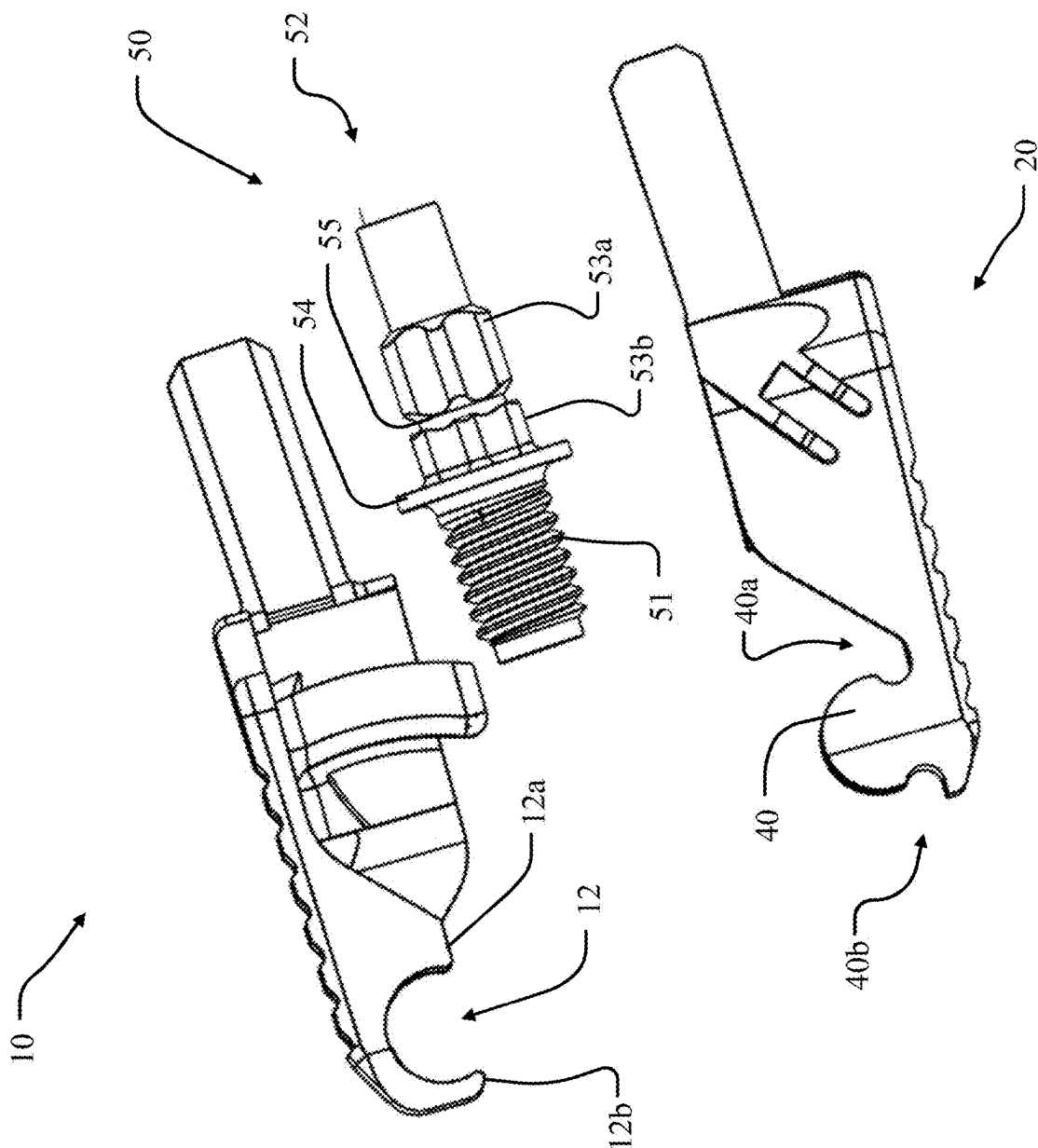


FIG. 7





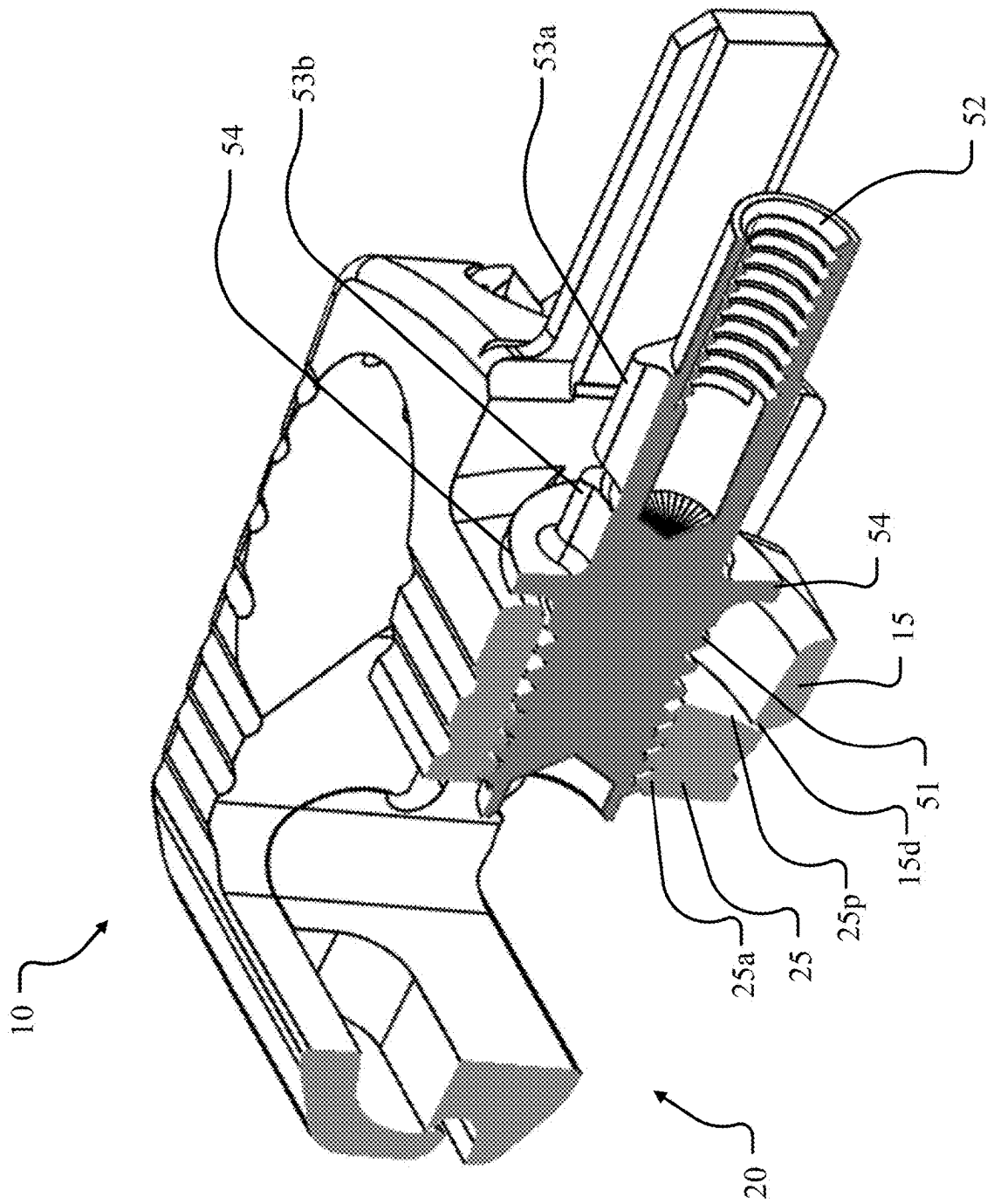


FIG. 10

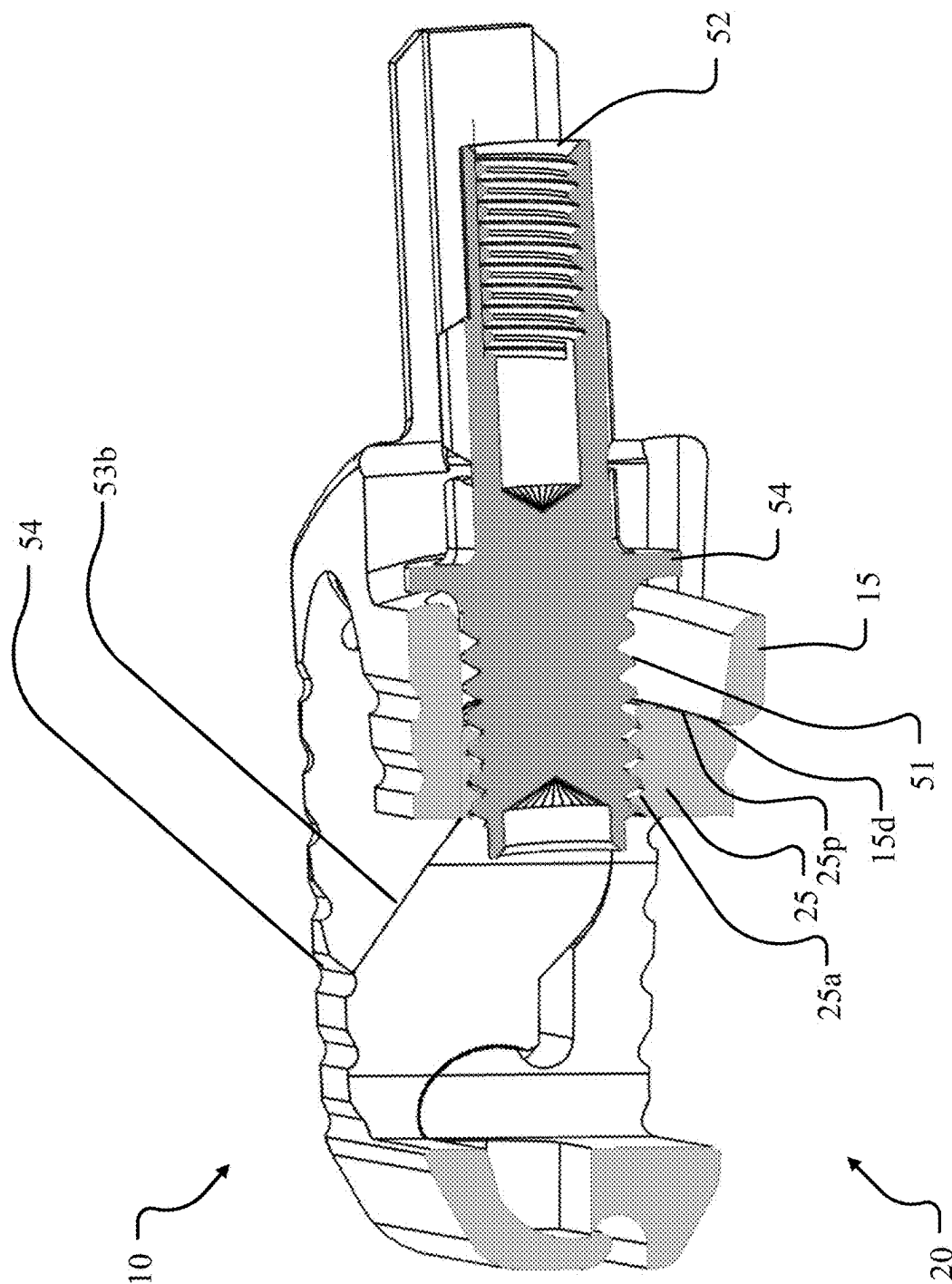


FIG. 11

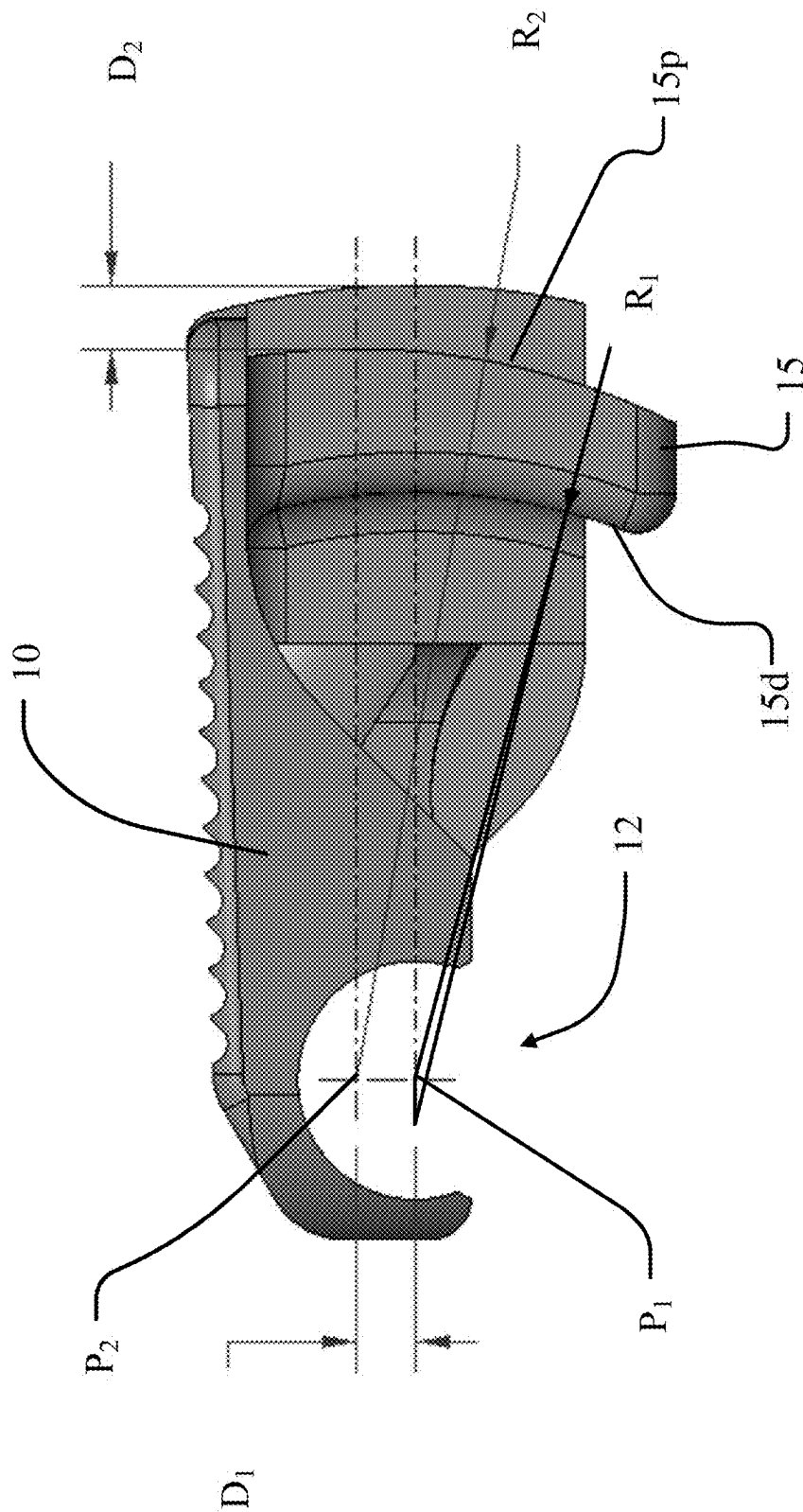
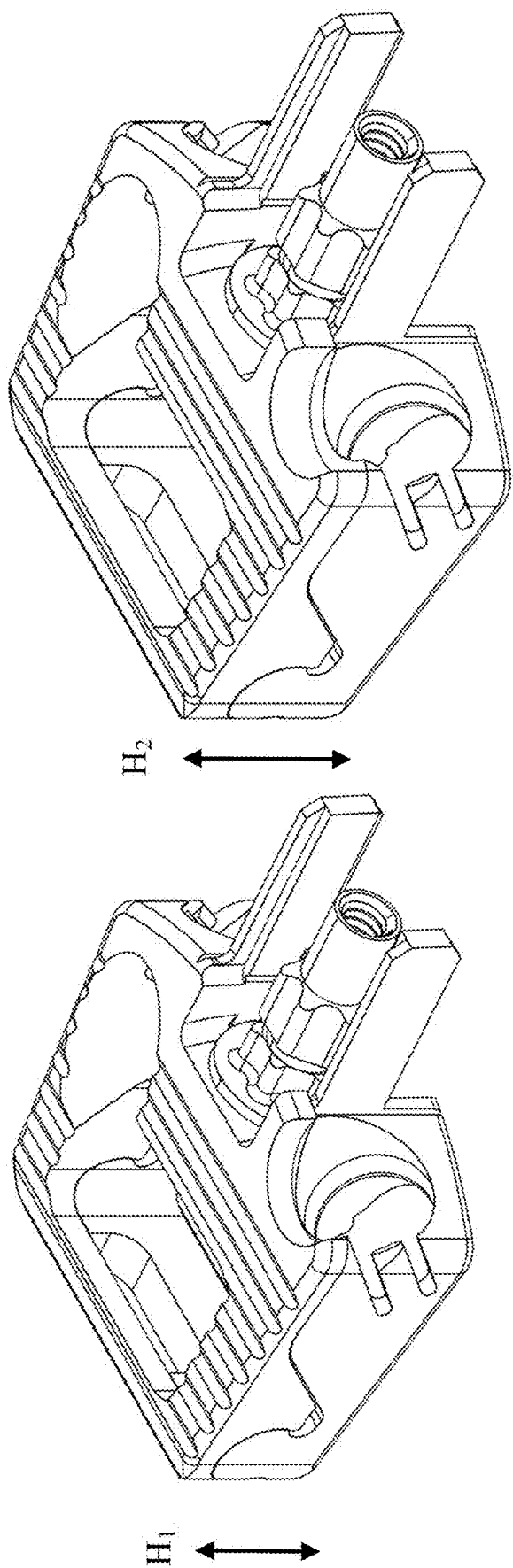
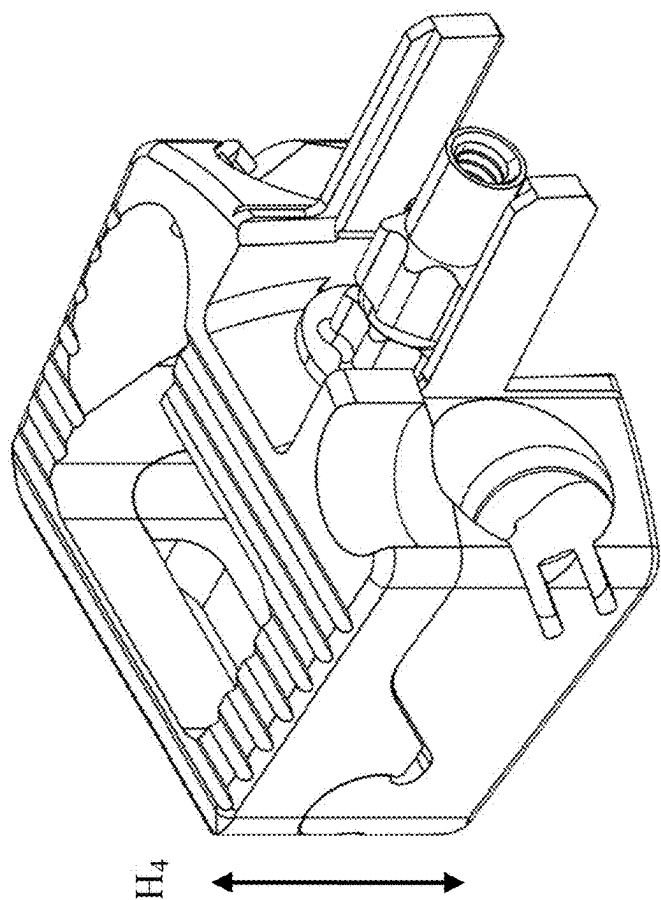


FIG. 12

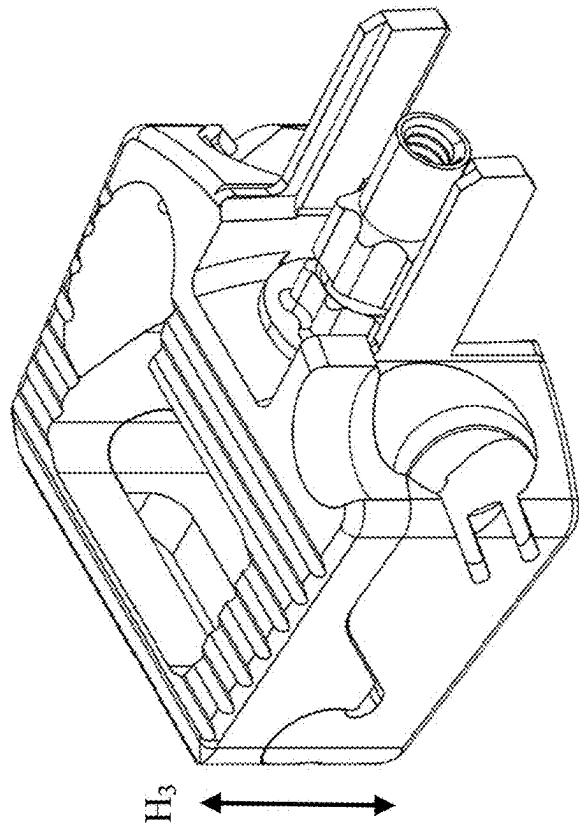


100
FIG. 13B

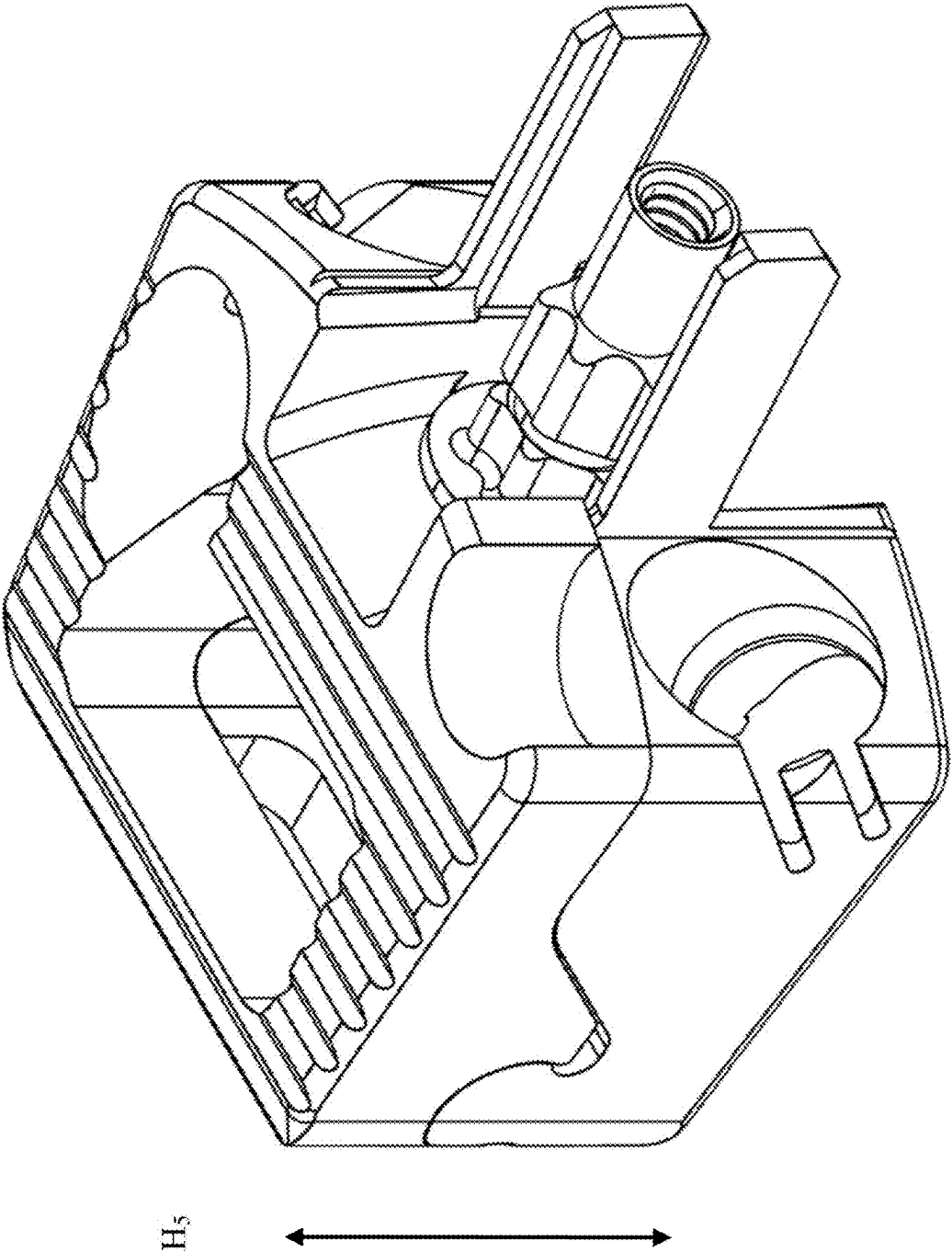
100
FIG. 13A



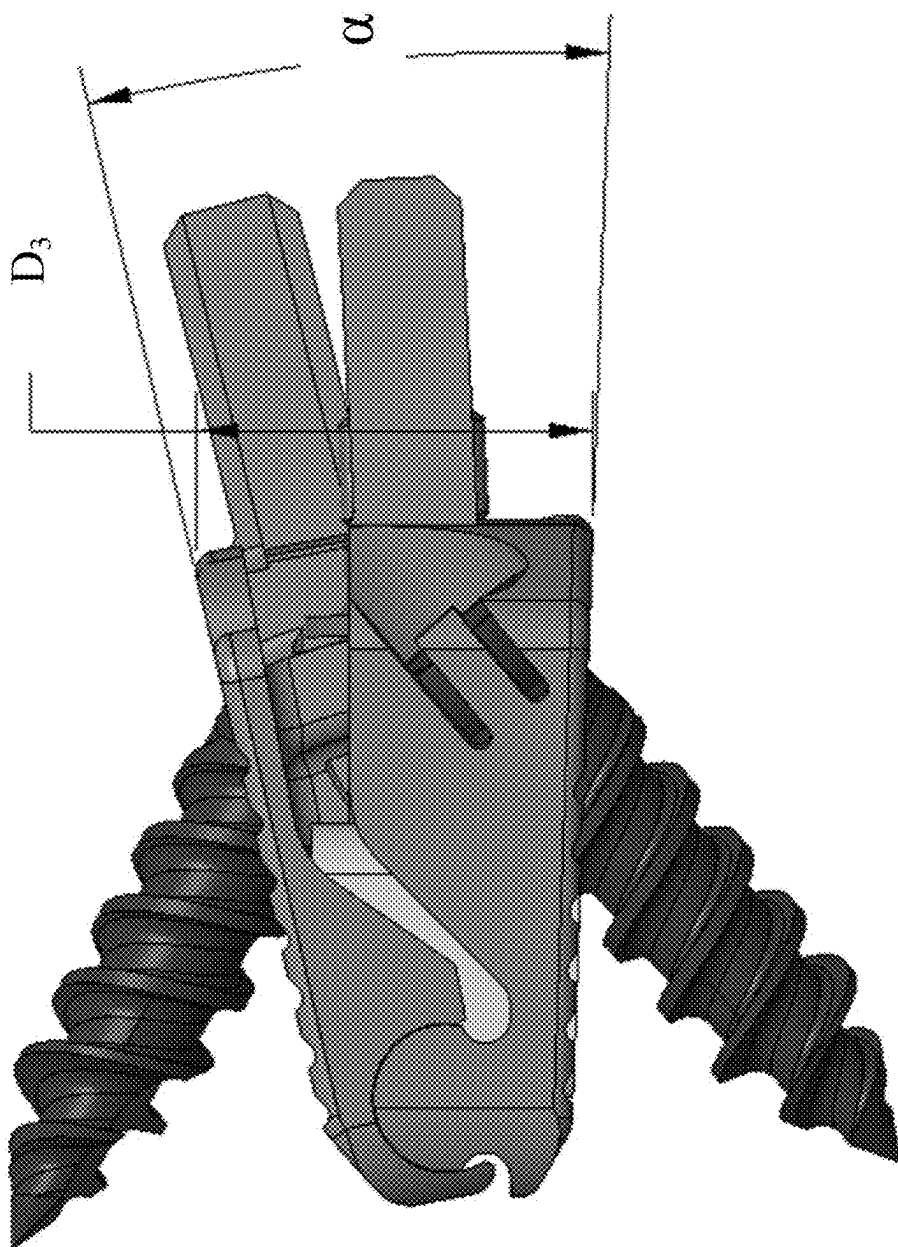
100
FIG. 14B



100
FIG. 14A



100
FIG. 15



100
FIG. 16

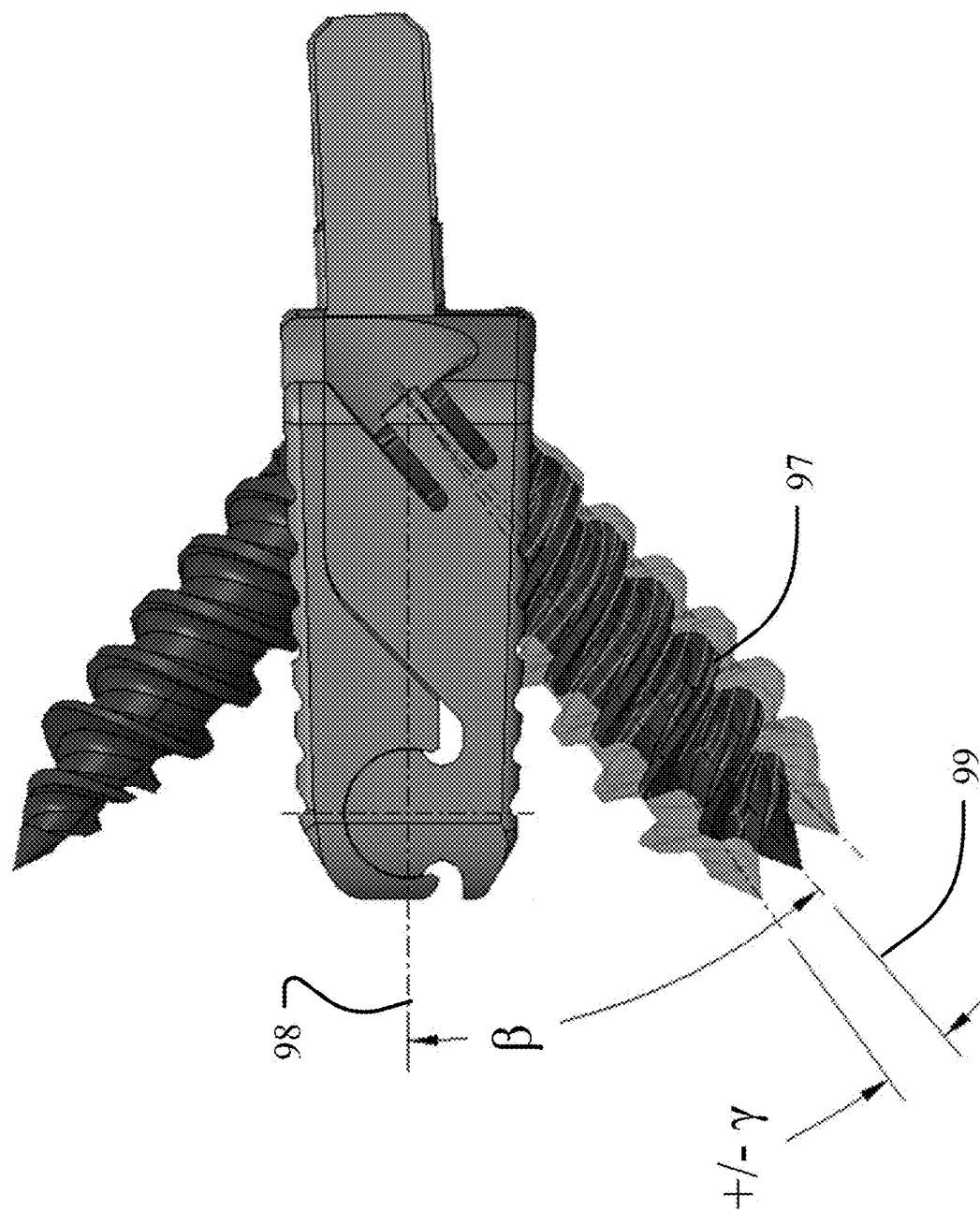


FIG. 17

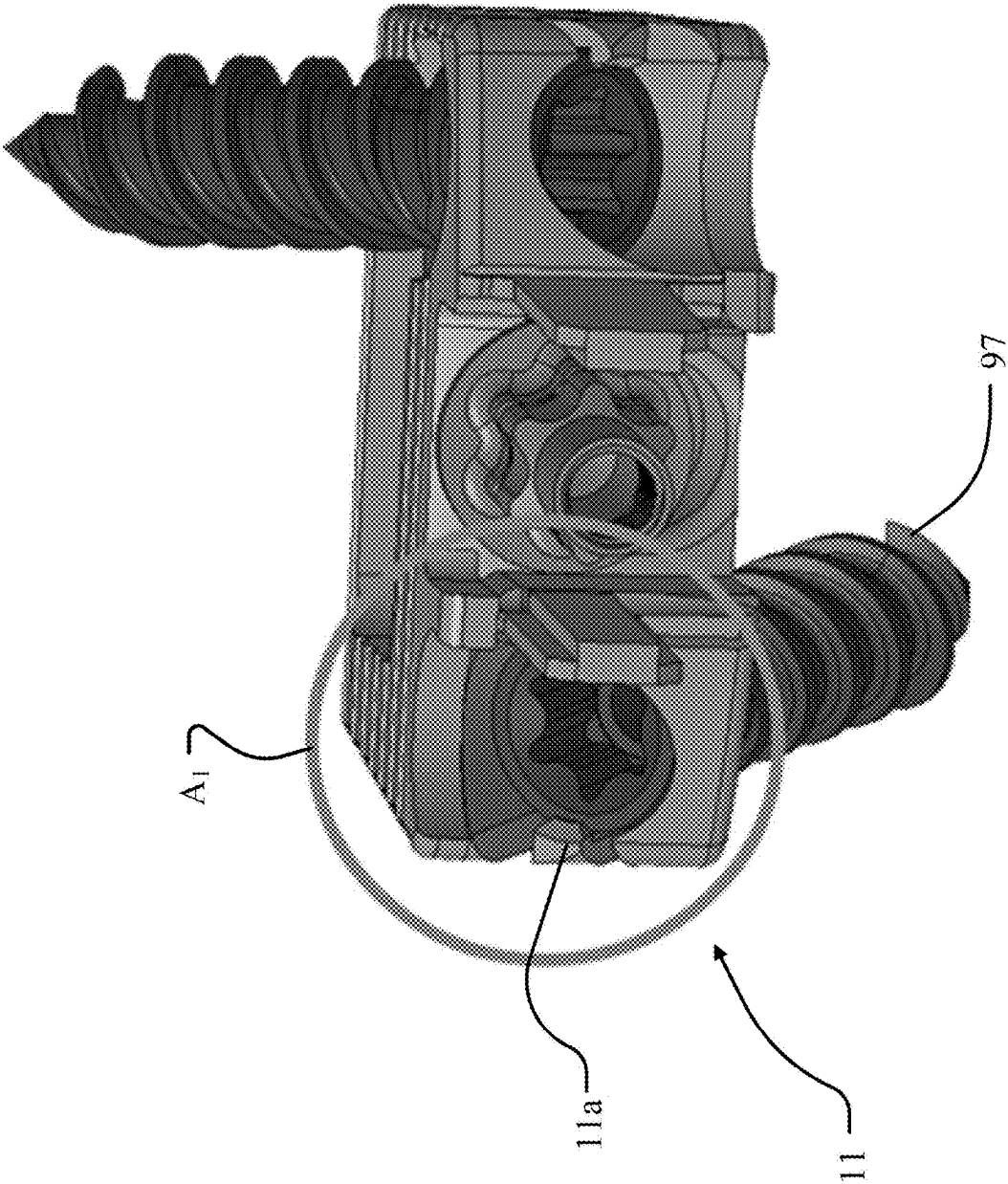


FIG. 18

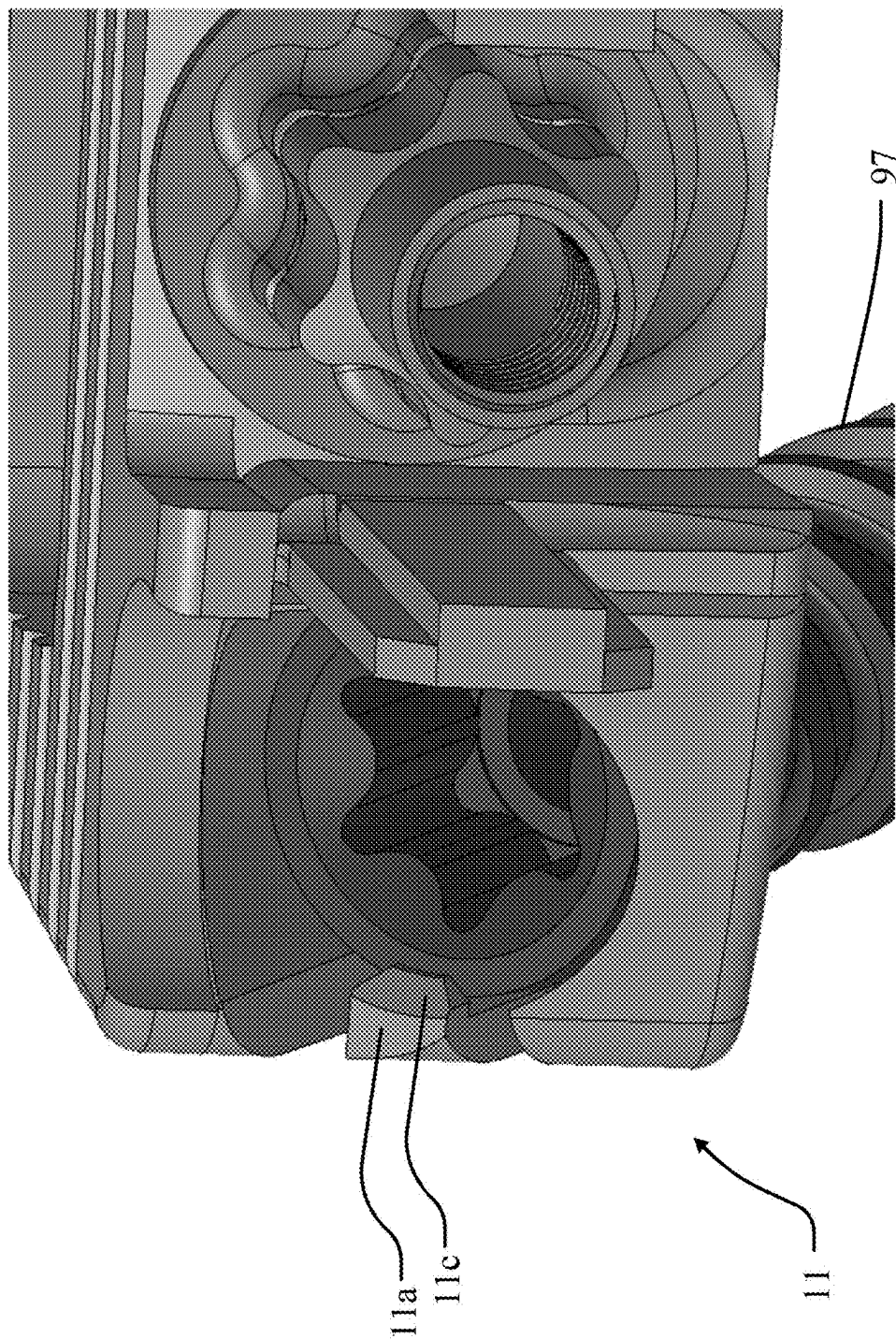
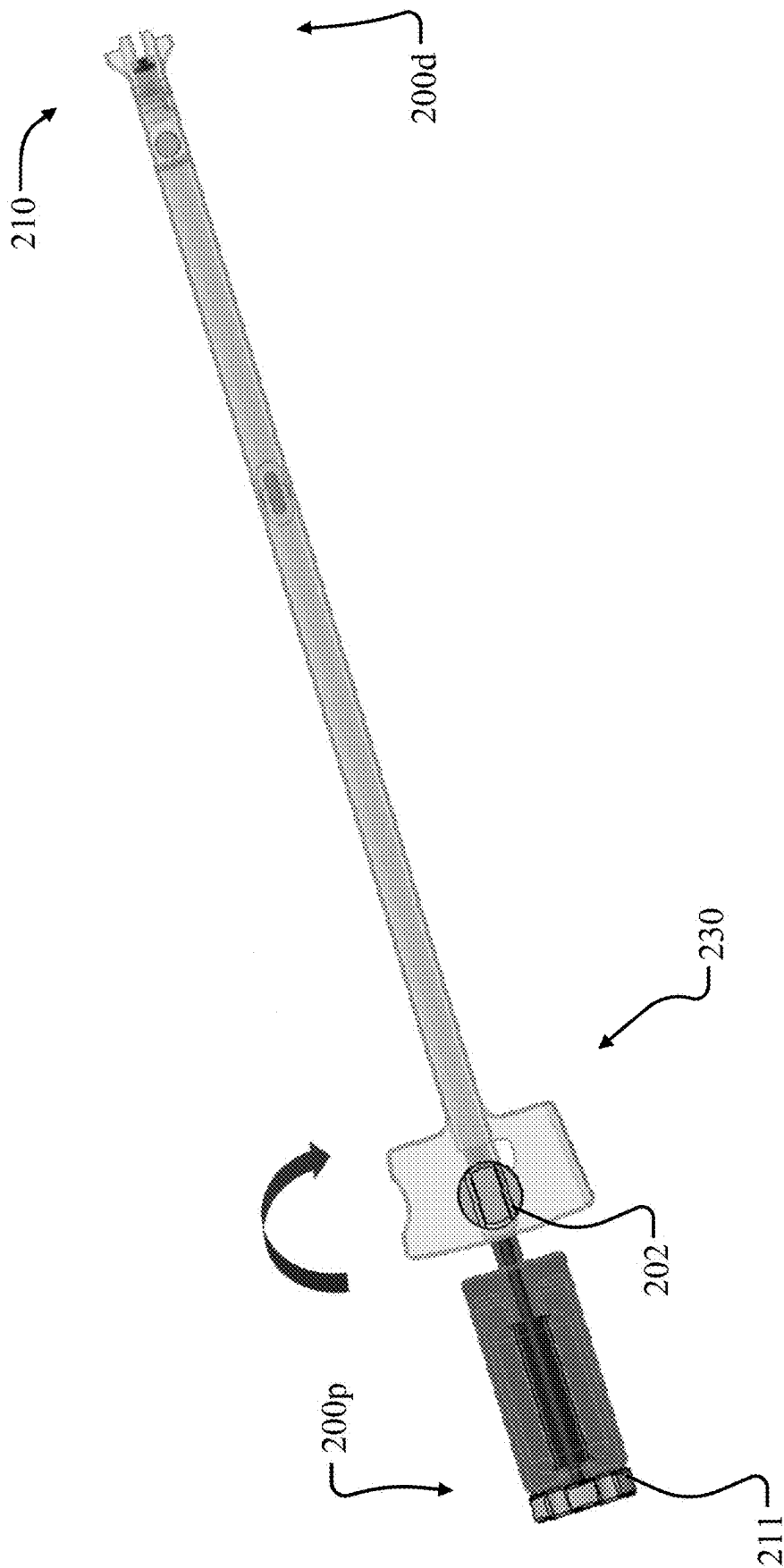


FIG. 19



200
FIG. 20

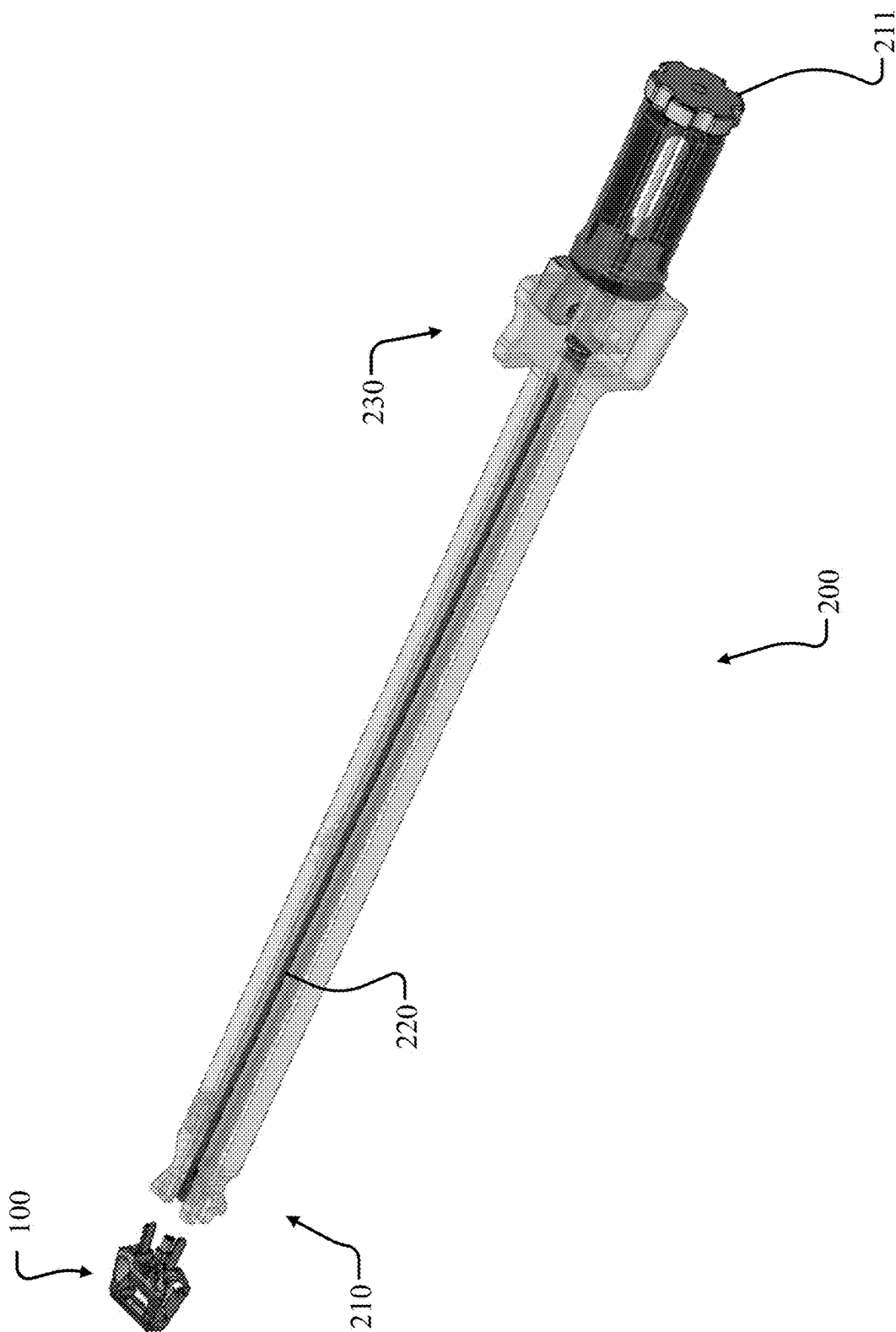


FIG. 21

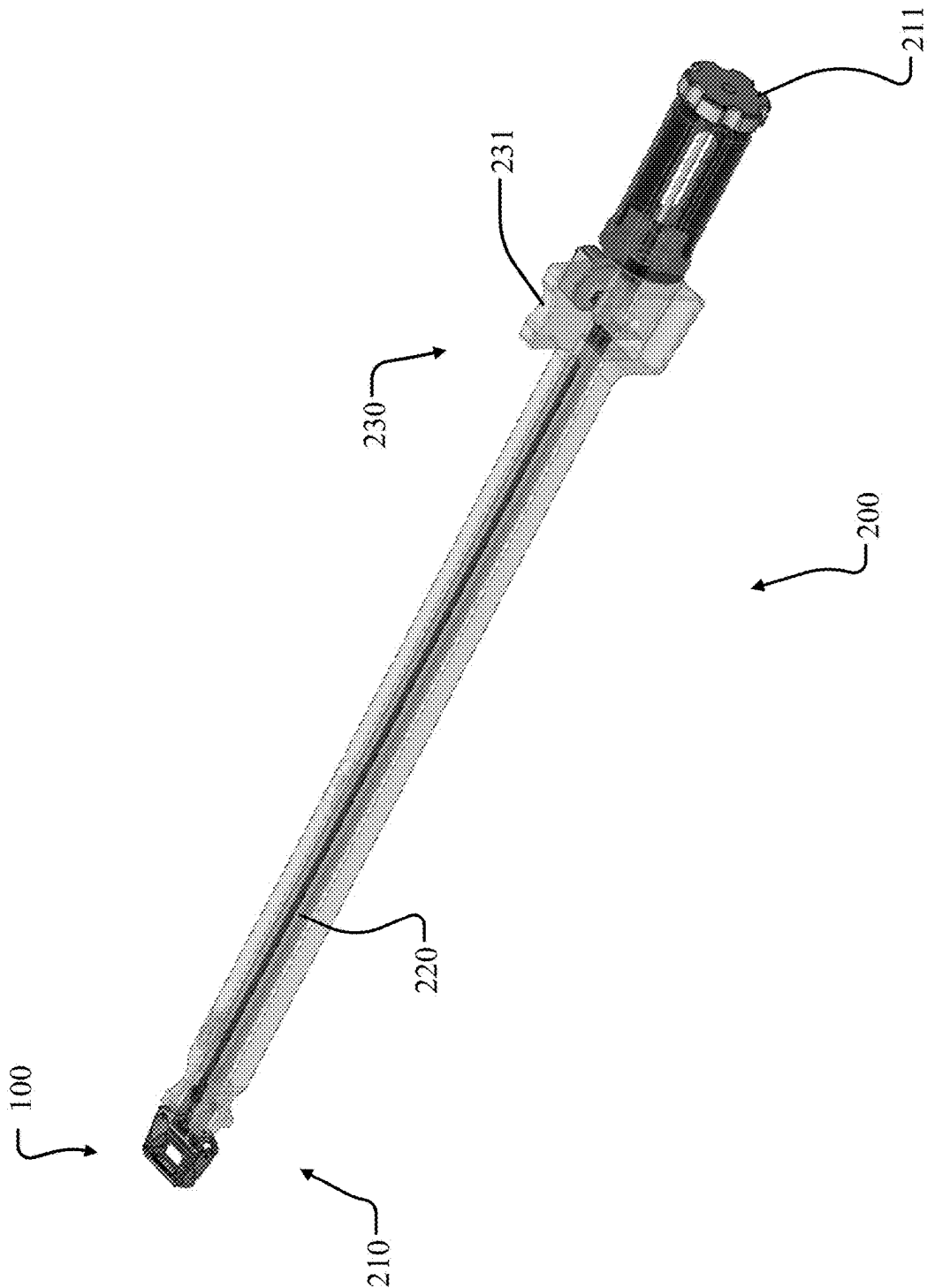
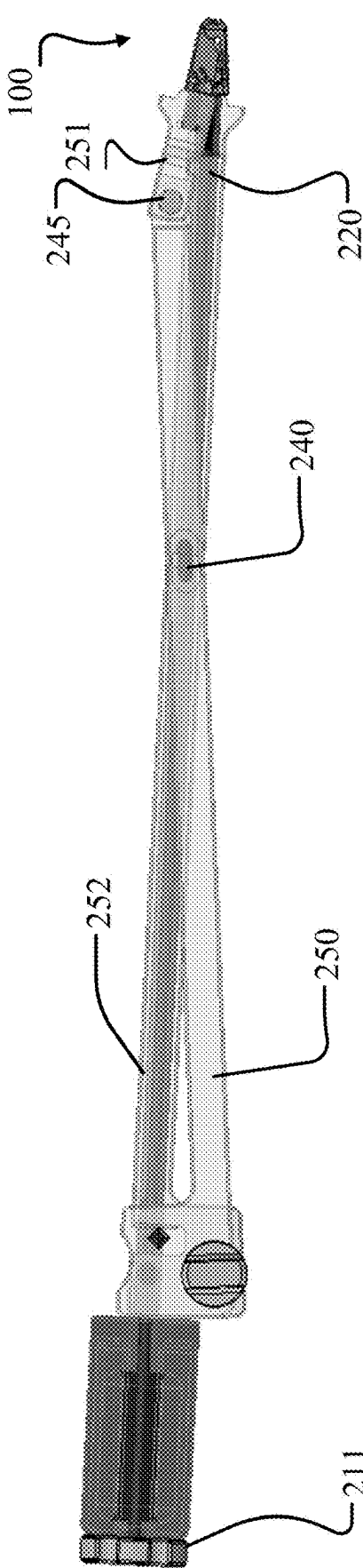
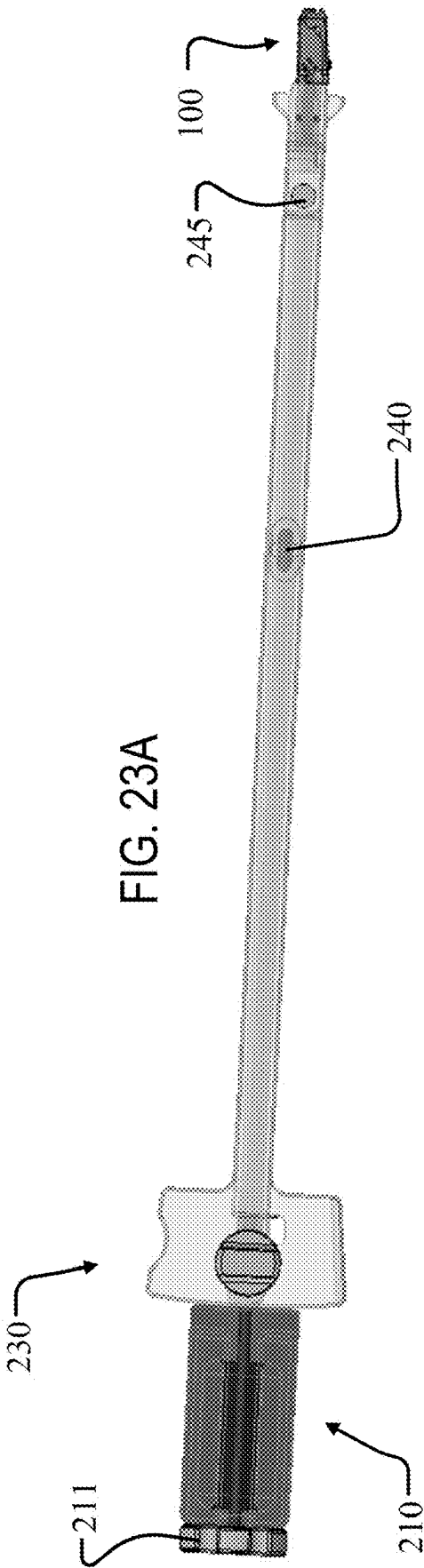


FIG. 22



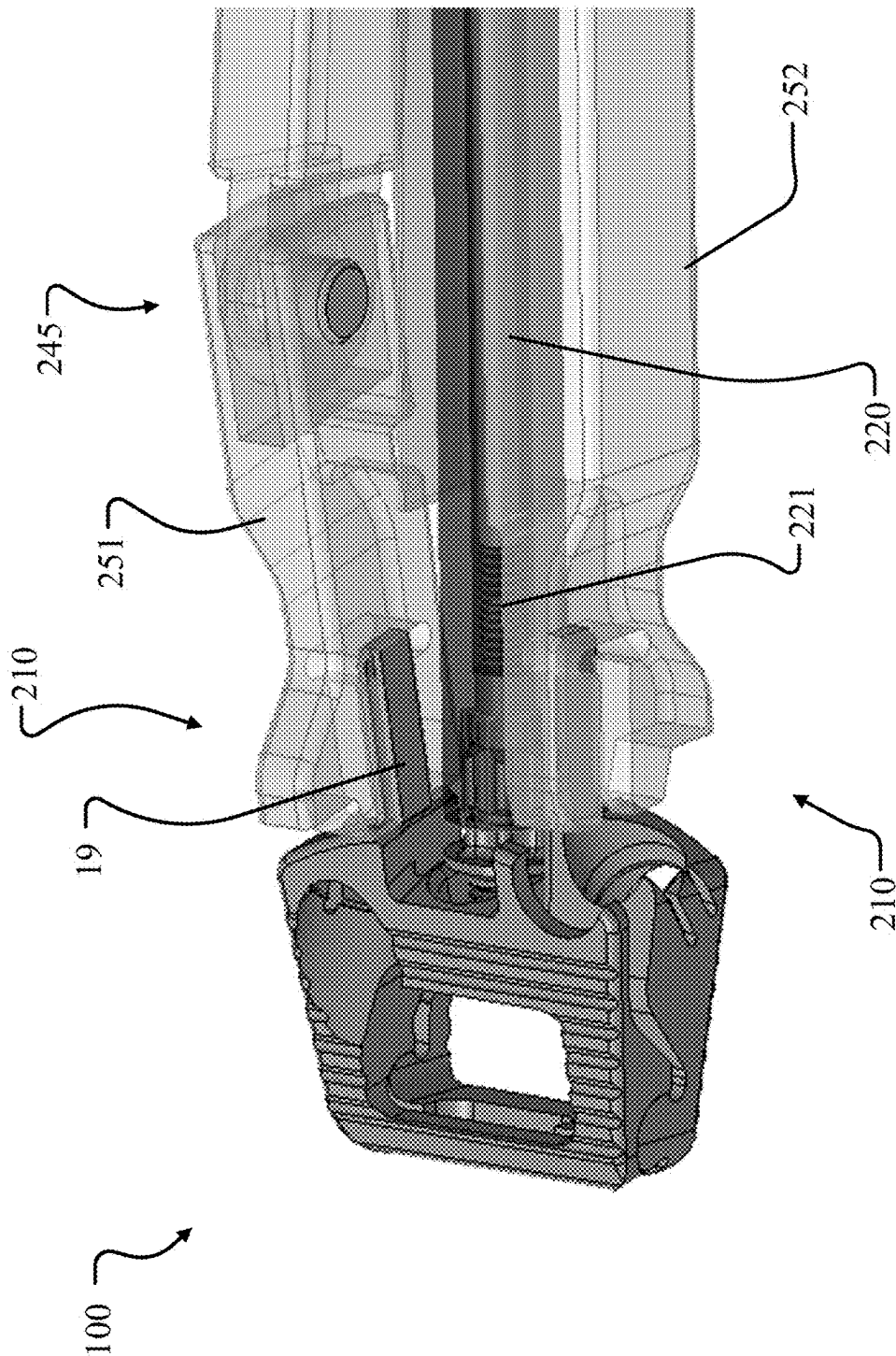


FIG. 24

100

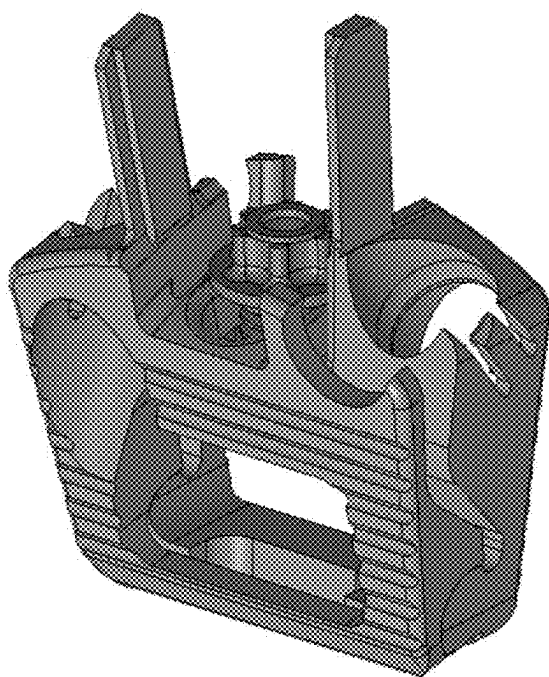
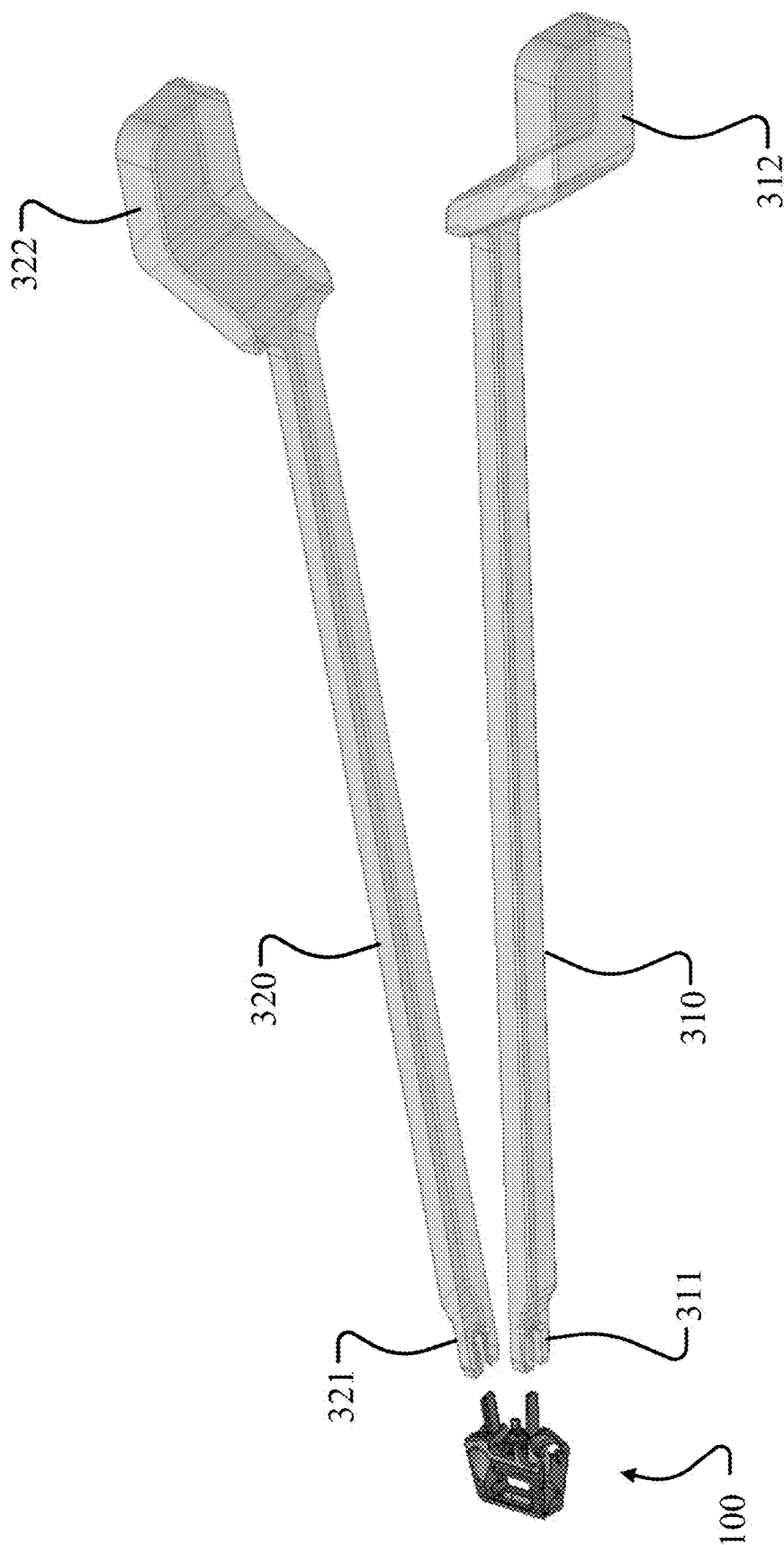
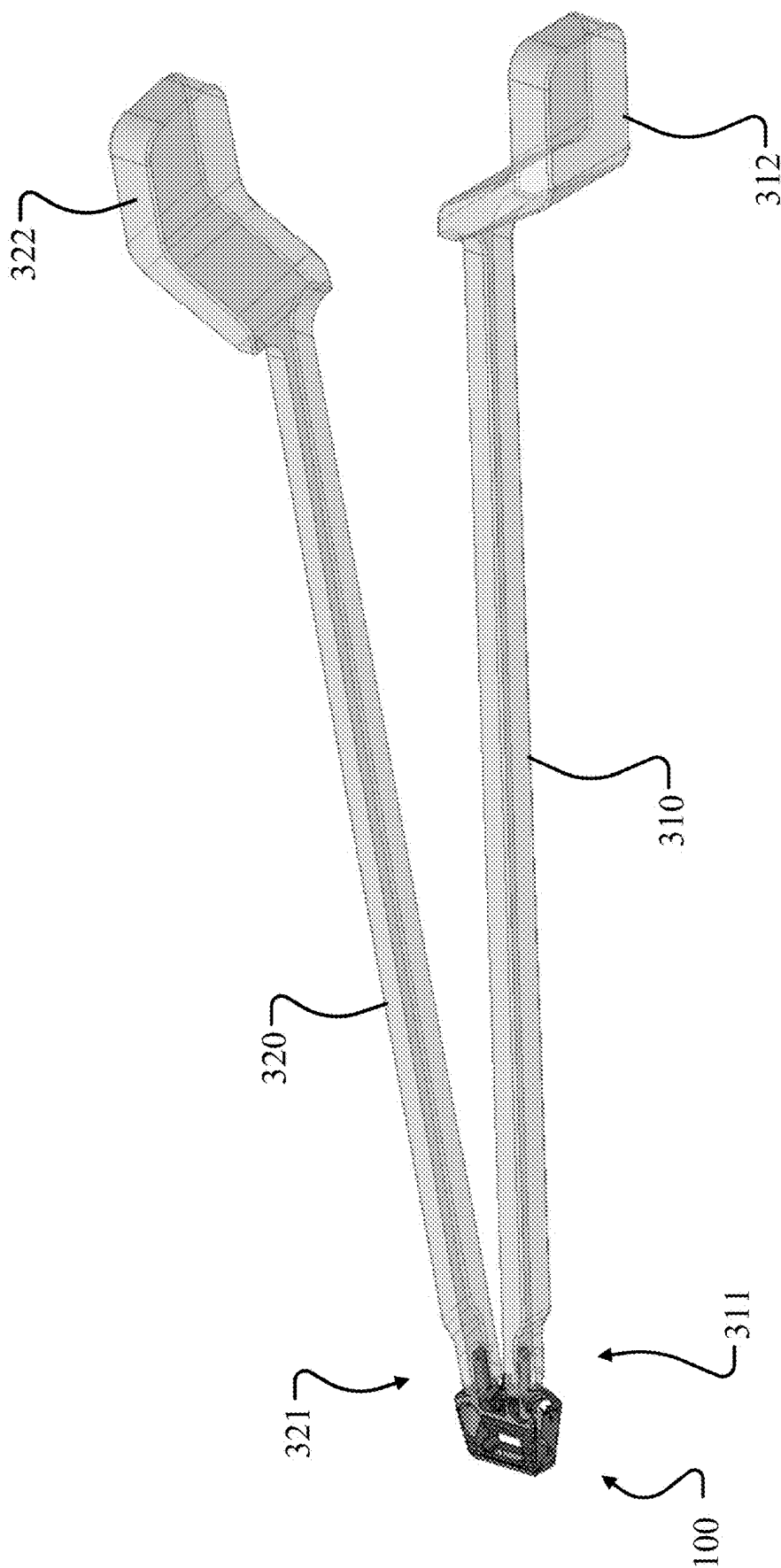


FIG. 25



300
FIG. 26



300
FIG. 27

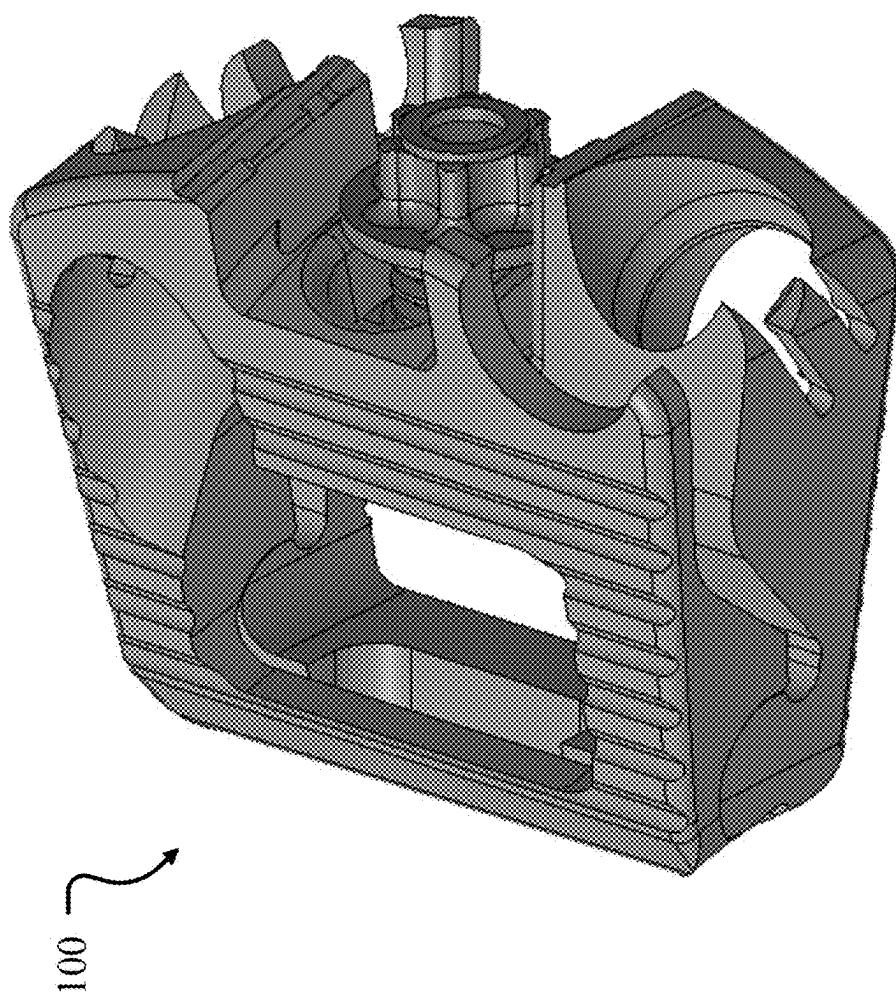


FIG. 28

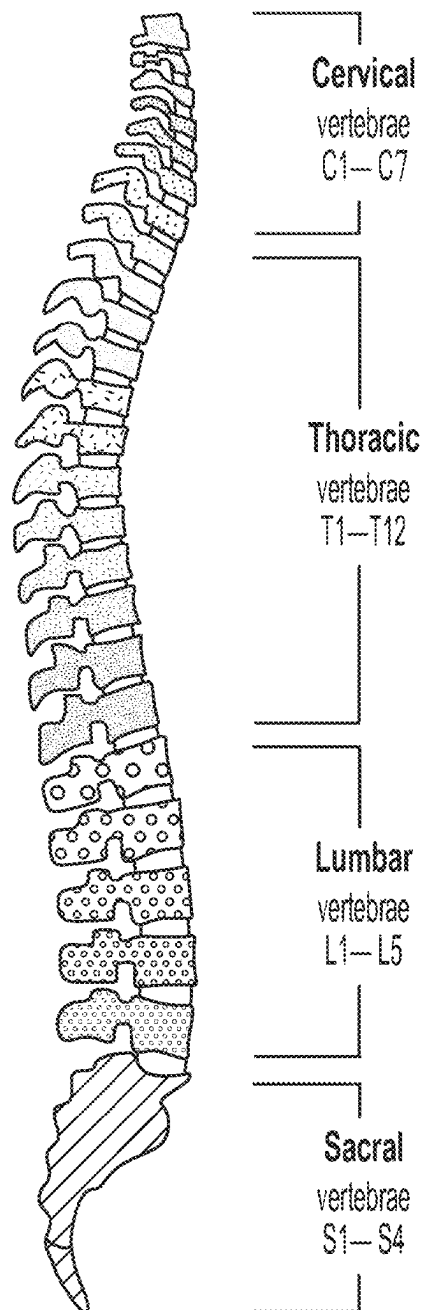


FIG. 29

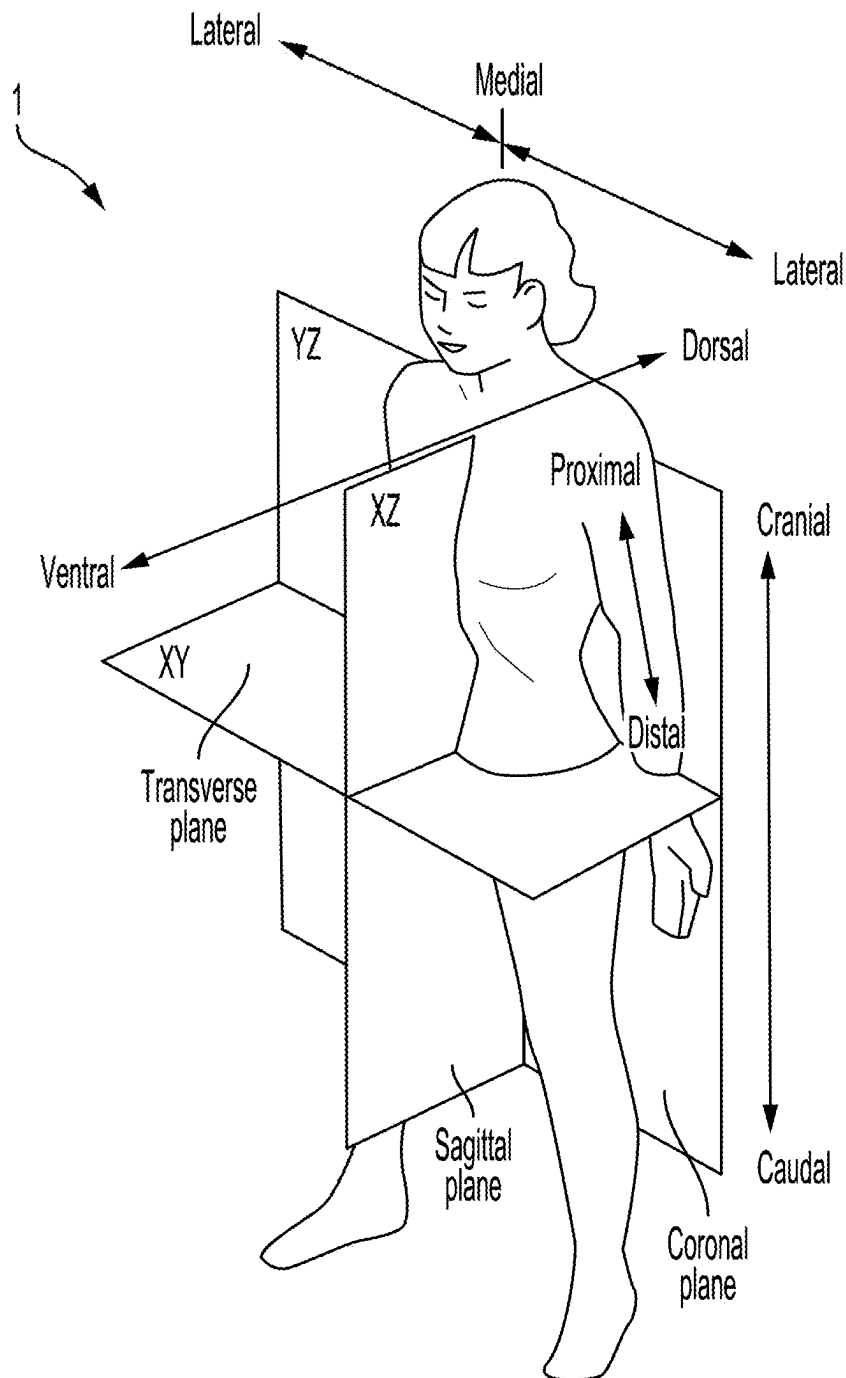


FIG. 30

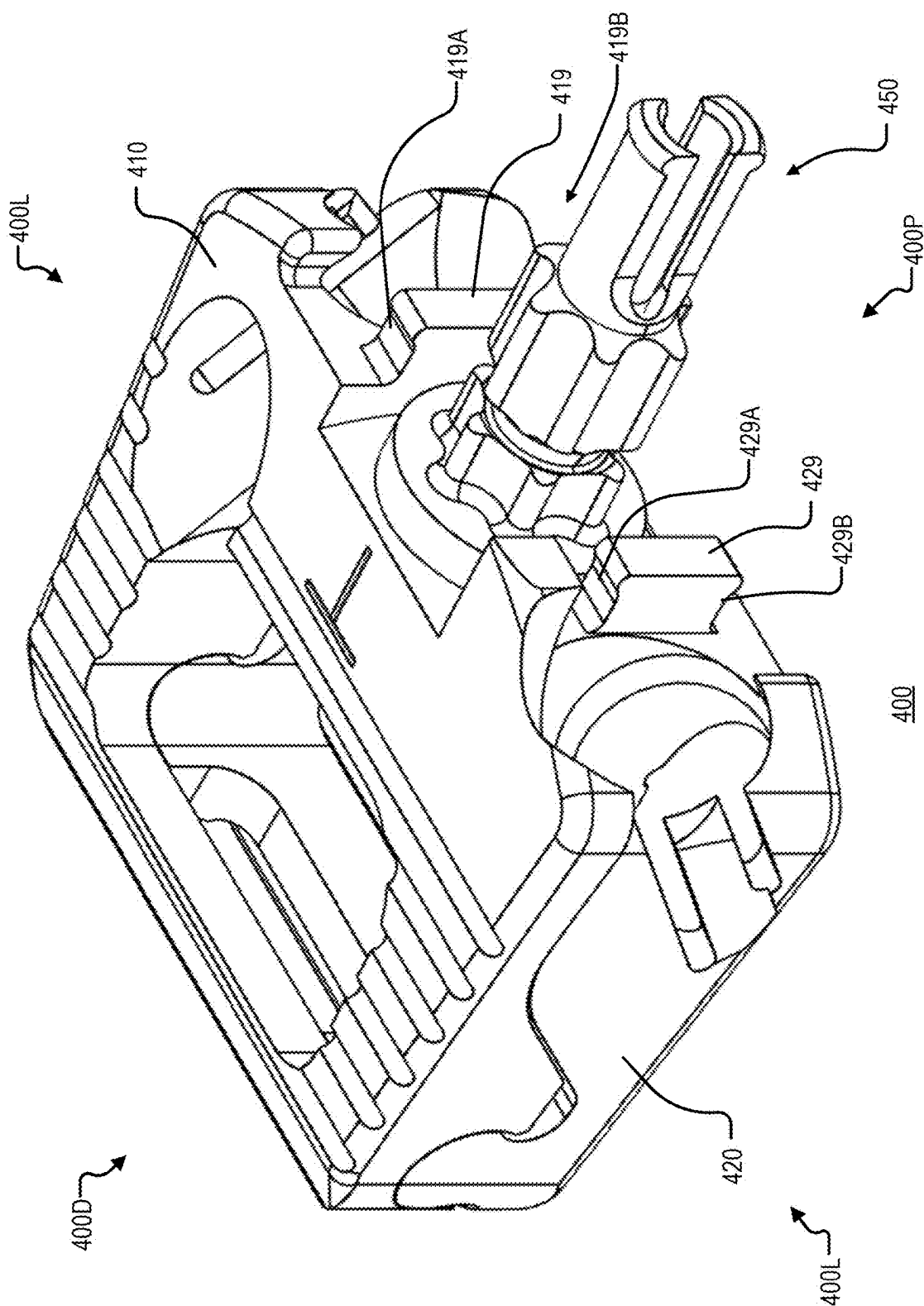
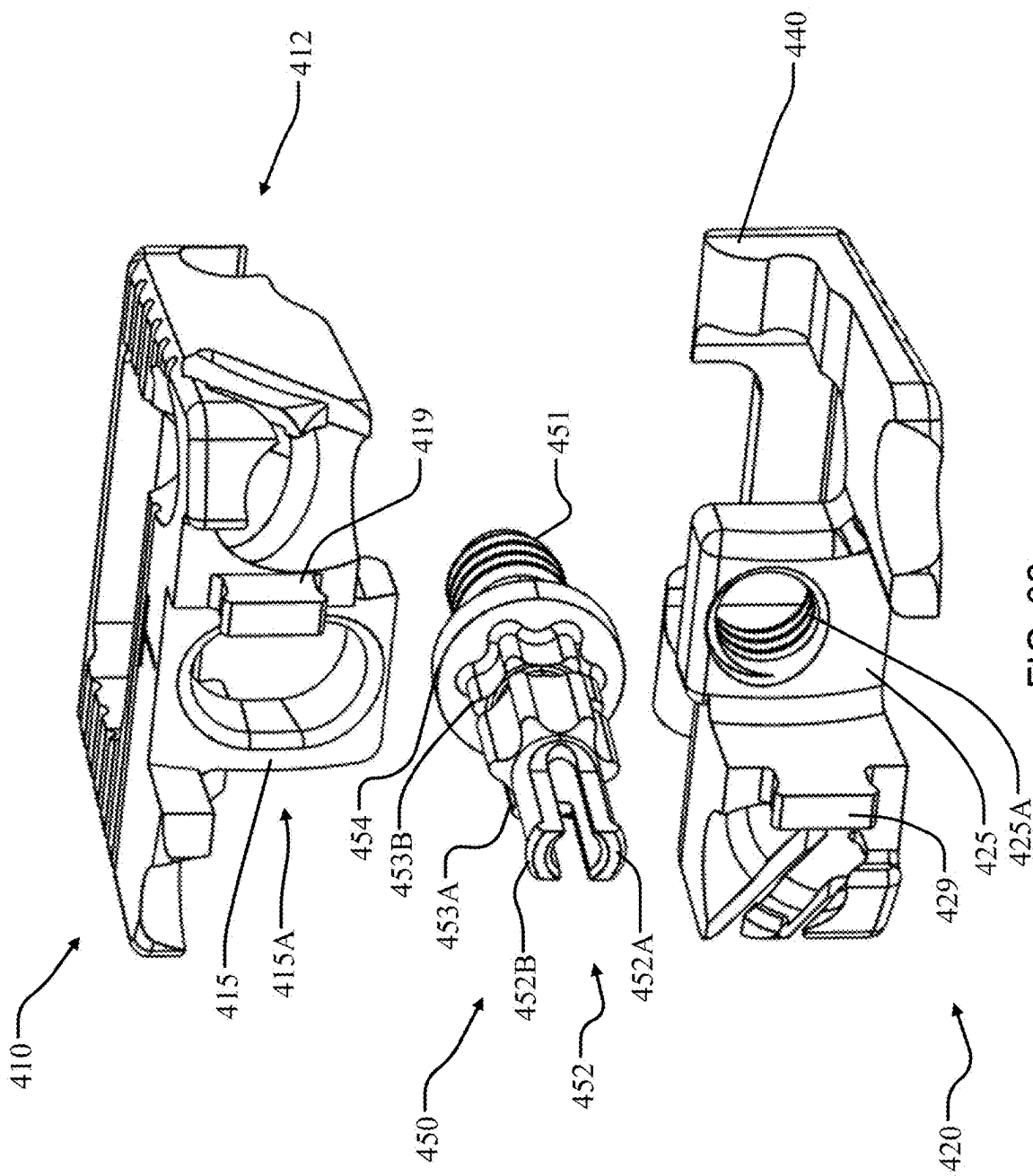


FIG. 31



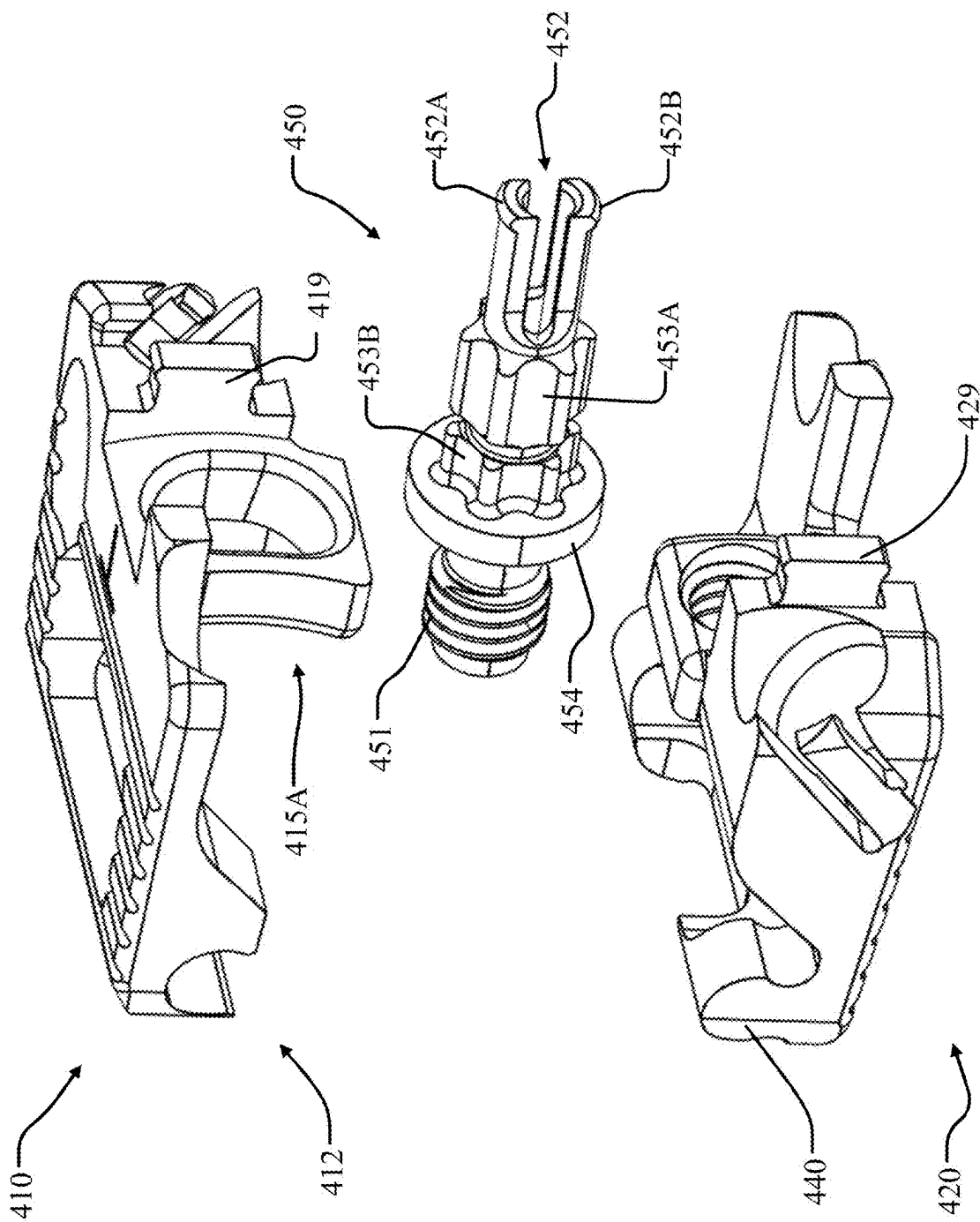


FIG. 33

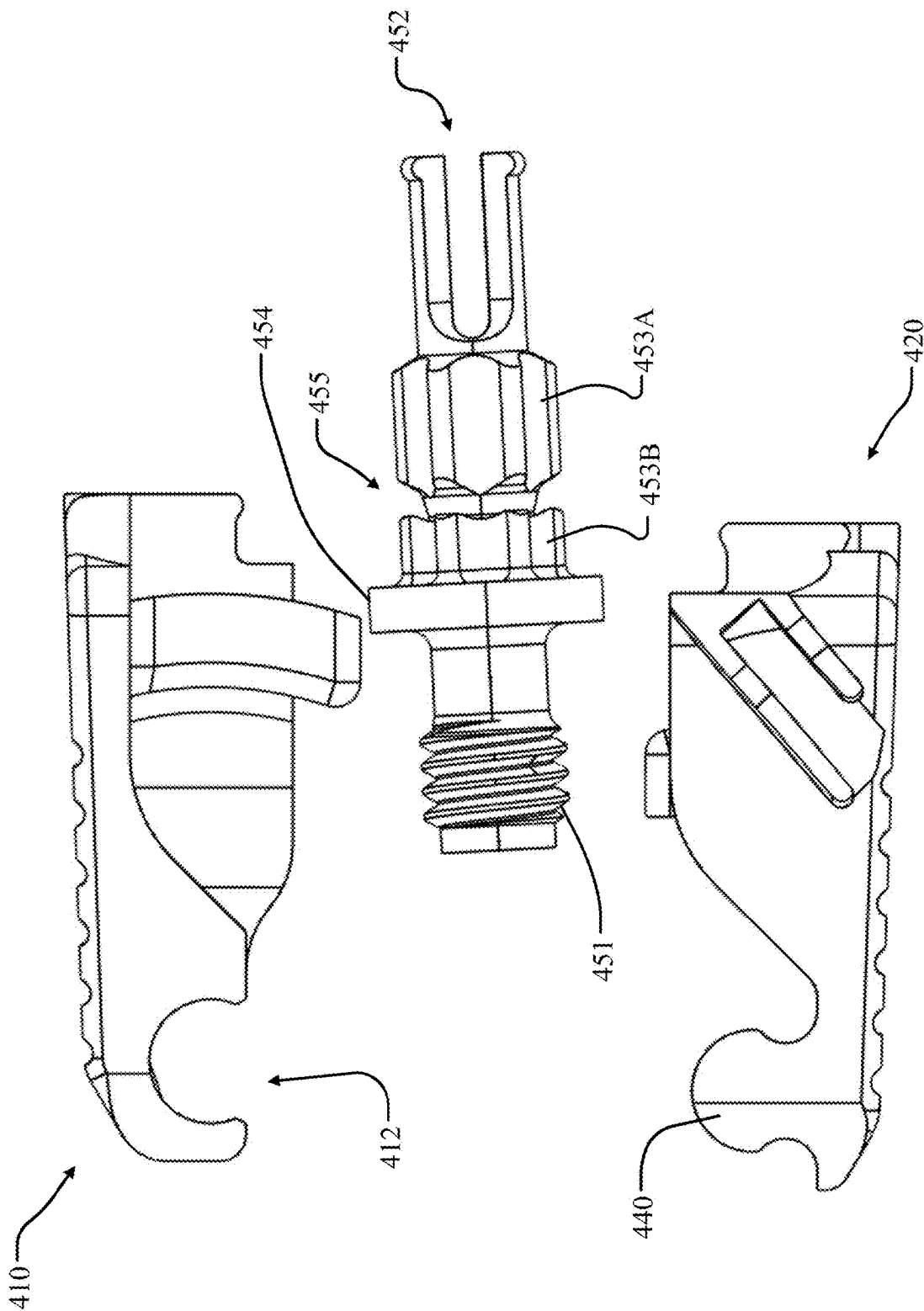


FIG. 34

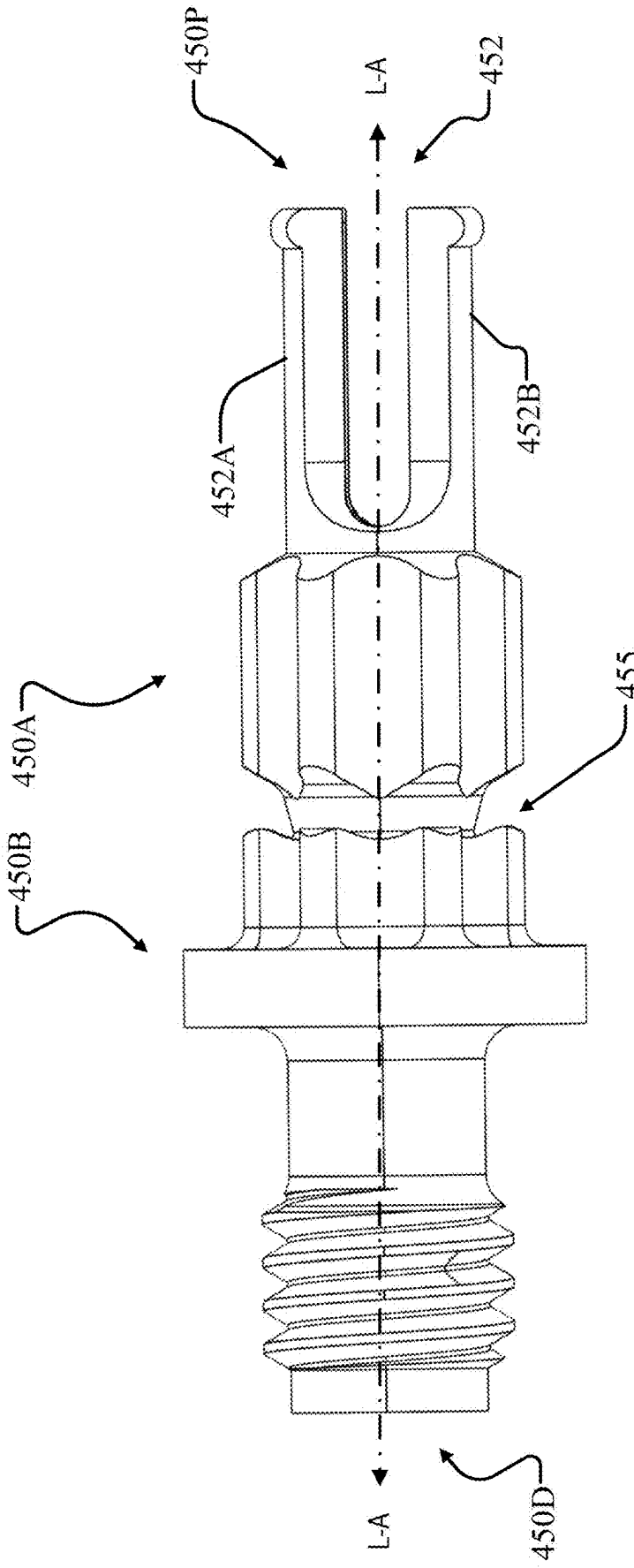


FIG. 35

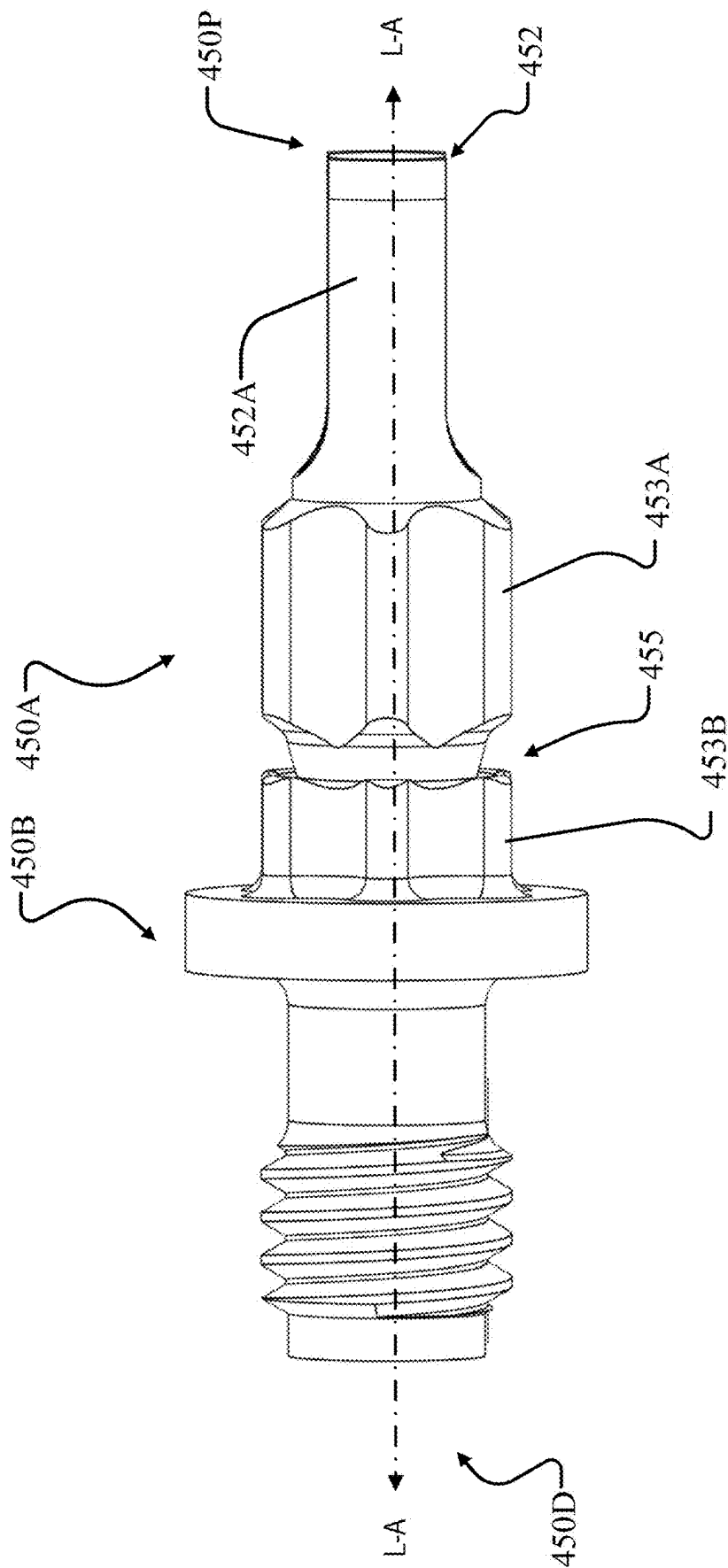


FIG. 36

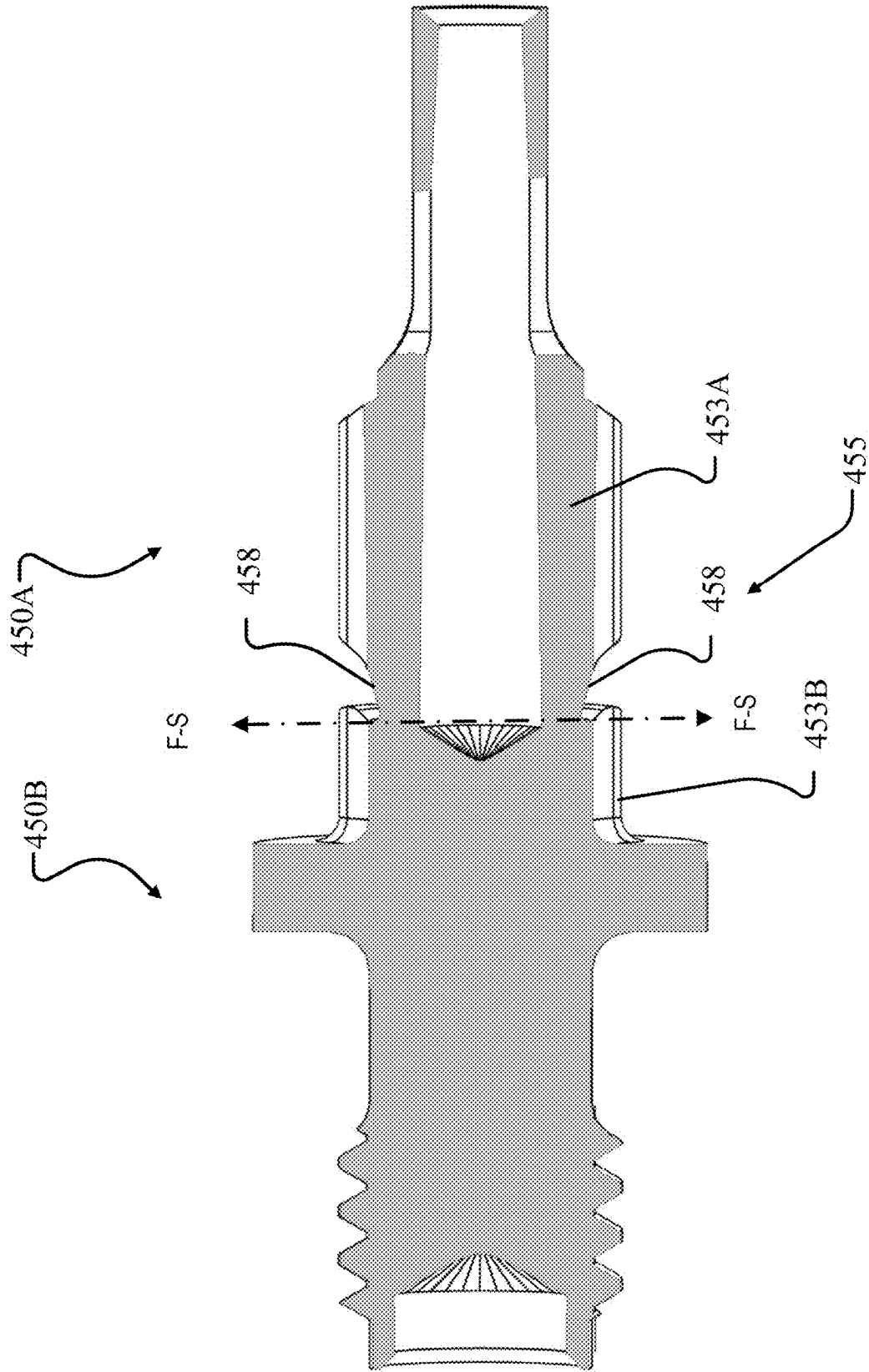


FIG. 37

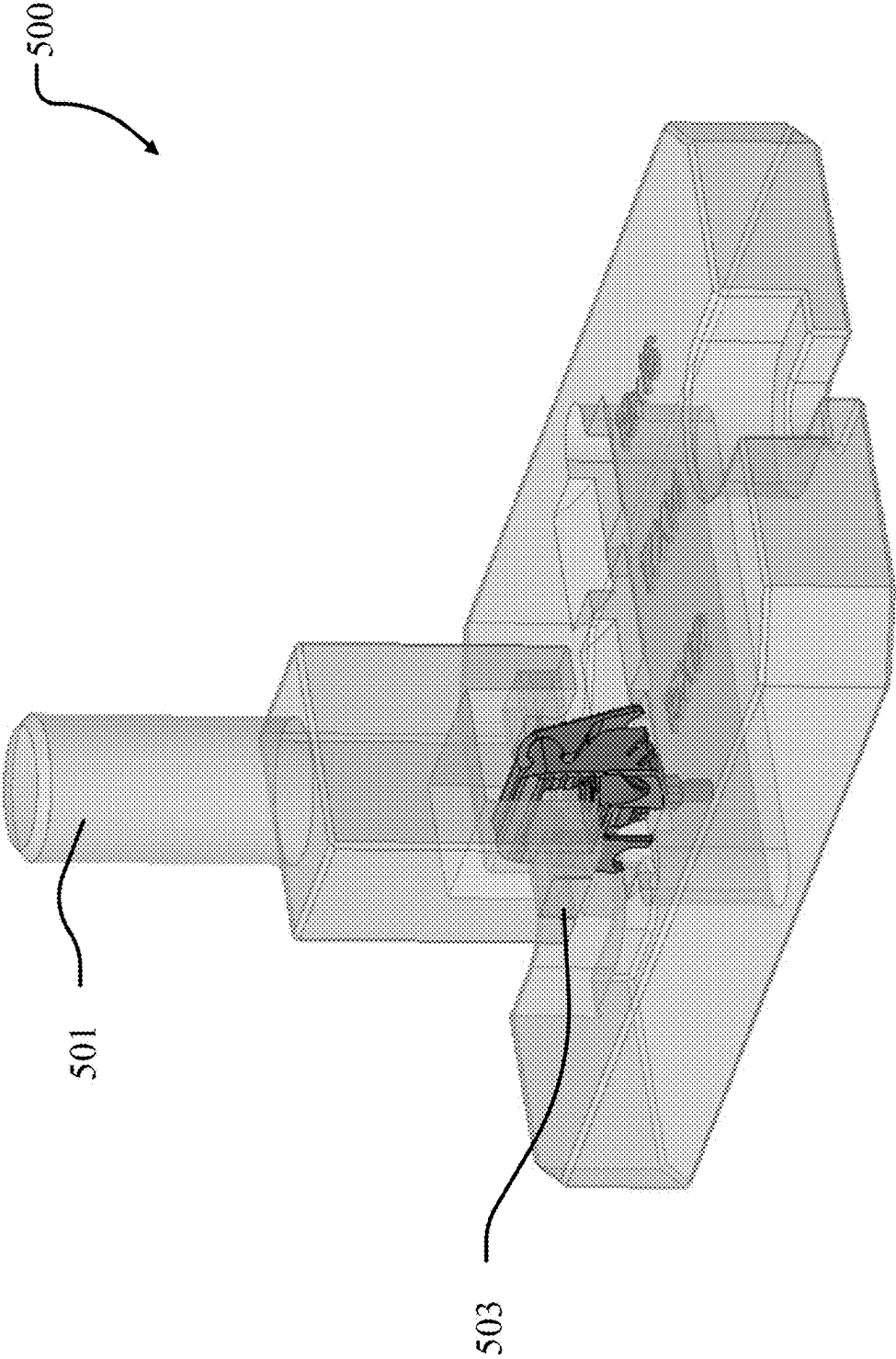


FIG. 38

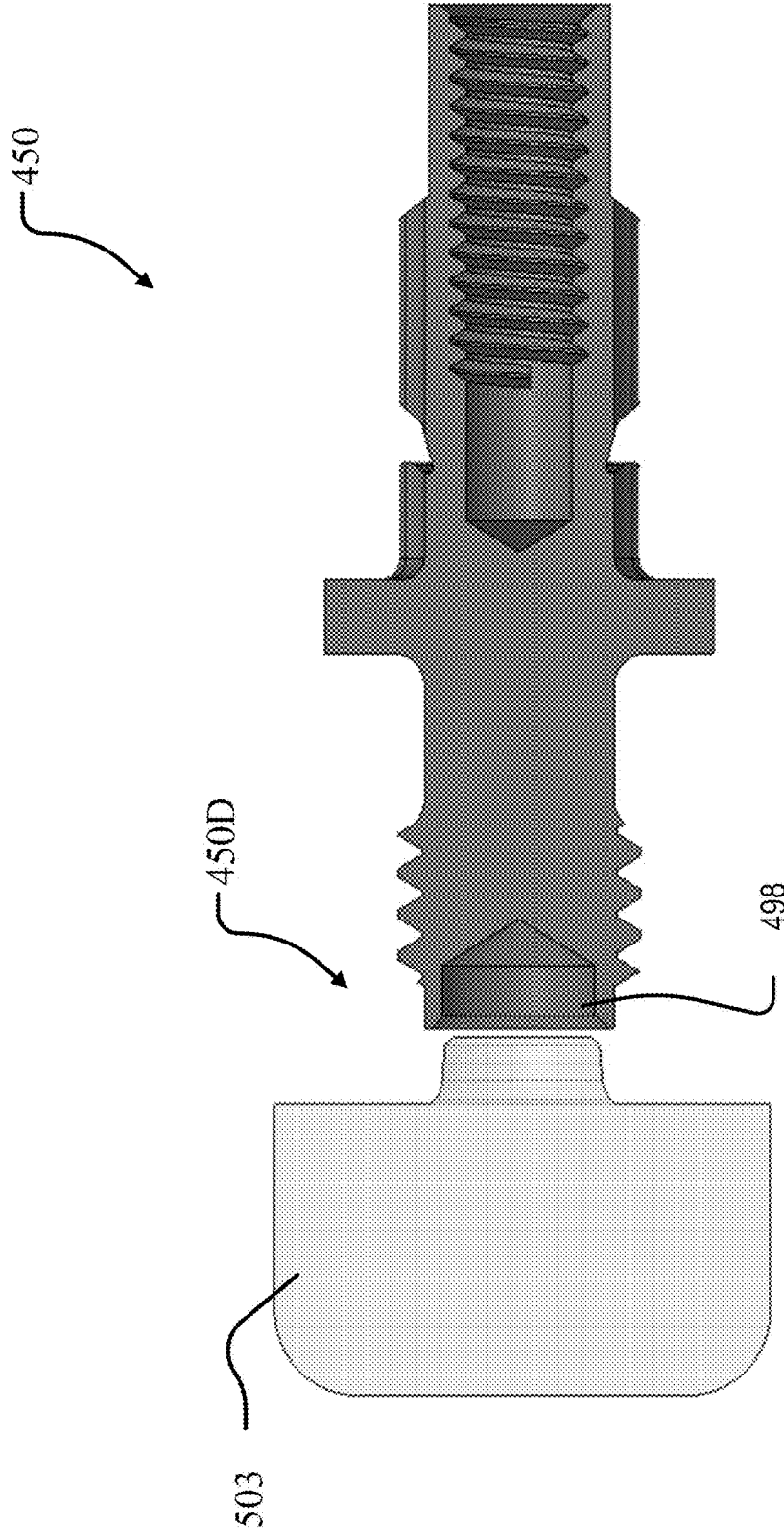


FIG. 39

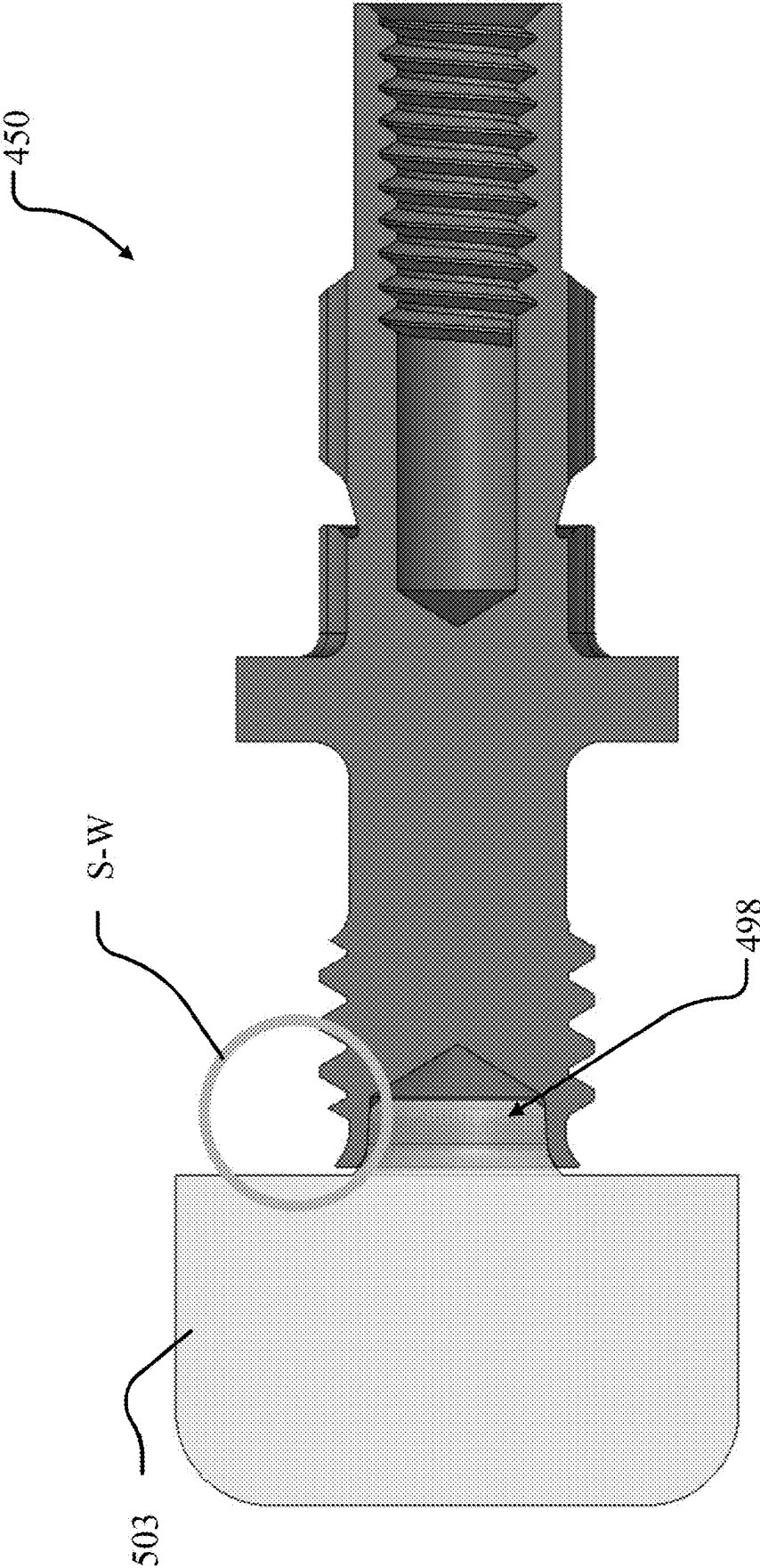
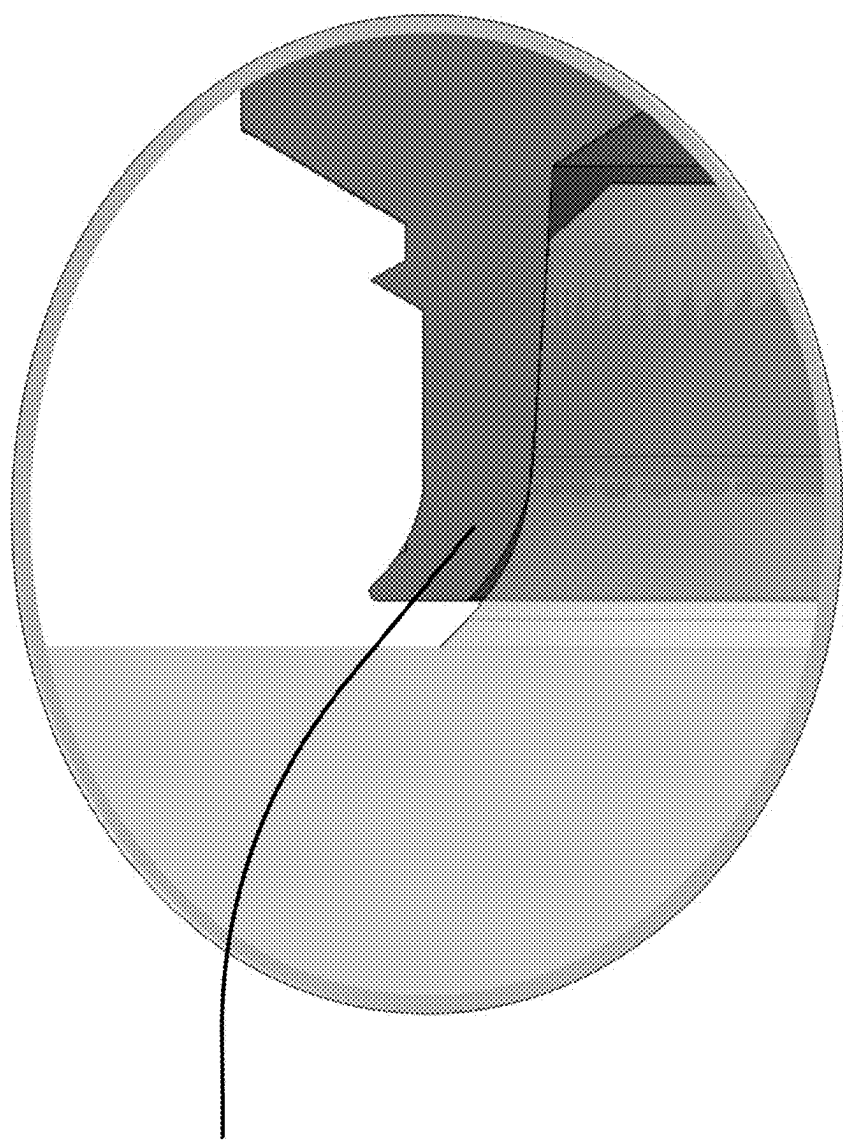
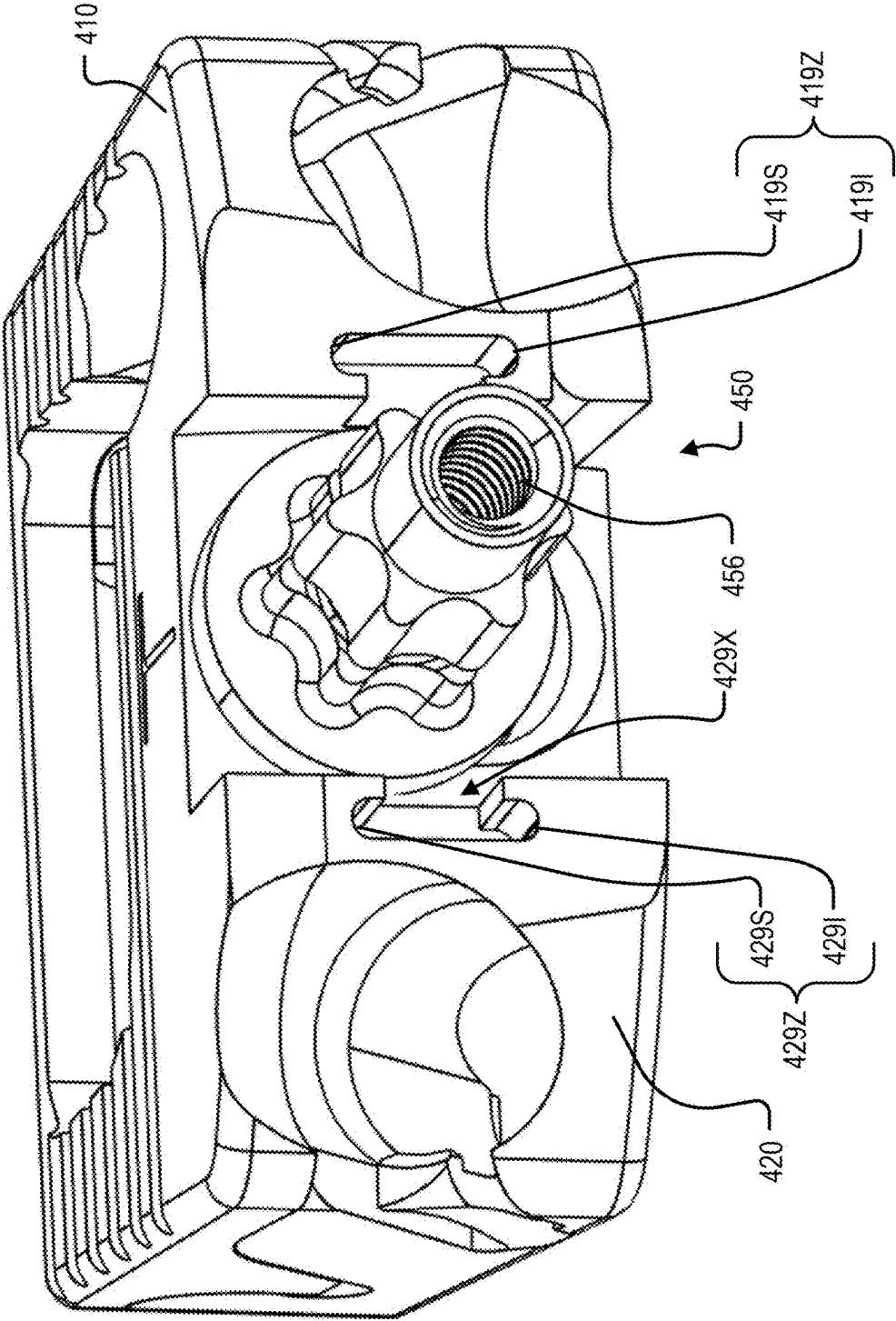


FIG. 40

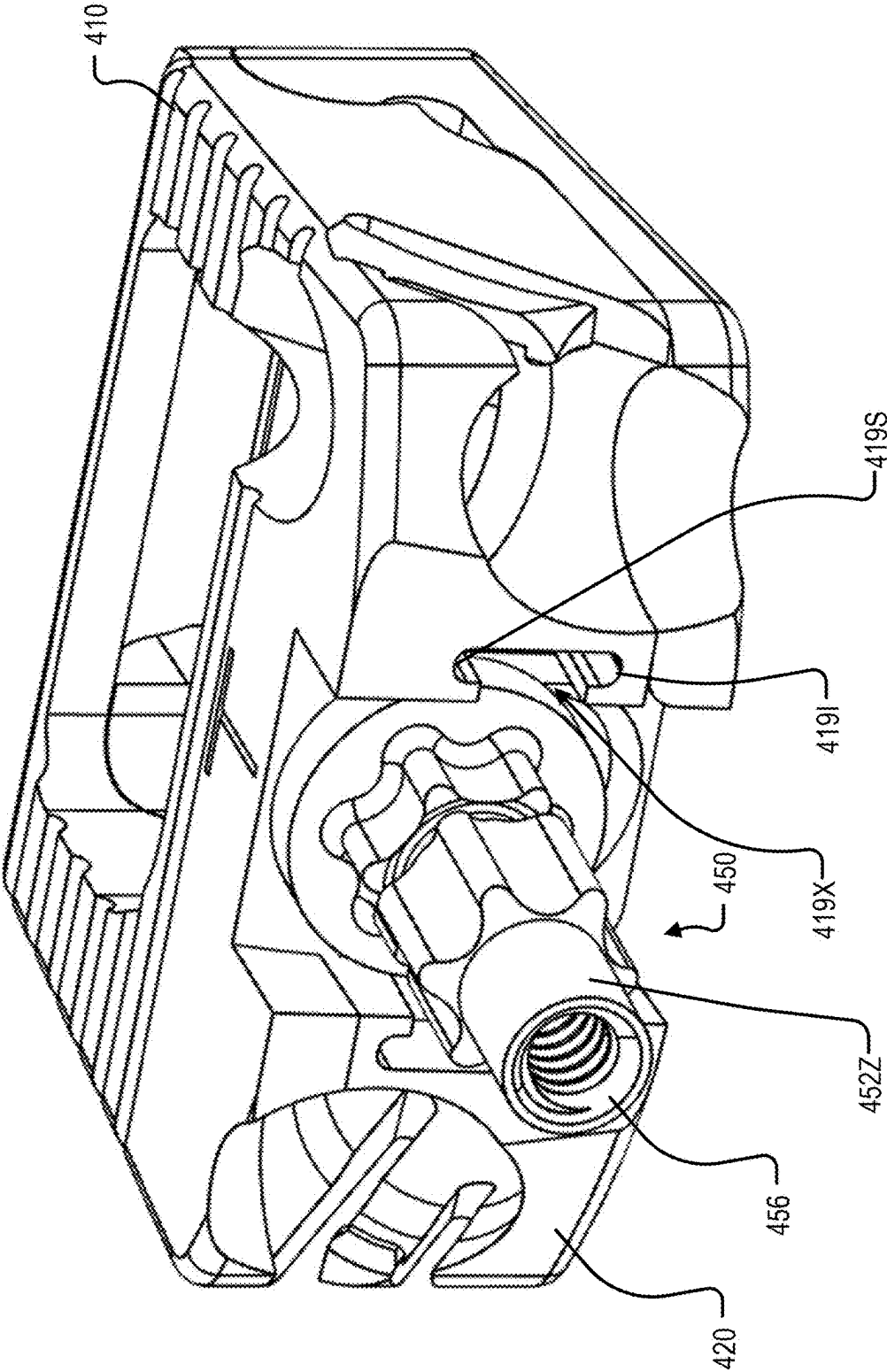


499

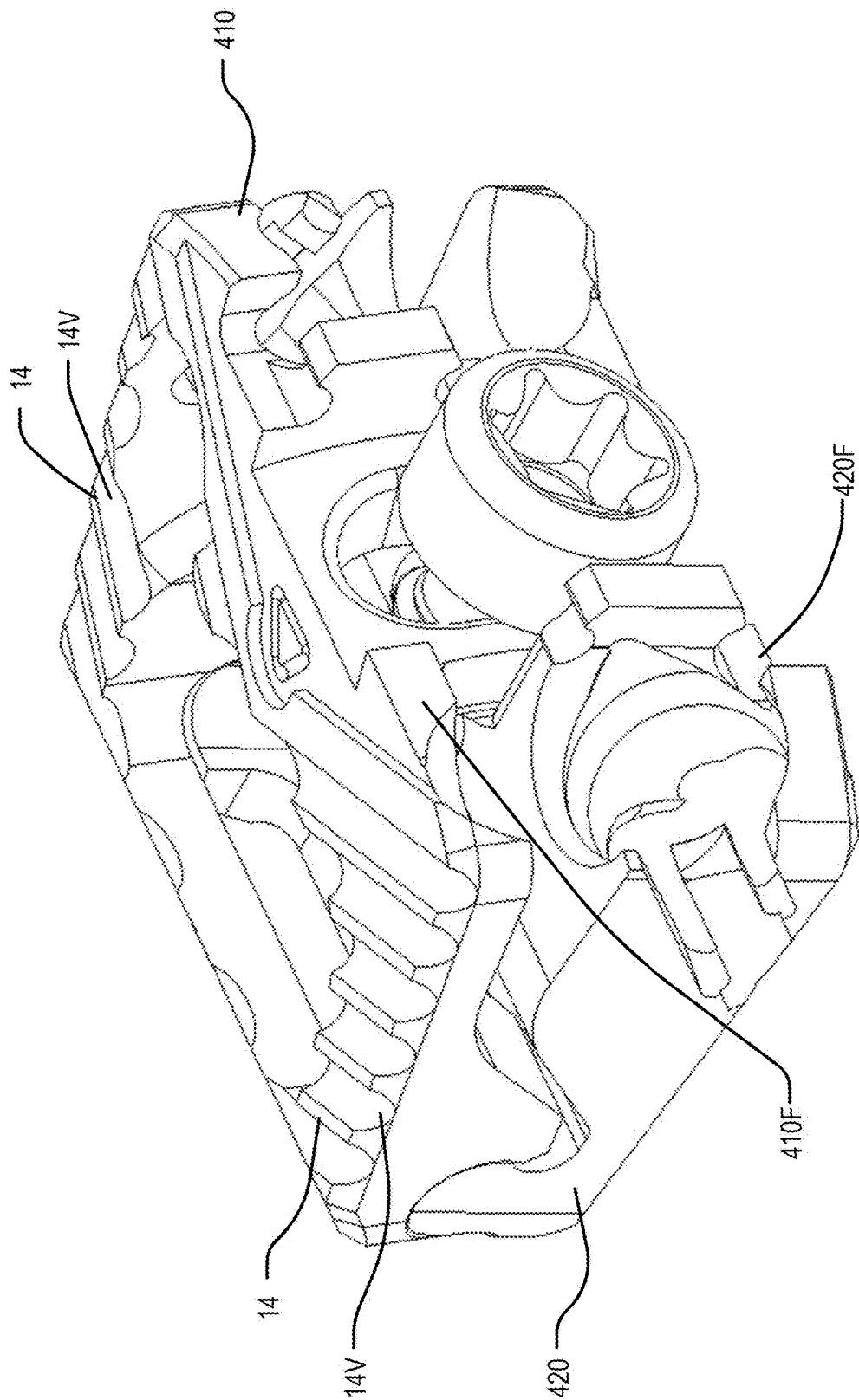
FIG. 41



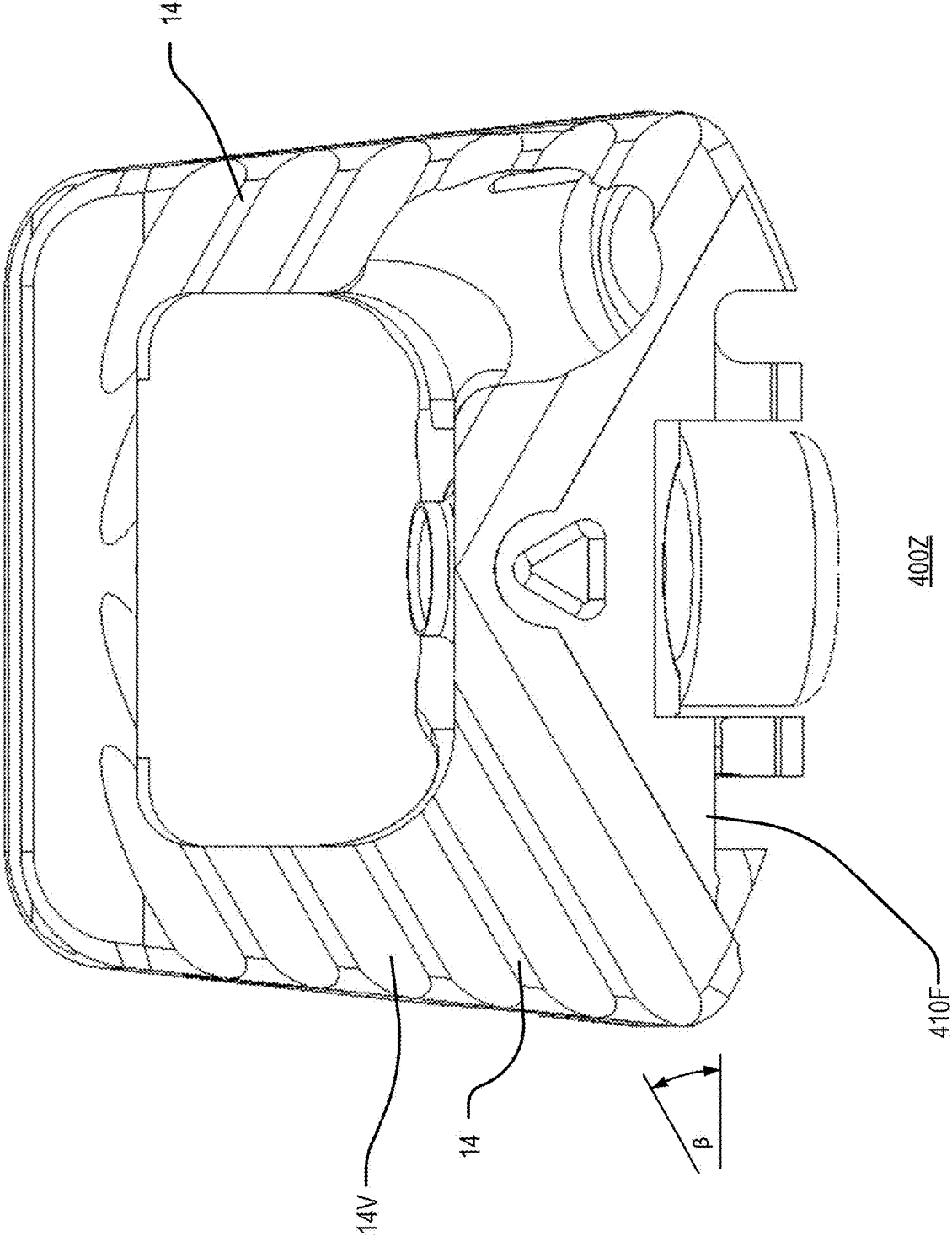
400Z
FIG. 42A



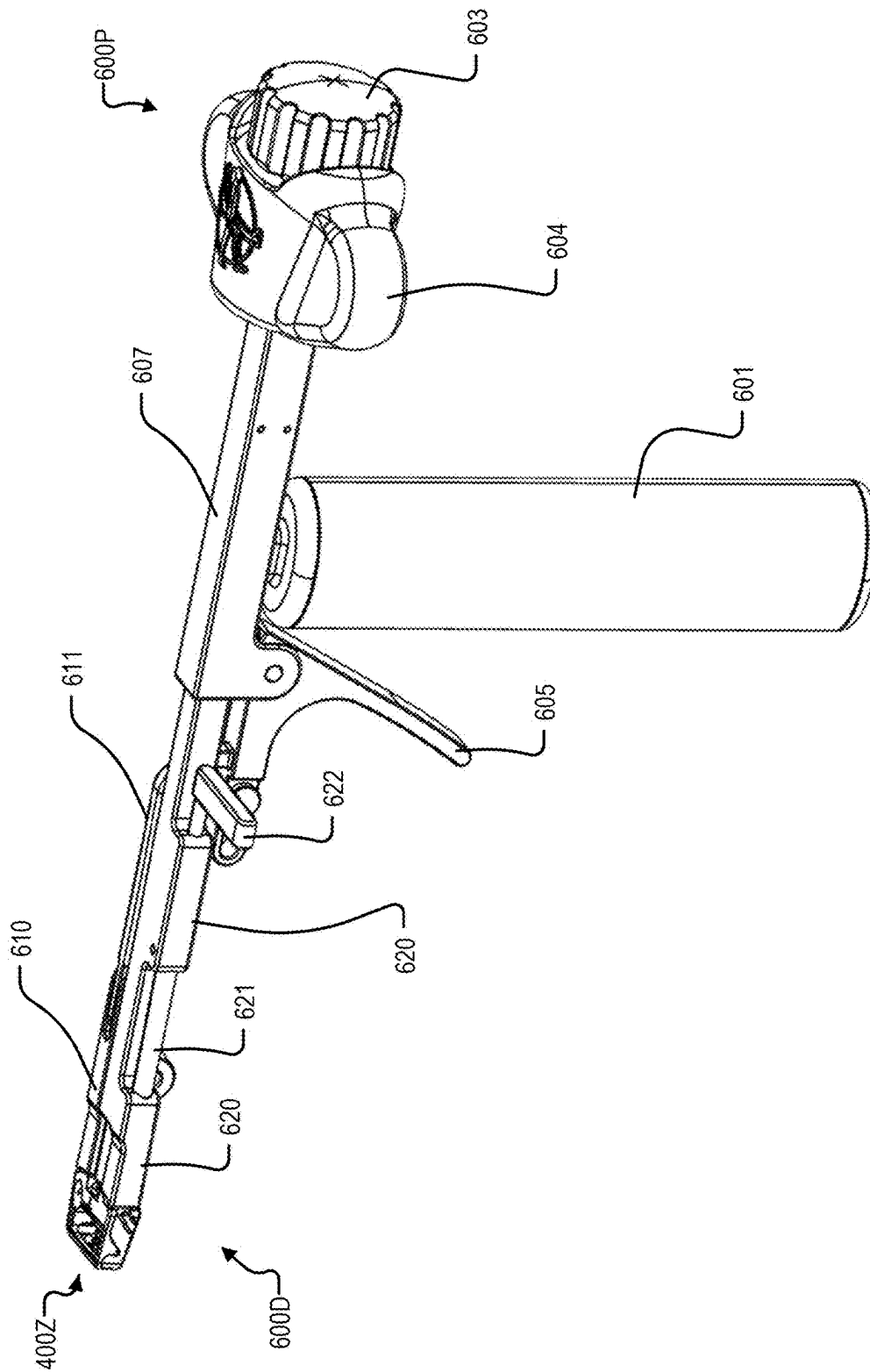
400Z
FIG. 42B



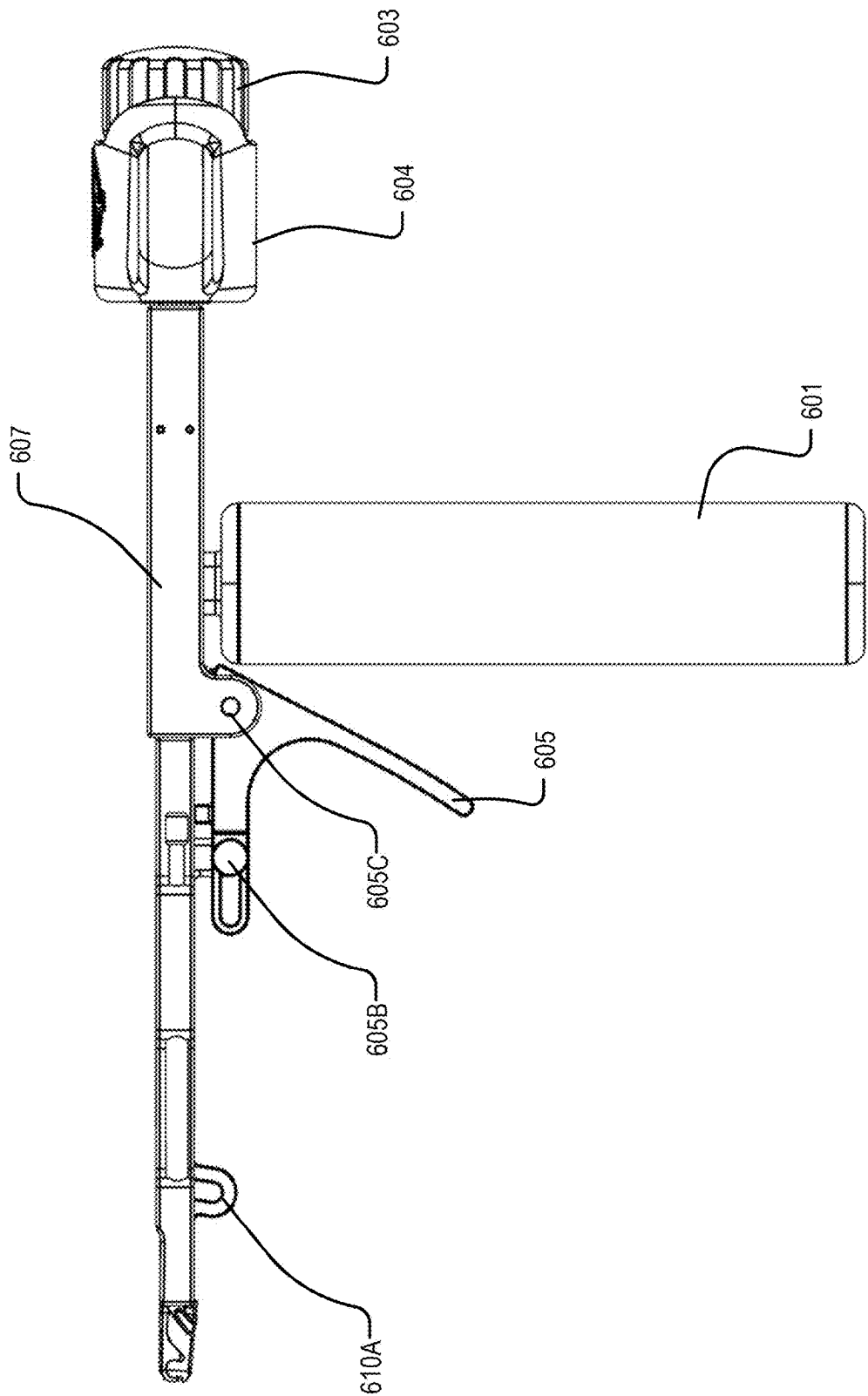
400Z
FIG. 43A



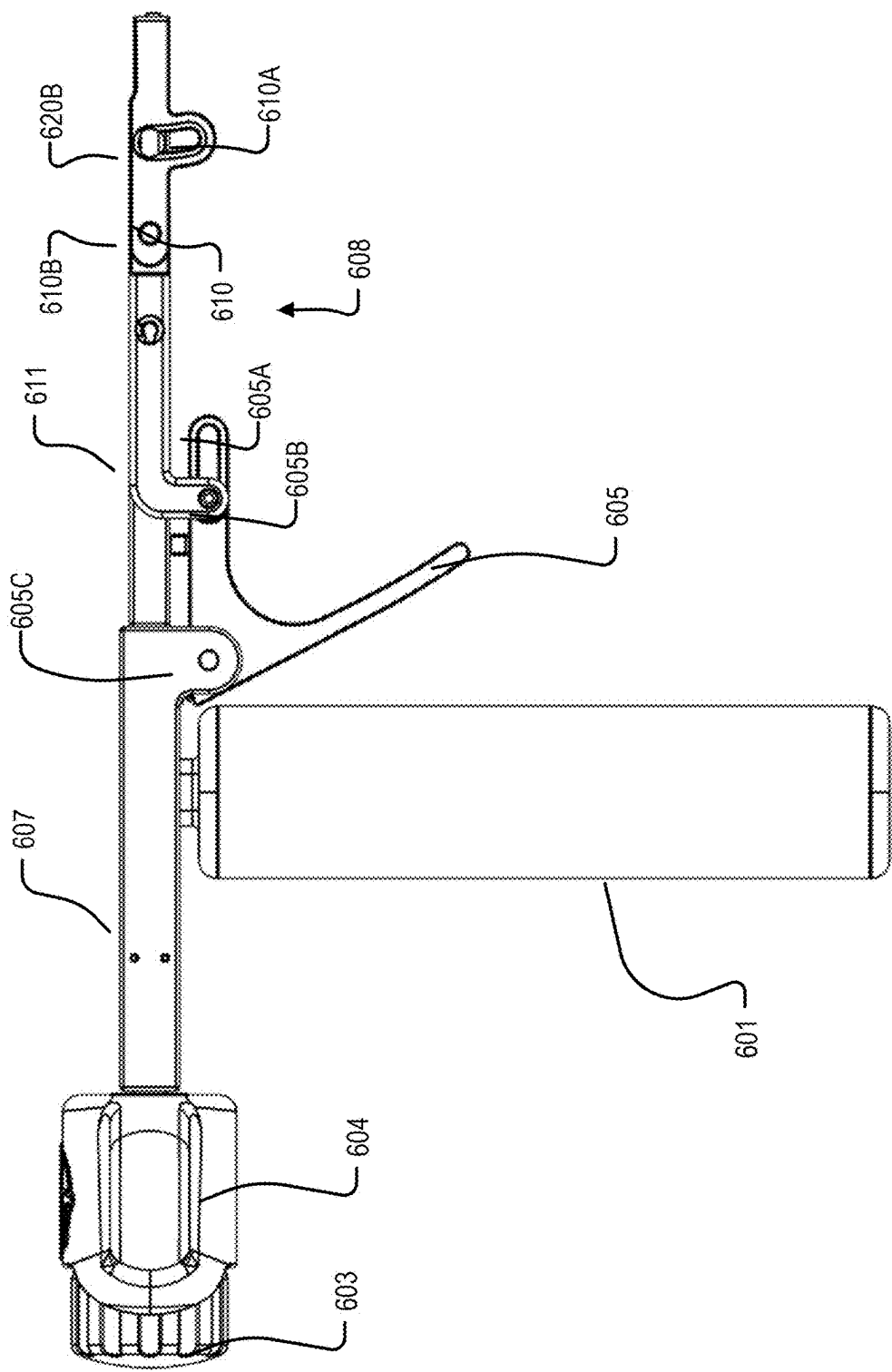
400Z
FIG. 43B



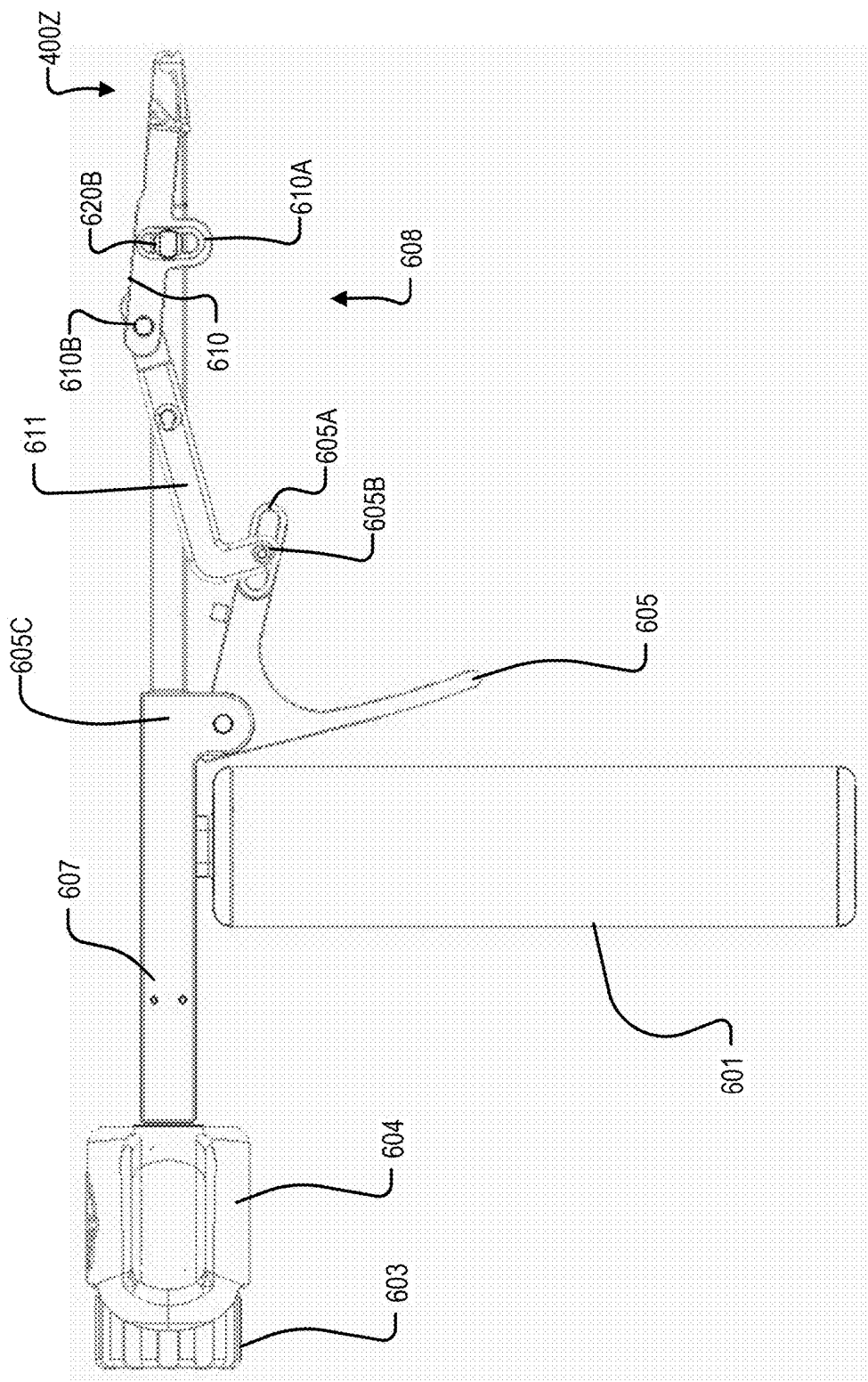
600
FIG. 44



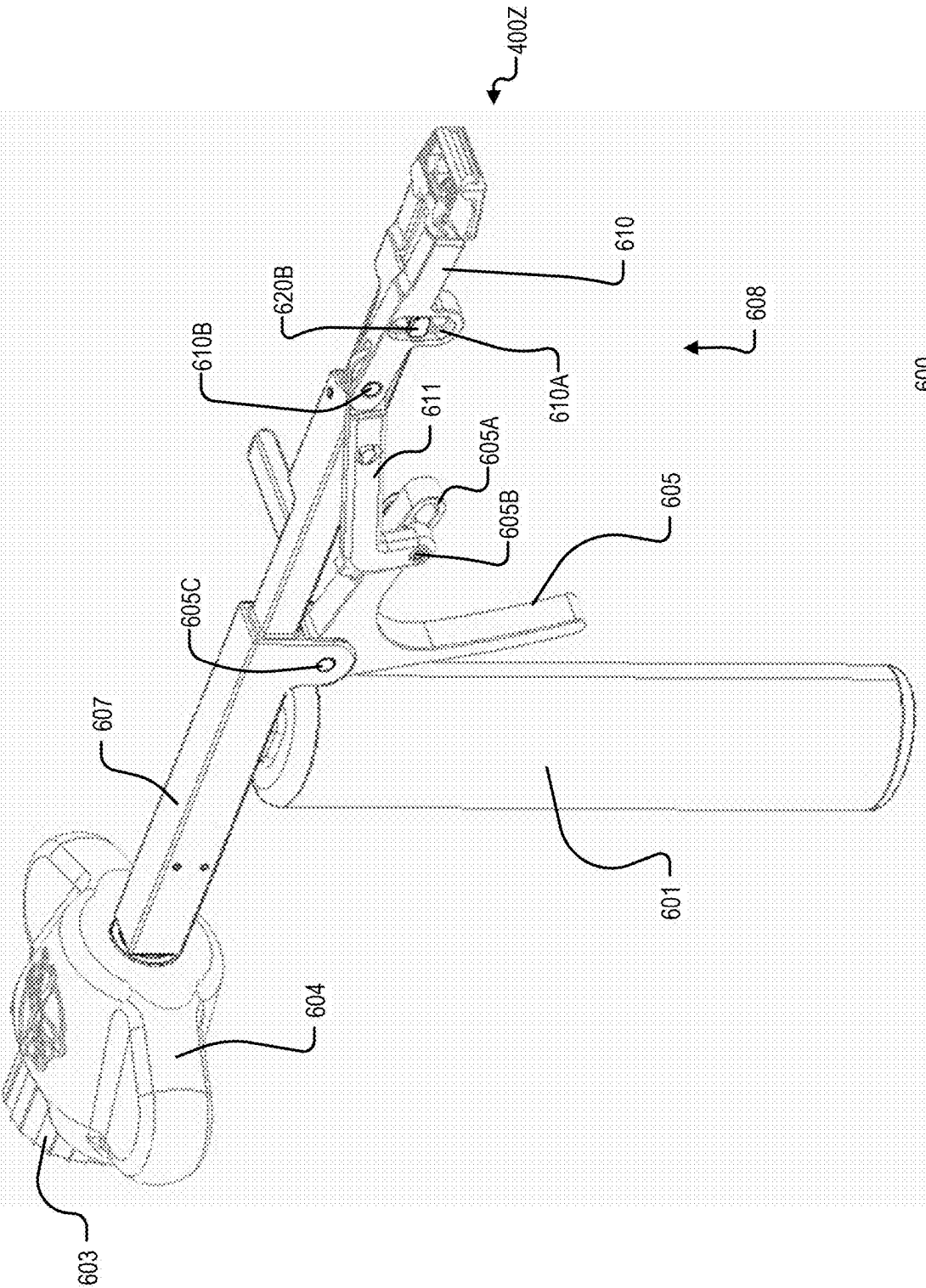
600
FIG. 45



600
FIG. 46A



600
FIG. 46B



600
FIG. 46C

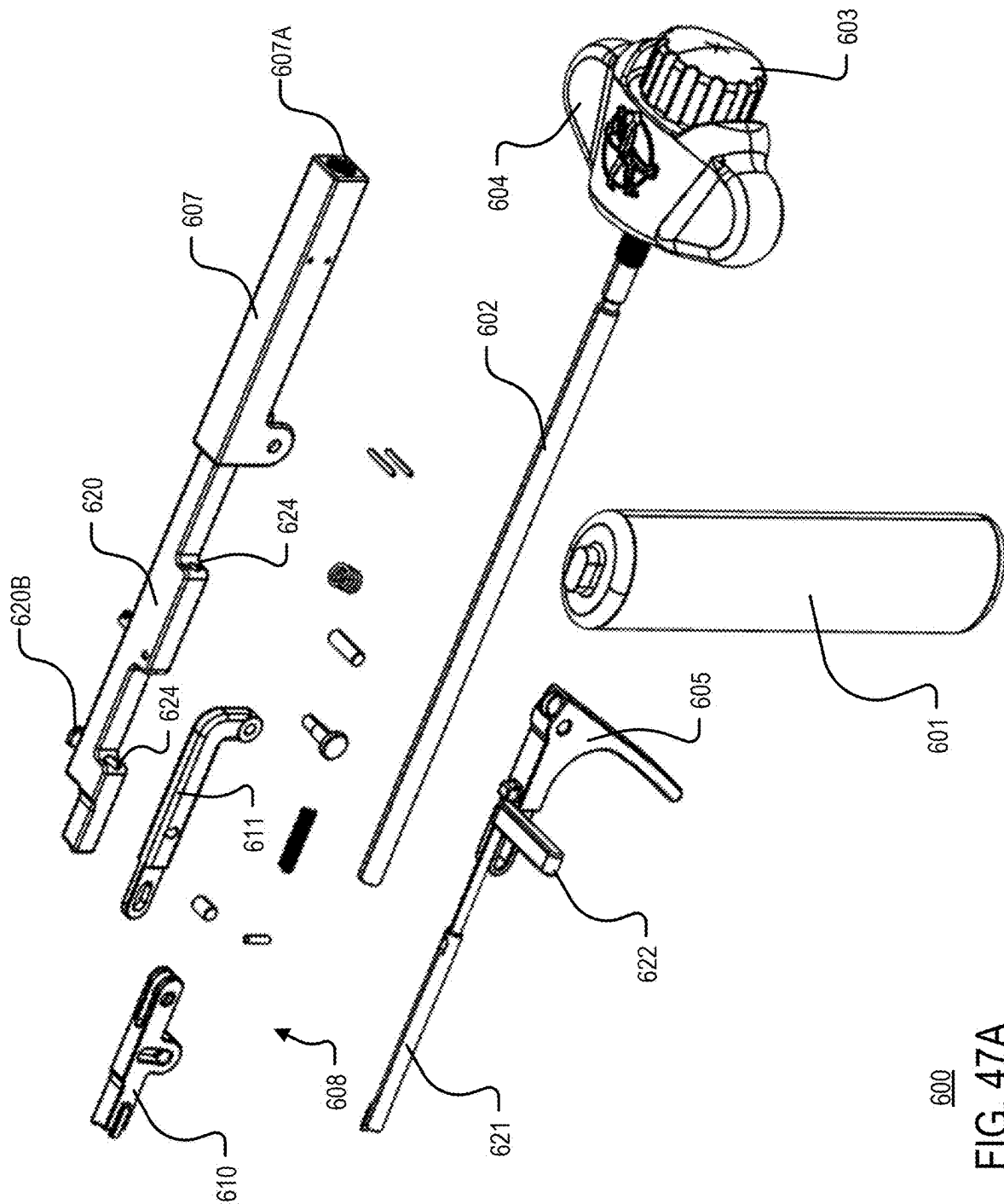
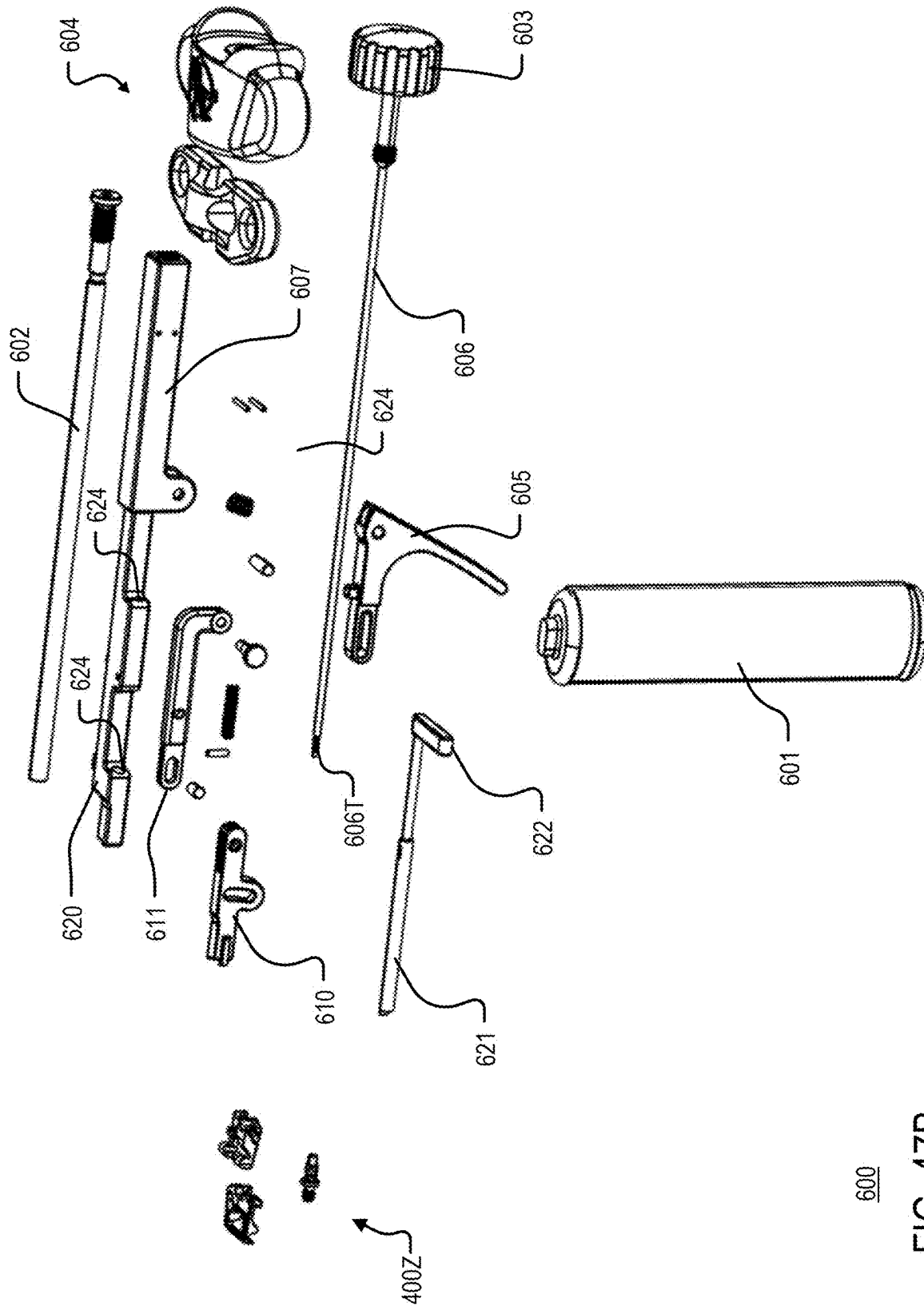


FIG. 47A



600

FIG. 47B

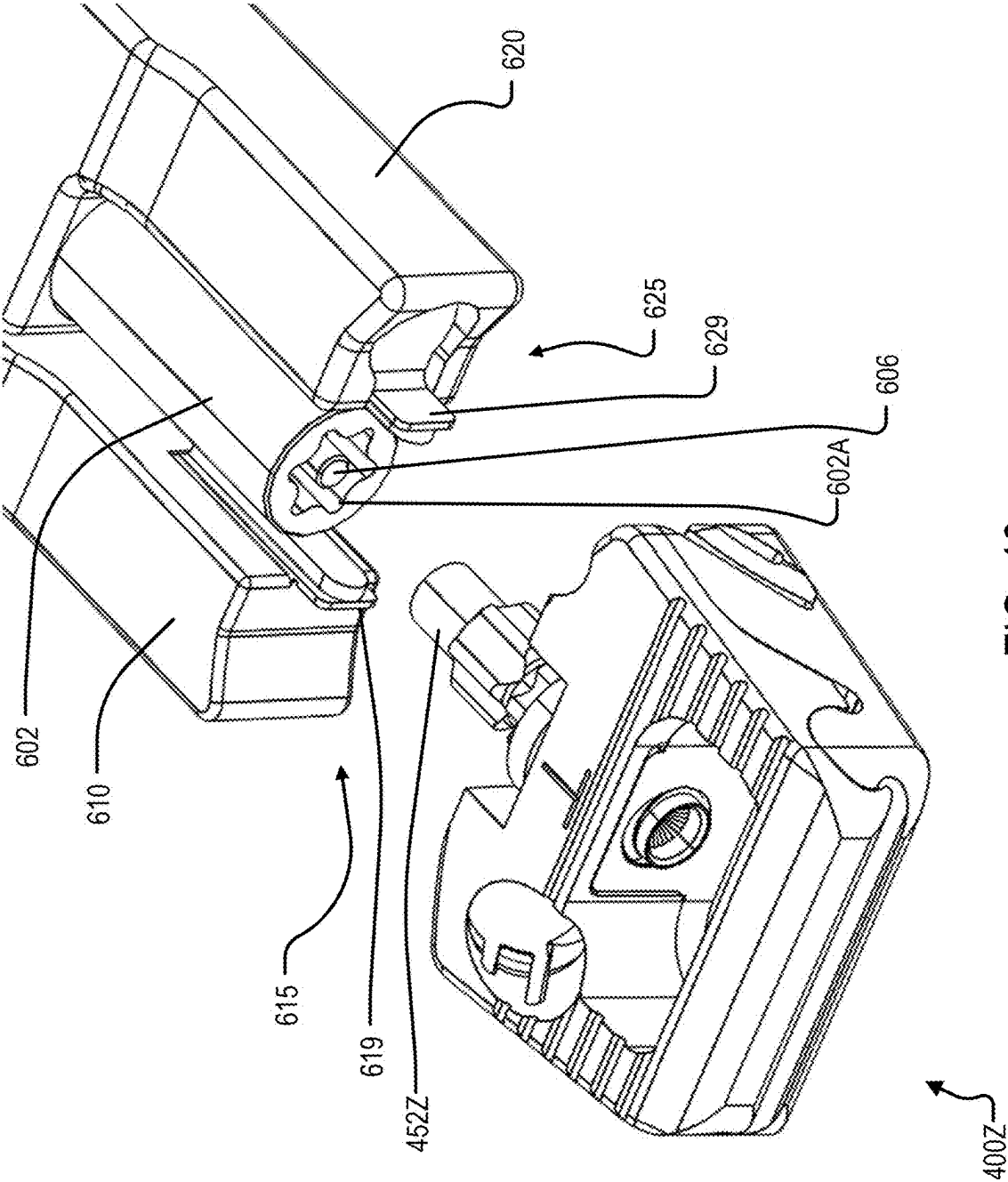


FIG. 48

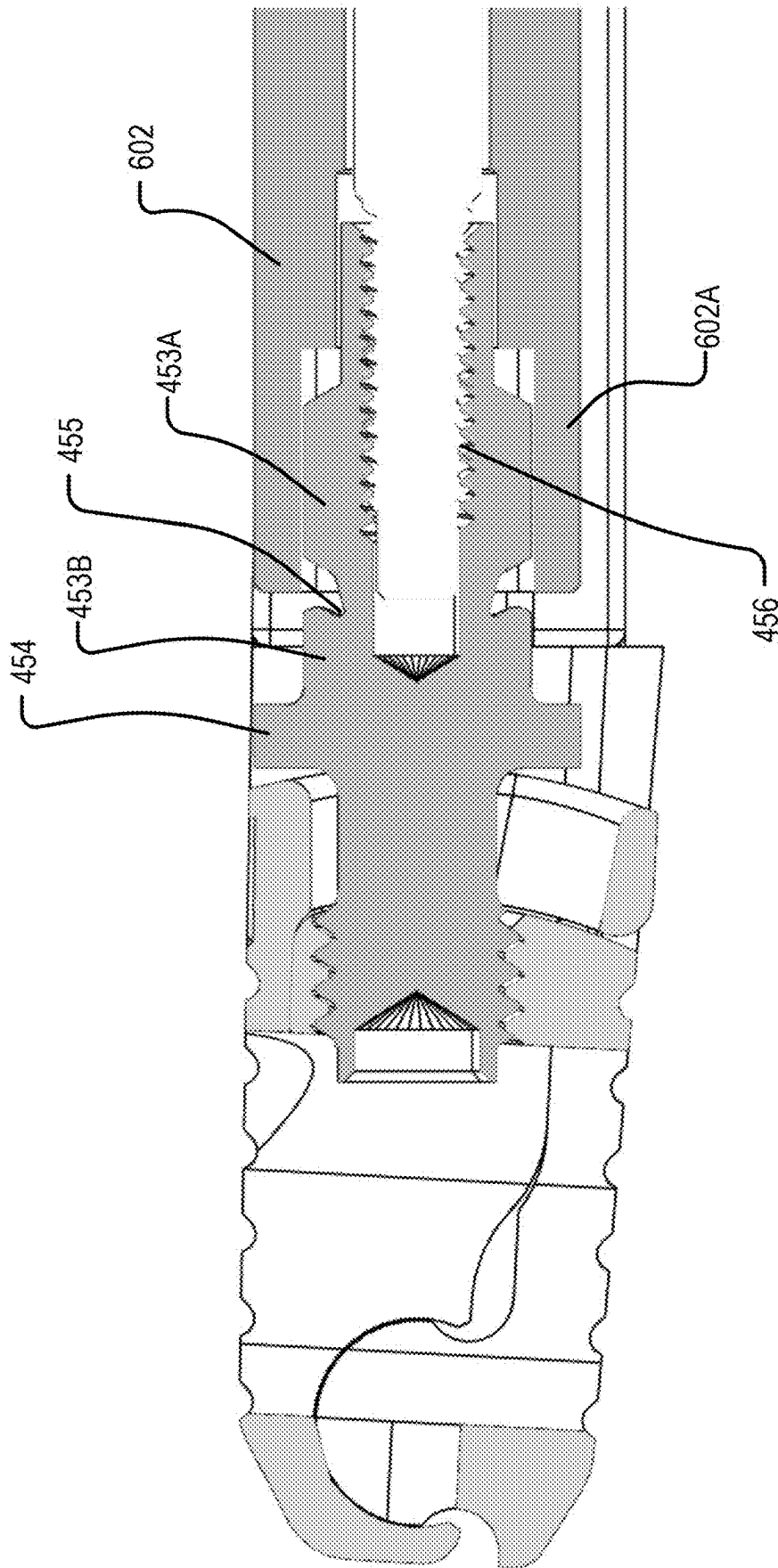


FIG. 49

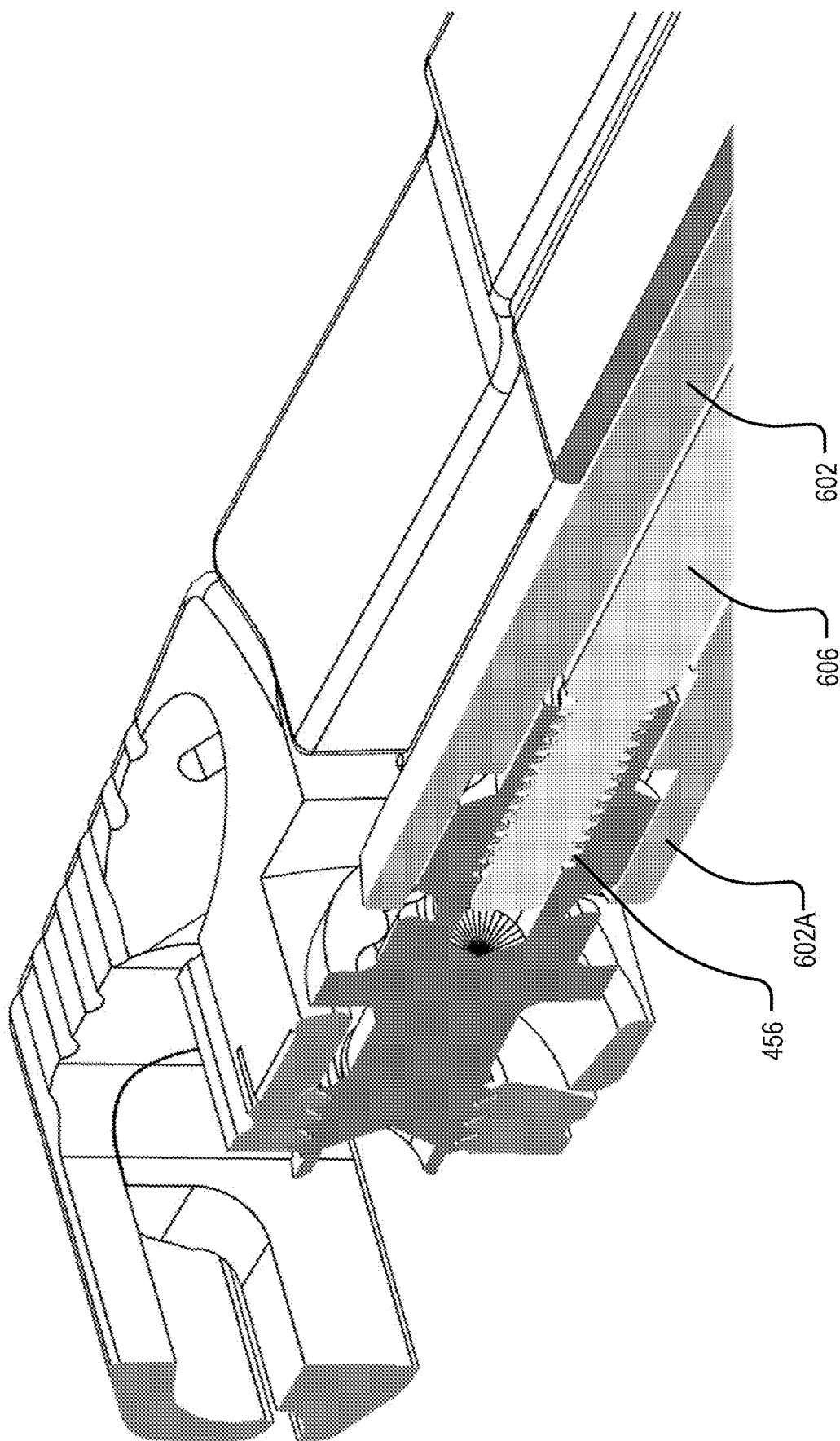


FIG. 50

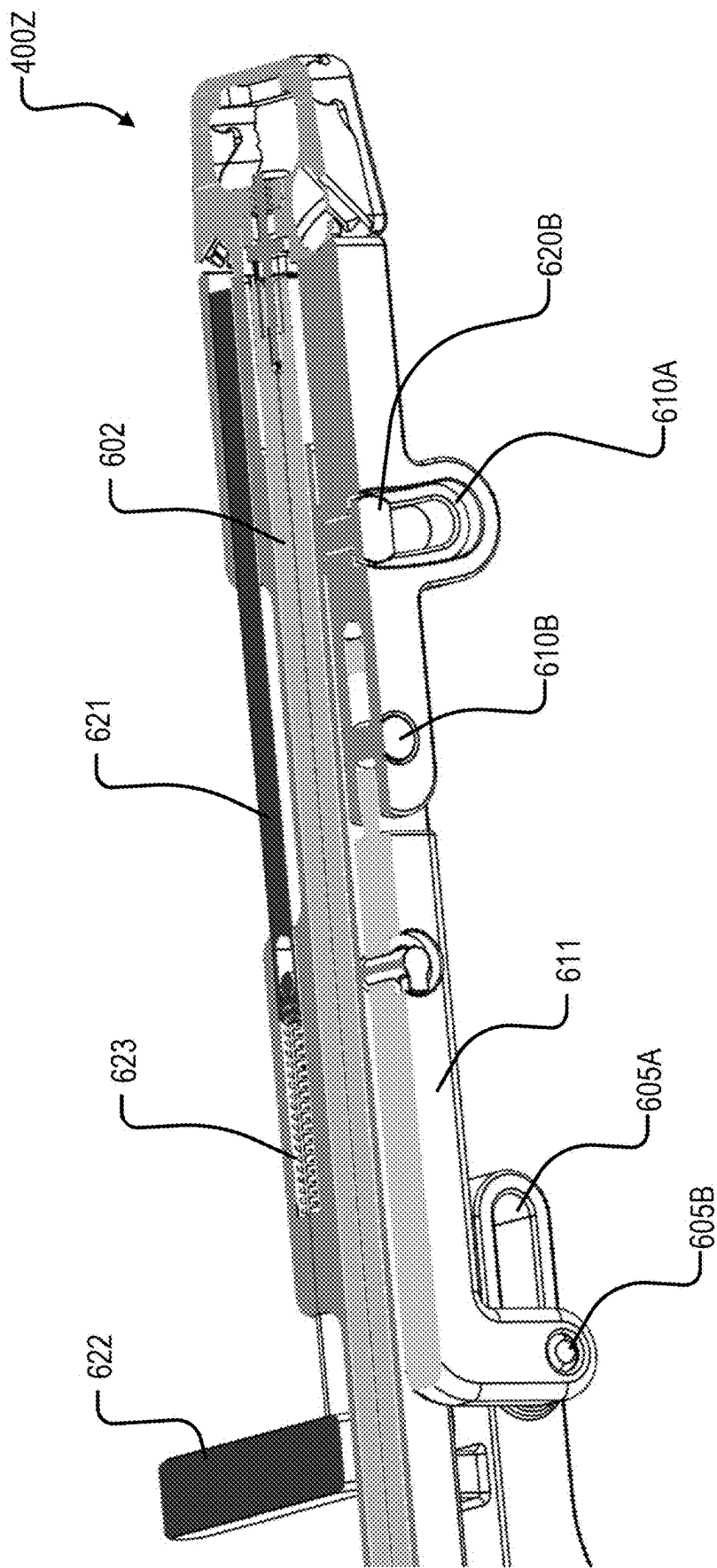


FIG. 51

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**EXPANDABLE INTERBODY IMPLANT AND
CORRESPONDING SURGICAL TOOL****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation in part of: U.S. patent application Ser. No. 17/665,449, titled EXPANDABLE IMPLANT AND CORRESPONDING INSERTER, filed Feb. 4, 2022; U.S. patent application Ser. No. 17/515,709, titled EXPANDABLE IMPLANT AND CORRESPONDING INSERTER, filed Nov. 1, 2021; and U.S. patent application Ser. No. 17/356,950, titled EXPANDABLE INTERBODY IMPLANT, filed Jun. 24, 2021 the entire disclosure of each application is incorporated herein by reference. This application also incorporates by reference U.S. application Ser. No. 17/307,578, titled EXTERNALLY DRIVEN EXPANDABLE INTERBODY AND RELATED METHODS, filed May 5, 2021; U.S. Pat. No. 11,096,796, titled INTERBODY SPINAL IMPLANT HAVING A ROUGHENED SURFACE TOPOGRAPHY ON ONE OR MORE INTERNAL SURFACES, and filed on Mar. 4, 2013; and U.S. Pat. No. 10,821,000, titled TITANIUM IMPLANT SURFACES FREE FROM ALPHA CASE AND WITH ENHANCED OSTEOINDUCTION, and filed Jun. 29, 2017.

FIELD

In one aspect, the present technology is generally related to an externally driven expandable interbody implant for use in a medical procedure related to the spine. In some embodiments, disclosed implants may be used in an anterior cervical discectomy and fusion (ACDF) procedure although other uses in other areas of the spine or between two bones are also contemplated.

BACKGROUND

Mechanically operated interbody implants may be used to align and/or realign a patient's spine during a medical procedure and/or for purposes of fusion, degenerative tissue and/or trauma/repair procedures. Conventional implants designed for the Thoracic and Lumbar region of the spine often include top and bottom endplates and a mechanical means to separate the top and bottom endplates. The mechanical mechanisms to separate the top and bottom endplates are often cumbersome and require a large footprint that is often unsuitable, for example, for ACDF type surgeries of the cervical portion of the spine. Many currently available ACDF type implants may be limited in an ability to optimize the adjustment of lordosis or sagittal alignment of the vertebral bodies because they may rely on a fixed lordotic angle between the superior/cephalad and inferior/caudad faces of the device.

SUMMARY

The techniques of this disclosure generally relate to an expandable interbody implant including a superior endplate and an inferior endplate hingedly coupled and which may further include a locking element to secure the inferior endplate and superior endplate in a particular configuration, for example. The superior and inferior endplates may be moved in a multitude of expanded and/or lordosed or kyphosed or otherwise angled configurations via an external inserter for example. In various embodiments, a locking

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screw may be a breakoff type screw. In various embodiments at least one breakoff tang on the implant may be used to grip the implant to insert it into a disc space and afterwards the breakoff tang may be broken off and removed. Additionally, in various embodiments the locking screw may be used to grip the implant and insert it into a disc space. Additionally, in various embodiments female recesses, rather than tangs (or male bosses/protrusions), may be used for gripping of the implant and inserting the implant into a disc space.

In one aspect, the present disclosure provides for an expandable implant movable between a contracted position (closed position) and an expanded position, for example. The expandable implant may include an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction (may also be referred to as an anterior-to-posterior direction depending on surgical technique), extending from a first lateral side to a second lateral side in a widthwise direction, and extend from a superior endplate to an inferior endplate in a heightwise direction (may also be referred to as a cephalad-to-caudal and/or vertical direction depending on surgical technique), for example. In various embodiments, the expandable body may be defined by a superior endplate and an inferior endplate that are hingedly connected, for example. In various embodiments, the superior endplate includes a first core having a distal engagement surface (may also be referred to as a posterior engagement surface depending on surgical technique) and the inferior endplate includes a second core having a proximal engagement surface (may also be referred to as an anterior engagement surface) and a threaded screw aperture, for example. In various embodiments, disclosed implants may include a threaded breakoff screw having a fracture surface that is disposed in the threaded screw aperture and movable between a locked position and an unlocked position, for example. In various embodiments, when in the locked position, the breakoff screw urges the distal engagement surface of the first core into direct contact with the proximal engagement surface of the second core, for example.

In another aspect, the disclosure provides for a system including a medical implant and a surgical tool, for example. The system may include an expandable implant movable between a contracted position and an expanded position, for example. In various embodiments, the expanded position may also refer to a distracted and angled orientation of the superior endplate and inferior endplate. The expandable implant may include an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction and extending from a first lateral side to a second lateral side in a widthwise direction, for example. In various embodiments, the expandable body may be defined by a superior endplate and an inferior endplate that are hingedly connected, for example. In various embodiments, the superior endplate includes a first core having a distal engagement surface and the inferior endplate includes a second core having a proximal engagement surface and a threaded screw aperture, for example. In various embodiments, disclosed implants may include a locking screw that is disposed in the threaded screw aperture and movable between a locked position and an unlocked position, for example. In various embodiments, when in the locked position, the locking screw urges the distal engagement surface of the first core into direct contact with the proximal engagement surface of the second core, for example. The system may also include a surgical tool for expanding the implant and tightening the locking screw while the implant is expanded at a desired height, position, and/or angle.

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The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view of an expandable implant.
 FIG. 2 is an alternate perspective view of an expandable implant.
 FIG. 3 is a top down view of an expandable implant.
 FIG. 4 is a side view of an expandable implant.
 FIG. 5 is a rear perspective view of an expandable implant.
 FIG. 6 is a perspective view of the interior of a superior endplate of an expandable implant.
 FIG. 7 is a perspective view of the interior of an inferior endplate of an expandable implant.
 FIG. 8 is a perspective exploded parts view of an expandable implant.
 FIG. 9 is an exploded parts view of an expandable implant from a side view perspective.
 FIG. 10 is a perspective cross section view of an expandable implant.
 FIG. 11 is a cross section view of an expandable implant.
 FIG. 12 is a side view of a superior endplate for use with at least some expandable implant embodiments.
 FIG. 13A is a perspective view of a first expandable implant.
 FIG. 13B is a perspective view of a second expandable implant.
 FIG. 14A is a perspective view of a third expandable implant.
 FIG. 14B is a perspective view of a fourth expandable implant.
 FIG. 15 is a perspective view of a fifth expandable implant.
 FIG. 16 is a side view of an expandable implant in the expanded configuration.
 FIG. 17 is a side view of an expandable implant showing a bone screw trajectory.
 FIG. 18 is a front view of an expandable implant.
 FIG. 19 is a front view of an enlarged area of FIG. 18.
 FIG. 20 is a perspective view of an inserter for use with disclosed expandable implants.
 FIG. 21 is a perspective view of an inserter for use with disclosed expandable implants shown in skeleton outlining for ease of understanding.
 FIG. 22 is a perspective view of an inserter for use with disclosed expandable implants shown in skeleton outlining for ease of understanding.
 FIG. 23A is a rear view of an inserter in a non-expanded position.
 FIG. 23B is a rear view of an inserter in an expanded position.
 FIG. 24 is an enlarged view of a distal end of an inserter in an expanded position coupled to an example expandable implant in a corresponding expanded position.
 FIG. 25 is a perspective view of an expandable implant in an expanded configuration after a breakoff portion of a locking screw has been broken off.
 FIG. 26 is a perspective view of a surgical instrument for use with disclosed expandable implants.
 FIG. 27 is a perspective view of a surgical instrument for use with disclosed expandable implants.

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FIG. 28 is a perspective view of an expandable implant after the mounting tangs have been broken off.

FIG. 29 is a reference drawing showing the human spine of which various disclosed implant embodiments may be installed in.

FIG. 30 is a reference drawing showing various planes and reference directions of which the various disclosed implant embodiments may move in or act in with respect to a patient.

FIG. 31 is a perspective view of a second implant embodiment.

FIG. 32 is a first perspective exploded parts view of the second implant embodiment.

FIG. 33 is a second perspective exploded parts view of the second implant embodiment.

FIG. 34 is a side exploded parts view of the second implant embodiment.

FIG. 35 is a first side view of a breakoff screw having a recessed fracture surface.

FIG. 36 is a second side view of a breakoff screw having a recessed fracture surface.

FIG. 37 is a cross section view of a breakoff screw having a recessed fracture surface.

FIG. 38 is a perspective view of a swaging fixture.

FIG. 39 is a cross section view of a swage mandrel and a distal end of a breakoff screw before the commencement of a swage process.

FIG. 40 is a cross section view showing a result of a swage process.

FIG. 41 is an enlarged view of region S-W of FIG. 40.

FIG. 42A is a front perspective view of a third implant embodiment.

FIG. 42B is an alternate front perspective view of a third implant embodiment.

FIG. 43A is a front perspective view of an implant having angled engagement features.

FIG. 43B is a top down view of an implant having angled engagement features.

FIG. 44 is a perspective view of a surgical tool for use with disclosed implant embodiments.

FIG. 45 is a first side view of a surgical tool for use with disclosed implant embodiments.

FIG. 46A is a second side view of a surgical tool for use with disclosed implant embodiments.

FIG. 46B is a third side view of a surgical tool in an operative position for use with disclosed implant embodiments.

FIG. 46C is a perspective view of a surgical tool in the operative position.

FIG. 47A is first exploded parts view of a surgical tool for use with disclosed implant embodiments.

FIG. 47B is second exploded parts view of a surgical tool for use with disclosed implant embodiments.

FIG. 48 is a perspective view showing a surgical tool immediately prior to coupling with a surgical implant.

FIG. 49 is a first cross section drawing of the surgical tool and implant in a coupled configuration.

FIG. 50 is a second cross section drawing of the surgical tool and implant in a coupled configuration.

FIG. 51 is a third cross section drawing of the surgical tool and implant in a coupled configuration.

DETAILED DESCRIPTION

Embodiments of the present disclosure relate generally, for example, to spinal stabilization systems, and more particularly, to surgical instruments for use with spinal stabili-

zation systems. Embodiments of the devices and methods are described below with reference to the Figures.

The following discussion omits or only briefly describes certain components, features and functionality related to medical implants, installation tools, and associated surgical techniques, which are apparent to those of ordinary skill in the art. It is noted that various embodiments are described in detail with reference to the drawings, in which like reference numerals represent like parts and assemblies throughout the several views, where possible. Reference to various embodiments does not limit the scope of the claims appended hereto because the embodiments are examples of the inventive concepts described herein. Additionally, any example(s) set forth in this specification are intended to be non-limiting and set forth some of the many possible embodiments applicable to the appended claims. Further, particular features described herein can be used in combination with other described features in each of the various possible combinations and permutations unless the context or other statements clearly indicate otherwise.

Terms such as “same,” “equal,” “planar,” “coplanar,” “parallel,” “perpendicular,” etc. as used herein are intended to encompass a meaning of exactly the same while also including variations that may occur, for example, due to manufacturing processes. The term “substantially” may be used herein to emphasize this meaning, particularly when the described embodiment has the same or nearly the same functionality or characteristic, unless the context or other statements clearly indicate otherwise.

Referring to FIGS. 1-43 generally, various embodiments and views of an expandable implant 100 are disclosed. The components of expandable implant 100 can be fabricated from biologically acceptable materials suitable for medical applications, including metals, synthetic polymers, ceramics and bone material and/or their composites. For example, the components, individually or collectively, can be fabricated from materials such as stainless steel alloys, commercially pure titanium, titanium alloys, Grade 5 titanium, super-elastic titanium alloys, cobalt-chrome alloys, superelastic metallic alloys (e.g., Nitinol, super elasto-plastic metals, such as GUM METAL™), ceramics and composites thereof such as calcium phosphate (e.g., SKELITE™), thermoplastics such as polyaryletherketone (PAEK) including polyetheretherketone (PEEK), polyetherketoneketone (PEKK) and polyetherketone (PEK), carbon-PEEK composites, PEEK-BaSO₄ polymeric rubbers, polyethylene terephthalate (PET), fabric, silicone, polyurethane, silicone-polyurethane copolymers, polymeric rubbers, polyolefin rubbers, hydrogels, semi-rigid and rigid materials, elastomers, rubbers, thermoplastic elastomers, thermoset elastomers, elastomeric composites, rigid polymers including polyphenylene, polyamide, polyimide, polyetherimide, polyethylene, epoxy, bone material including autograft, allograft, xenograft or transgenic cortical and/or corticocancellous bone, and tissue growth or differentiation factors, partially resorbable materials, such as, for example, composites of metals and calcium-based ceramics, composites of PEEK and calcium based ceramics, composites of PEEK with resorbable polymers, totally resorbable materials, such as, for example, calcium based ceramics such as calcium phosphate, tri-calcium phosphate (TCP), hydroxyapatite (HA)-TCP, calcium sulfate, or other resorbable polymers such as polylactide, polyglycolide, polytyrosine carbonate, polycaprolactone, polylactic acid or polylactide and their combinations.

In various embodiments, components may be coated with a ceramic, titanium, and/or other biocompatible material to

provide surface texturing at (a) the macro scale, (b) the micro scale, and/or (c) the nano scale, for example. Similarly, components may undergo a subtractive manufacturing process providing for surface texturing configured to facilitate osseointegration and cellular attachment and osteoblast maturation. Example surface texturing of additive and subtractive manufacturing processes may comprise (a) macro-scale structural features having a maximum peak-to-valley height of about 40 microns to about 500 microns, (b) micro-scale structural features having a maximum peak-to-valley height of about 2 microns to about 40 microns, and/or (c) nano-scale structural features having a maximum peak-to-valley height of about 0.05 microns to about 5 microns. In various embodiments, the three types of structural features may be overlapping with one another, for example. Additionally, such surface texturing may be applied to any surface, e.g., both external exposed facing surfaces of components and internal non exposed surfaces of components. Further discussion regarding relevant surface texturing and coatings is described in, for example, U.S. Pat. No. 11,096,796, titled Interbody spinal implant having a roughened surface topography on one or more internal surfaces, and filed on Mar. 4, 2013—the entire disclosure of which is incorporated herein by reference in its entirety. Accordingly, it shall be understood that any of the described coating and texturing processes of U.S. Pat. No. 11,096,796, may be applied to any component of the various embodiments disclosed herein, e.g., the exposed surfaces and internal surfaces of endplates. Another example technique for manufacturing an orthopedic implant having surfaces with osteoinducting roughness features including micro-scale structures and nano-scale structures is disclosed in U.S. Pat. No. 10,821,000, the entire contents of which are incorporated herein by reference. Additionally, an example of a commercially available product may be the Adaptix™ Interbody System sold by Medtronic Spine and comprising a titanium cage made with Titan nanoLOCK™.

Referring generally to FIGS. 1-5 various views of an expandable implant 100 in a collapsed position are illustrated. FIGS. 1-2 are various perspective views of an expandable implant 100. FIG. 3 is a top down view of an expandable implant 100. In the example embodiment, expandable implant 100 may include a proximal end 100P, a distal end 100D, and first and second lateral sides 100L. Additionally, a pair of bone screw apertures 11, 21 may be positioned on the proximal end 100P, for example. In various embodiments, bone screw apertures 11, 21 may comprise a corresponding bone screw retention mechanism 11a, 21a (may also be referred to as an anti-backout locking mechanism). In the example embodiment, the bone screw retention mechanisms 11a, 21a, comprise a flexible tang member having a hook portion at an end thereof that allows the flexible tang member to flex outward in a lateral direction away from the corresponding bone screw aperture 11, 21 during initial installation of the bone screw and to flex back inward towards the corresponding bone screw aperture 11, 21 to prevent a corresponding bone screw from backing out. For example, as the bone is installed in bone screw aperture 11, 21, the bone screw retention mechanism 11a, 21a, may flex outward as the underside of the head portion contacts inclined surface 11c (see FIG. 19).

In various embodiments, and as illustrated in FIGS. 1-2, mounting tangs 19, 29 may extend in a proximal direction (may also be referred to as an anterior direction depending on surgical technique and orientation), for example. In various embodiments, implant 100 may be referred to as an externally driven expandable implant because an end user or

surgeon may use a surgical tool to open and close implant 100, e.g. expand implant 100. For example, an external tool may adjust the lordotic angle of implant 100 as will be explained in detail with respect to FIGS. 20-25. Once implant 100 is expanded to an appropriate lordotic angle (also referred to as angle of inclination), an end user may fix the relative angle of the superior endplate 10 relative to the inferior endplate 20 by tightening locking screw 50, for example. In some embodiments, superior endplate 10 may be referred to as a “cephalad” endplate and inferior endplate 20 may be referred to as a “caudal” endplate.

Locking screw 50 may also be used in other embodiments, such as fixation of posterior rods, fixation of pedicle screws, and other set screw constructs. Additionally, locking screw 50 may be referred to as a “breakoff screw” in some embodiments.

At least one advantage of relying on an external tool to adjust a lordotic angle of implant 100 may be the reduction of internal components within implant 100 relative to other forms of implants relying on various moving mechanisms and/or expansion mechanisms, for example. Accordingly, in various embodiments, implant 100 may have a relatively large void space in the interior thereof, which may facilitate a fusion process during an ACDF procedure. For example, implant 100 may have a relatively large internal volume 101 that is open through the superior endplate 10 and inferior endplate 20 which may be packed with bone graft material, for example.

As illustrated in FIG. 3, implant 100 may extend in a proximal-to-distal direction (may also referred to as a longitudinal direction and/or an anterior-to-posterior direction depending on surgical technique and final orientation) from the proximal end 100P (may be referred to as anterior end depending on surgical technique and final orientation) to the distal end 100D (may be referred to as posterior end depending on surgical technique and final orientation) though axis P-D through the center of the implant 100, for example. Implant 100 may extend in a widthwise direction (also referred to as lateral direction) from the first lateral side 100L to the second lateral side 100L through axis W-W through the center of the implant 100 and the center of locking screw 50, for example. The axis P-D may be perpendicular and/or substantially perpendicular to the axis W-W. For example, the proximal-to-distal direction may be perpendicular to the widthwise direction. Additionally, a width of the implant may taper from a proximal end 100P where it is widest towards a distal end 100D where it is narrowest. In various embodiments, implant 100 may extend from a superior endplate 10 to an inferior endplate 20 in a heightwise direction (may also be referred to as a cephalad-to-caudal and/or vertical direction depending on surgical technique and final orientation).

FIG. 4 is a side view of an expandable implant 100. In the example illustration, it is shown that a superior endplate 10 is connected to an inferior endplate 20 such that the superior endplate may pivot about a hinge member 40. In the example embodiment, hinge member 40 comprises an arcuate rail portion of inferior endplate 20 that extends in the widthwise direction, for example. In other embodiments, the hinge member 40 may be reversed relative to the superior and inferior endplates than as illustrated in FIG. 4. In the example embodiment, hinge member 40 may be nested into a corresponding arcuate cavity of the superior endplate 10 such that superior endplate 10 may expand and/or otherwise rotate about hinge member 40. Additionally, in various embodiments the superior endplate 10 and/or inferior endplate 20 may include various engagement elements 14 for

engaging with an adjacent boney structure such as a vertebrae, for example. In the example embodiment, the engagement elements comprise a series of alternating rails and valleys therebetween that extend in the widthwise direction. In some embodiments, the valleys may be angled about 20-40 degrees and in at least one embodiment the valleys may be angled at about 30 degrees relative to the corresponding rail (see FIGS. 42A and 42B). At least one advantage of this orientation may be a relatively greater resistance and/or suppression of expulsion of the implant 100 in both the lateral direction and in the proximal-to-distal direction. However, claws, hooks, dimples, spikes, etc. are also contemplated as example engagement elements 14. In some embodiments, an acid etching process may be utilized to form a roughened or textured surface to facilitate securing the implant between boney portions and/or suppressing expulsion of implant 100.

FIG. 5 is a rear perspective view of an expandable implant 100. In the example illustration it is shown that the distal end 100D is narrower than the proximal end 100P. FIG. 6 is a perspective view of the interior of a superior endplate 10. In the example illustration, it is shown that the distal end of superior endplate 10 includes an arcuate channel 12 of which the hinge member 40 may be disposed inside of. The proximal end of superior endplate 10 may include a bone screw aperture cutout 21b to allow a relief area for a corresponding bone screw to be insert through bone screw aperture 21 of inferior endplate 20, for example. Superior endplate 10 may also include a core 15 comprising an aperture 15a that extends from a proximal surface 15p thereof to a distal surface 15d thereof, for example. In various embodiments, aperture 15a may be referred to as a “slot” or “screw slot”. In some embodiments, core 15 may be referred to as a support frame and take a generally rectangular shape. In various embodiments, the distal surface 15d may be curved and generally face the distal end 100D of implant 100. FIG. 7 is a perspective view of the interior of an inferior endplate 20. In the example illustration, it is shown that the distal end of inferior endplate 20 includes a hinge member 40 in the form of an arcuate rail that may be disposed inside of the arcuate channel 12 of the superior endplate 10, for example. The proximal end of inferior endplate 20 may include a bone screw aperture cutout 11b to allow a relief area for a corresponding bone screw to be insert through bone screw aperture 11 of superior endplate 10, for example. Inferior endplate 20 may also include a core 25 comprising a threaded aperture 25a that extends from a proximal surface 25p thereof to a distal surface 25d thereof, for example. In some embodiments, core 25 may be referred to as a support frame and take a generally rectangular shape. In various embodiments, the superior endplate 10 and inferior endplate 20 may each be formed by a unitary single piece, respectively.

FIG. 8 is a perspective exploded parts view and FIG. 9 is an exploded parts view from a side view perspective of an expandable implant 100. In the example embodiment, a locking screw 50 is illustrated. Locking screw 50 may include an external thread pattern 51 on an outside circumferential surface thereof, for example. The external thread pattern 51 of locking screw 50 may have a size and shape generally corresponding to the threaded aperture 25a of core 25 of inferior endplate 20, for example. In various embodiments, an engagement surface 54 may be disposed adjacent and proximal of external thread pattern 51. In the example embodiment, engagement surface 54 is shaped like a washer and is directly connected to locking screw 50. However, in other embodiments, engagement surface 54 may be a washer

or separated element, for example. In some embodiments, engagement surface **54** may be conically shaped and/or spherically shaped. Engagement surface **54** may include a relatively planar and/or flat distal surface and/or proximal surface (anterior/ventral surface). In various embodiments, a proximal end of set screw **50** may include an aperture having an internal threaded surface **52**. For example, a cylindrical shaped proximal end may include an aperture having a thread pattern disposed on an internal circumferential surface of the cylindrical shaped proximal end. In the example embodiment, a first drive feature **53a** and a second drive feature **53b** may be disposed adjacent to and distally with respect to a proximal most end of set screw **50**. Additionally, first and second drive features **53a**, **53b** may be disposed proximally with respect to engagement surface **54**. In the example embodiment, drive features **53a**, **53b** take a hexalobular shape, although various other shapes such as hexagonal, polygonal, Torx, etc. are also contemplated. In some embodiments, a surgical drive tool having a corresponding socket may be coupled to drive features **53a**, **53b** to cause rotation of locking screw **50**. Similarly, in some alternative embodiments, a drive tool with a protruding threaded member having a thread pattern with a corresponding size and shape to internal threaded surface **52** may also cause rotation of set screw **50**.

As seen best in FIG. 9, set screw **50** may also include a breakoff location **55**, for example. In the example embodiment, breakoff location **55** is disposed directly between drive features **53a**, **53b** and is designed to shear off when a sufficient rotational force (torque) is applied to a proximal end of set screw **50** while a distal end of set screw **50** is stationary, e.g., when set screw **50** is secured in a locked position and a continued rotational force (torque) is applied to the proximal end of set screw **50** the drive feature **53a** and cylindrical end having the internal threaded surface **52** may breakoff. In various embodiments, the internal threaded surface **52** may also be utilized to ensure that the broken off portion is removed from the patient and remains connected to a surgical tool/breakoff tool. As also seen best in FIG. 9, the inferior endplate **20** may include a first relief **40a** and a second relief **40b** on opposite sides of hinge member **40**. The first relief **40a** may have a size and shape corresponding to a size and shape of a first portion **12a** of superior endplate **10** and the second relief **40b** may have a size and shape corresponding to a size and shape of a second portion **12b** of superior endplate **10**, for example. In various embodiments, portions **12a**, **12b** may comprise a hook shape, outdent, and/or protrusion, for example. In the example embodiment, portions **12a**, **12b** may be disposed on opposite sides of channel **12** and cup hinge member **40** such that the superior endplate **10** and inferior endplate **20** may rotate relative to one another without becoming uncoupled.

FIG. 10 is a perspective cross section view and FIG. 11 is a cross section view of expandable implant **100**. In the example embodiment, the superior endplate **10** and inferior endplate **20** are coupled together by hinge member **40**, and core **25** may be positioned behind of core **15**, e.g., core **25** may be positioned distally with respect to core **15**. Additionally, the outside external thread pattern **51** of locking screw **50** may engage with the threaded aperture **25a** of the core **25** and extend through aperture **15a** of core **15**. In this way, when locking screw **50** is rotated, the distal surface **15d** of core **15** may engage with the proximal surface **25p** of core **25**. For example, by tightening locking screw **50** the engagement surface **54** of locking screw **50** pushes against the

proximal surface **15p** of core **15** thereby bringing the superior endplate **10** and inferior endplate **20** into frictional engagement.

FIG. 12 is a side view of a superior endplate **10** for use with at least some expandable implant **100** embodiments. In the example embodiment, superior endplate **10** may include an arcuate channel **12** of which the hinge member **40** may be disposed inside of. In various embodiments, arcuate channel **12** may be defined by a first circle having a center point at P_1 and/or a segment of the circle having the center point at P_1 . The center point P_1 may define an axis of rotation that superior endplate **10** may rotate and/or pivot with respect to inferior endplate **20**. For example, superior endplate **10** may be hingedly coupled to hinge member **40** as explained above and rotatable about an axis of rotation defined by center point P_1 , for example. Additionally, in various embodiments a distal surface **15d** of core **15** may be a curved surface defined (in part or in total) by a second circle having a center point at P_1 and a radius R_1 . The proximal surface **15p** of core **15** may also be a curved surface defined (in part or in total) by a segment of a circle having a radius R_2 and a center point P_2 . In the example embodiment, P_2 is located a distance D_1 above point P_1 and radius R_2 is greater than radius R_1 . Additionally, the proximal surface **15p** of core **15** is offset a distance D_2 from the proximal most face of the superior endplate **10**. In the example embodiment, center point P_2 is vertically above center point P_1 however, in other embodiments, center point P_2 may be offset by a greater amount or even a lesser amount than illustrated. In some examples, P_2 may not be aligned vertically above P_1 . In various embodiments, R_1 may be about 7-9 mm+/-about 1 mm and R_2 may be about 8-10 mm+/-about 1 mm although these numbers may be modified in some embodiments having a larger or smaller footprint. In various embodiments D_1 is about 0.25 mm to about 1.0 mm and D_2 is about 0.25 mm to about 1.25 mm. In at least one embodiment, D_1 is about 0.75 mm and D_2 is about 0.8 mm and R_2 is about 9.2 mm.

The above explained geometrical relationship between the offset center points P_1 and P_2 and R_1 and R_2 may have several advantages in terms of operability and functionality. At least one advantage is that the superior endplate **10** may have a natural tendency to apply a force against the engagement surface **54** of locking screw **50** such that locking screw **50** may function similar to a wedge preventing implant **100** from fully collapsing. Another advantage is that a biasing force may be applied that naturally urges the superior endplate **10** and inferior endplate **20** into an expanded position which may assist with expanding the implant **100** when positioned between a superior vertebrae and an inferior vertebrae, for example. For example still, an end user such as a surgeon may expand implant **100** and the offset arrangement explained above may facilitate the function of keeping implant **100** lordosed at the chosen angle.

FIG. 13A is a perspective view of a first expandable implant, FIG. 13B is a perspective view of a second expandable implant, FIG. 14A is a perspective view of a third expandable implant, FIG. 14B is a perspective view of a fourth expandable implant, and FIG. 15 is a perspective view of a fifth expandable implant. In the series of illustrations it is shown that various embodiments in accordance with the principles of this disclosure may be variously sized depending on the particular location in a human body and the particular patient specific human anatomy. For example, FIG. 13A illustrates a first expandable implant **100** having a first height H_1 or thickness between the superior endplate **10** and inferior endplate **20**, FIG. 13B illustrates a second expandable implant **100** having a second height H_2 or

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thickness, FIG. 14A illustrates a third expandable implant 100 having a third height H_3 or thickness, FIG. 14B illustrates a fourth expandable implant 100 having a fourth height H_4 or thickness, and FIG. 15 illustrates a fifth expandable implant 100 having a fifth height H_5 or thickness. In at least some embodiments, H_1 may be about 5 mm, H_2 may be about 6 mm, H_3 may be about 7 mm, H_4 may be about 8 mm, H_5 may be about 9 mm, for example. In various embodiments, an angle of inclination between the superior endplate 10 and inferior endplate 20 may be about 4 degrees to about 15 degrees in an expanded configuration, e.g., an angled and/or inclined configuration.

FIG. 16 is a side view of an expandable implant 100 in the expanded configuration. In an expanded position, a distance D_3 between the superior endplate 10 and inferior endplate 20 at the proximal end 100P may be relatively greater than in the closed configuration, for example. Additionally, an angle of inclination α may be relatively greater in an expanded position than in the closed configuration, for example. In this embodiment, implant 100 may have a height H_1 corresponding to FIG. 13A and be about 5 mm in a closed configuration. In the illustrated expanded configuration of FIG. 16, D_3 may be about 8 mm to 9 mm and α may be about 10 degrees to about 20 degrees. In at least one embodiment, D_3 may be 8 mm in a fully expanded position and α may be about 15 degrees, for example.

FIG. 17 is a side view of an expandable implant 100 showing a bone screw trajectory 99. In the example embodiment, it is shown that a centered bone screw trajectory 99 of bone screw 97 is at an angle β with respect to a plane 98 that crosses through a center of the implant from a first lateral side to a second lateral side, for example. Additionally, the bone screw trajectory 99 may be varied $\pm$ by a degree γ , for example. In various embodiments, β may be about 30 degrees to about 50 degrees and γ may be about 2 degrees to about 10 degrees. In the example embodiment, β may be about 40 degrees and γ may be about 5 degrees.

FIG. 18 is a front view of an expandable implant showing an area A_1 and FIG. 19 is a front view of an enlarged area A_1 of FIG. 18. In the example embodiment, bone screw 97 is in a position extending through bone screw aperture 11 where it cannot backout due to bone screw retention mechanism 11a. The bone screw retention mechanism 11a includes an inclined surface 11c such that when bone screw 97 is being installed, an underside of the head portion of bone screw 97 directly contacts the inclined surface 11c thereby pushing the bone screw retention mechanism 11a laterally outward and away from bone screw aperture 11, for example. Thereafter, when bone screw 97 is installed and the head portion of bone screw 97 is beneath inclined surface 11c the bone screw retention mechanism may flex back towards bone screw aperture 11 such that it will prevent bone screw 97 from backing out, e.g., a blocking surface of bone screw retention mechanism 11a may contact an upper surface of the head portion of bone screw 97. In the example embodiment, bone screw retention mechanism 11a comprises flexible arm (or spring tab) having an inclined surface 11c (or ramp) that is disposed on a lateral end of implant 100 adjacent bone screw aperture 11.

Referring generally to FIGS. 20-24 an inserter 200 for use with disclosed expandable implants 100 is illustrated. Inserter 200 may extend in a longitudinal direction from a proximal end 200p to a distal end 200d, for example. Inserter 200 may include a pair of handles 230, a handle lock 202, and mounting arms 210 for securely coupling to mounting tangs 19, 29 of implant 100, for example. Inserter 200 may further include a tightening knob 211 that is connected to a

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drive shaft 220 having a drive end 221. Drive end 221 may have a size and shape generally corresponding to a size and shape of the various drive features of locking screw 50, for example the internal threaded surface 52, first drive feature 53a, and/or second drive feature 53b. In the example embodiment shown in FIG. 24, drive end 221 includes an end portion having an outside threaded surface with a corresponding size and shape to the internal threaded surface 52 of locking screw 50. In various embodiments, tightening knob 211 may rotate drive shaft 220 and drive end 221 to engage drive end 221 with locking screw 50 and pull implant 100 towards inserter tool 200 such that mounting tangs 19, 29 are securely nested within corresponding channels of mounting arms 210. Additionally, an end user may rotate inserter 200 thereby translating a rotational force through drive end 221 to locking screw 50, e.g., via first drive feature 53a, and/or second drive feature 53b.

As seen best in FIGS. 23A and 23B, inserter 200 may include a pair of handles 230, a stationary arm 252, a primary pivoting arm 250, and a secondary pivoting arm 251. For example, to expand implant 100 an end user may toggle handle lock 202 to an unlocked position and push down on thumb indentation 231 of the handle 230 that is connected to the primary pivoting arm 250 and secondary pivoting arm 251. In doing so, primary pivoting arm 250 may pivot with respect to medial pivot point 240 and secondary pivoting arm 251 may pivot with respect to distal pivot point 245, for example. The path of travel of secondary pivoting arm 251 may lift up on the corresponding mounting tang 19 or 29 and cause the superior endplate 10 and inferior endplate 20 to separate from one another at the proximal end, for example. In doing so, an end user can lordose implant 100 to a desired angle. For example, as seen best in FIG. 24, secondary pivoting arm 251 has lifted up on mounting tang 19, which is nested in a corresponding channel of mounting arm 210.

Once implant 100 is lordosed to a desired configuration, an end user may rotate drive shaft 220 and drive end 221 to tighten locking screw 50 as explained previously. After locking screw 50 is relatively tight, the end user may continue to apply a rotational force to locking screw 50 until a proximal portion comprising the cylindrical part having an internal threaded surface 52, and the first drive feature 53a breaks off at breakoff location 55. For example, once locking screw 50 is tightened to a designed torque, the locking screw 50 may shear off as explained previously. At least one advantage of utilizing the locking screw 50, is that it may prevent over tightening which can cause deformation to implant 100. As shown in FIG. 25, implant 100 has been expanded to a desired position and/or lordotic angle. Additionally, locking screw 50 has locked the relative position of the superior endplate 10 with respect to the inferior endplate 20 and the proximal portion of locking screw 50 has been broken off as explained above.

FIGS. 26 and 27 are various views of a surgical instrument 300 for use with disclosed expandable implants 100. In some embodiments, surgical instrument 300 may be referred to as a breakoff instrument and may be used to breakoff the mounting tangs 19, 29 of implant 100. In the example embodiment, surgical instrument 300 comprises a first instrument 310 and a second instrument 320. First instrument 310 may extend in a longitudinal direction from handle 312 towards gripping end 311. Similarly, second instrument 320 may extend in a longitudinal direction from handle 322 to gripping end 321. Gripping ends 311, 321 may comprise a channel having a size and shape generally corresponding to mounting tangs 19, 29. For example, as seen best in FIG.

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27, the mounting tangs 19, 29 may be insert inside of the corresponding channels of gripping ends 311, 321. After the mounting tangs 19, 29 are nested within gripping ends 311, 321 an end user may push laterally outward and/or inward against handles 312, 322 to breakoff the corresponding mounting tang 19, 29. For example, as shown in FIG. 28 expandable implant 100 is in an expanded and lordosed configuration and the proximal portion of locking screw 50 and tangs 19, 29 have been broken off.

FIG. 29 is a reference drawing showing the human spine of which various disclosed implant embodiments may be installed in. FIG. 30 is a reference drawing showing various planes and reference directions of which the various disclosed implant embodiments may move in or act in with reference to a patient 1.

Referring generally to FIGS. 31-37 a second implant 400 embodiment is disclosed. Implant 400 may include the same, similar, and/or substantially the same components and functionality as explained above with respect to implant 100. Accordingly, duplicative description will be omitted. It shall be understood that various components and functionality of implant 100 are readily combinable with implant 400 and vice versa unless the context clearly indicates otherwise.

FIG. 31 is a perspective view of a second implant 400 embodiment. In this embodiment, implant 400 extends in a proximal-to-distal direction between a proximal end 400P and a distal end 400D and extends in a width-wise direction between a first lateral end 400L and a second lateral end 400R. Additionally, implant 400 includes a superior endplate 410 and an inferior endplate 420 having substantially similar features and functionality as explained above with respect to superior endplate 10 and inferior endplate 20 of implant 100, for example. However, in this embodiment, superior endplate 410 may include a first gripping protrusion 419 extending in a proximal direction from the proximal end 400P of the superior endplate 410. Similarly, inferior endplate 420 may include a second gripping protrusion 429 extending in a proximal direction from the proximal end 400P of the inferior endplate 420. In this embodiment, a size and shape of first gripping protrusion 419 is substantially the same as a size and shape of the second gripping protrusion 429. However, in other embodiments the first and second gripping protrusions 419, 429 may be differently sized and shaped, e.g., to bias the implant towards a surgical instrument and/or orientation for insertion. In this embodiment, and in the closed position, each gripping protrusion 419, 429 is disposed at approximately the same distance from an axis of rotation of breakoff set screw 450. In this embodiment, gripping protrusions 419, 429 may replace the need for the tangs 19, 29 of implant 100, for example. However, the concepts of utilizing breakoff tangs 19, 29 and gripping protrusions 419, 429 are not necessarily mutually exclusive and attributes of one may be combined and/or modified in view of the other.

In various embodiments, gripping protrusions 419, 429 may include various types of contouring to facilitate grasping of gripping protrusions 419, 429 with a corresponding inserter, for example surface indentations, surface outdents, channeling, etc. In the example embodiment, gripping protrusion 419 comprises a superior gripping surface 419A including an indented portion and an outdented chamfered portion at a proximal most end thereof, for example. Additionally, gripping protrusion 419 comprises an inferior gripping surface 419B including an indented portion and an outdented chamfered portion at a proximal most end thereof, for example. Likewise, gripping protrusion 429 comprises a superior gripping surface 429A including an indented por-

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tion and an outdented chamfered portion at a proximal most end thereof, for example. Additionally, gripping protrusion 429 comprises an inferior gripping surface 429B including an indented portion and an outdented chamfered portion at a proximal most end thereof, for example. In this way, gripping protrusions 419 and 429 are shaped like dovetails and a corresponding inserter tool may comprise a corresponding shaped dovetail groove which may grasp onto and/or slip over gripping protrusions 419 and 429 (not illustrated).

Referring generally to FIGS. 32, 33, and 34 various exploded parts views of implant 400 are illustrated. FIG. 32 is a first perspective exploded parts view of implant 400, FIG. 33 is a second perspective exploded parts view of implant 400, and FIG. 34 is a side exploded parts view of implant 400. In the example embodiment, the superior and inferior endplates 410, 420 of implant 400 may be hingedly coupled together by hinge member 440 and arcuate channel 412 having similar attributes as explained above with respect to hinge member 40 and channel 12 of implant 100, for example. Additionally, superior endplate 410 may also include a core 415 having an aperture 415A and inferior endplate 420 may include a core 425 having a threaded aperture 425A having similar attributes to core 15 and core 25 as explained above with respect to implant 100, for example. In the example embodiment, implant 400 utilizes a breakoff screw 450 for locking of a position of the superior and inferior endplate 410, 420. Breakoff screw 450 may include an external thread pattern 451 on an outside circumferential surface thereof, for example. The external thread pattern 451 of breakoff screw 450 may have a size and shape generally corresponding to the threaded aperture 425A of core 425 of the inferior endplate 420, for example. In various embodiments, an engagement surface 454 may be disposed adjacent and proximal of external thread pattern 451. In the example embodiment, engagement surface 454 is shaped like a washer and is directly connected to breakoff screw 450. However, in other embodiments, engagement surface 454 may be a washer or separated element, for example. In some embodiments, engagement surface 454 may be conically shaped. In the example embodiment, engagement surface 454 may include a relatively planar and/or flat distal surface and/or proximal surface.

In various embodiments, a proximal end of breakoff screw 450 may include a first flexible tang 452A and a second flexible tang 452B defining a discontinuous cylindrical shaped aperture 452 therebetween. Additionally, the first flexible tang 452A and second flexible tang 452B may each include an outdent at a proximal end thereof that is shaped like a segment of an annular ring. In the example embodiment, the first flexible tang 452A may flex inward towards the second flexible tang 452B under loading and vice versa due to the gap between them. At least one advantage of this configuration is that it may facilitate securing breakoff screw 450 to a corresponding drive tool (not illustrated) and the retention of the broken off part. For example, a drive tool may comprise a drive end having a female cavity with a corresponding size and shape to the drive features 453A, 453B. In various embodiments, the cavity may include a pair of indentations corresponding in size and shape to the outdents of flexible tangs 452A and 452B, for example. In use, an end user may align the flexible tangs 452A, 452B with the cavity, push down against the flexible tangs 452A, 452B which may cause them to flex inward towards each other such that they may slide within the cavity until the outdents of flexible tangs 452A and 452B are seated within the corresponding indents of the drive tool. Thereafter, an

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end user may continue to rotate and or tighten breakoff screw **450**. In this way, after a proximal portion of breakoff screw **450** is broken off it may remain retained by the inserter due to the flexible tangs **452A** and **452B** being seated within the corresponding indents.

In some embodiments, aperture **452** may be understood as a cylindrical protrusion extending in a proximal direction and having a first slit and a second slit extending along the length thereof such that the cylindrical protrusion is compressible. In the example embodiment, a first drive feature **453A** and a second drive feature **453B** may be disposed adjacent to and distally with respect to a proximal most end of breakoff screw **450**. Additionally, first and second drive features **453A**, **453B** may be disposed proximally with respect to engagement surface **454**. In various embodiments, a breakoff location may be positioned between and/or adjacent to drive features **453A**, **453B**, which will be explained in further detail below. In the example embodiment, drive features **453A**, **453B** take a hexalobular shape, although various other shapes such as hexagonal, polygonal, torx, etc. are also contemplated. In some embodiments, a surgical drive tool having a corresponding socket may be coupled to drive features **453A** and/or **453B** to cause rotation of breakoff screw **450**. Similarly as explained above with respect to locking screw **50**, once breakoff screw **450** has been sufficiently tightened a proximal portion may breakoff and/or shear off while the distal portion may remain coupled to implant **400** locking a relative orientation of superior endplate **410** and inferior endplate **420** in place.

As seen best in FIGS. **35-37**, breakoff screw **450** may extend in a longitudinal direction along a longitudinal axis L-A that is coaxially aligned with breakoff screw **450**. FIG. **35** is a first side view of breakoff screw **450** in which the superior surface **452A** and inferior surface **452B** of discontinuous aperture **452** are visible. FIG. **36** is a second side view of a breakoff screw **450** that is rotated about 90 degrees with respect to FIG. **35** in which only the superior surface **452A** is visible. With reference to FIG. **37**, in various embodiments, breakoff location **455** may comprise a recessed fracture surface F-S that is inset with respect to a leading edge (proximal most edge) of drive feature **453A**. At least one advantage of the recessed fracture surface may be that delicate tissue is prevented and/or suppressed from coming into contact with relatively sharp ends of the fracture surface. In the example illustration, a relative location of the recessed fracture surface is represented by dashed lines F-S. In the example embodiment, breakoff location **455** may be considered as the boundary between a proximal portion **450A** and a distal portion **450B** of breakoff screw **450**, for example. In this embodiment, the boundary between drive features **453A** and **453B** comprises a necked down portion **458** extending from a distal end of drive feature **453A** to an inset portion of drive feature **453B** that is inset with respect to an outermost and/or proximal most surface of drive feature **453B** thereby defining a portion of breakoff set screw **450** having a minimum cross sectional diameter. Accordingly, when breakoff screw **450** is sufficiently tightened within threaded aperture **425A** of core **425** such that the breakoff location **455** experiences a sufficient torque the proximal portion **450A** may breakoff from the distal portion **450B**. For example, when a sufficient rotational force is applied to the proximal end of breakoff screw **450** while a distal end of breakoff screw **450** is stationary, i.e., when breakoff screw **450** is secured in a locked position and a continued rotational force (torque) is applied to the proximal end of breakoff screw **450** the drive feature **453A** and cylindrical end having the discontinuous aperture **452** may

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breakoff. For further explanation in the similar context of implant **100**, see FIGS. **10** and **11** and the corresponding discussion thereof.

Referring to FIGS. **38-42** an example swaging process is performed to a distal most end of breakoff screw **450**. FIG. **38** is a perspective view of a swaging fixture **500**, and FIG. **39** is a cross section view of a swage mandrel and a distal end **450D** of a breakoff screw **450** before the commencement of a swage process. FIG. **40** is a cross section view showing a result of a swage process, and FIG. **41** is an enlarged view of region S-W of FIG. **40**. In the example embodiment, swaging fixture **500** comprises a swaging ram **501** and a swaging mandrel **503** that are supported by the base of the apparatus. The swaging mandrel **503** may include an outdent that corresponds to and is slightly larger than the distal most indent **498** (swage bore) of breakoff screw **450**, for example. As seen in region S-W of FIG. **40**, when swaging mandrel **503** is advanced into the distal most indent **498** (swage bore) a flared out portion **499** (swaged portion) is formed at a distal most end of breakoff screw **450**. An example advantage of a swaged end may be that it facilitates retention of the breakoff screw **450** such that it serves as a stopping structure preventing breakoff screw **450** from backing out of implant **100**.

It shall be understood that although breakoff screw **450** is explained concurrently with implant **400** and in the context of an intervertebral implant, the concepts of breakoff screw **450** may be applied to other embodiments used for alternate purposes, for example for use in a pedicle screw to insert in a tulip to tighten down a rod. It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. For example, features, functionality, and components from one embodiment may be combined with another embodiment and vice versa unless the context clearly indicates otherwise. Similarly, features, functionality, and components may be omitted unless the context clearly indicates otherwise. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques).

Referring generally to FIGS. **42-51**, a third implant **400Z** and a surgical tool **600** for use with various implants is disclosed. FIGS. **42A-42B** illustrate a third embodiment of an implant **400Z**. Implant **400Z** may have the same, similar, and/or substantially the same components and functionality as explained above with respect to implant **400**, for example. Accordingly, duplicative description will be omitted. A difference may be that breakoff screw **450** does not include the discontinuous cylindrical shaped aperture **452** (see FIG. **34**) having tangs **452A**, **452B**, for example. Rather, breakoff screw **450**, to the extent included, may include a continuous cylindrical shaped aperture **452Z** without the discontinuity. In some embodiments, a similar screw may be used that is not a breakoff screw but rather may be referred to as a locking screw. Additionally, in this embodiment, implant **400Z** may include a first gripping indentation **419Z** and a second gripping indentation **429Z** (see FIG. **42A**) in lieu of the gripping protrusions **419**, **429** (as shown in FIG. **33**), for example. In various embodiments, first gripping indentation **419Z** may be formed as a part of the superior endplate **410** and second gripping indentation **429Z** may be formed as a part of the inferior endplate **420**, for example.

In various embodiments, gripping indentations **419Z**, **429Z** may include various types of contouring to facilitate coupling with a corresponding surgical tool **600**. In this embodiment, gripping indentations **419Z**, **429Z** each comprise a slotted indentation extending in a proximal to distal direction with a curved superior surface **419S**, **429S** and a curved inferior surface **419I**, **429I**. Additionally, gripping indentations **419Z**, **429Z** each have an open channel portion **419X**, **429X** adjacent the breakoff screw **450**, to the extent a breakoff screw is included. As will be explained in further detail below, corresponding gripping protrusions of a surgical tool **600** may have a substantially similar size and shape to the gripping indentations **419Z**, **429Z**, for example.

FIGS. **43A-43B** illustrate a similar embodiment as FIGS. **42A-42B**. As shown, the implant **400Z** includes angled engagement features **14**. In the example embodiment, engagement features **14** extend diagonally across the exposed uppermost surface of superior endplate **410** and across the exposed lowermost surface of inferior endplate **420**. The engagement features **14** comprise flattened top rails that are sequentially spaced apart with rounded bottom valleys **14V** therebetween. As seen best in the top down view of FIG. **43B**, the engagement features **14** of the superior endplate **410** are oriented at an angle β with respect to a proximal face **410F** of the superior endplate **410**. Similarly, the engagement features **14** of the inferior endplate **420** are oriented at an angle β with respect to a proximal face **420F** of the inferior endplate **420** (see FIG. **43A**). In various embodiments, the angle β may be about 20-40 degrees and in at least one embodiment the angle β may be angled at about 30 degrees. At least one advantage of this orientation may be a relatively greater resistance and/or suppression of expulsion of the implant **400Z** relative to embodiments in which engagement features solely extend horizontally across the implant **400Z**. In the example embodiment of FIGS. **43A-43B**, by orienting the engagement features **14** diagonally, the implant **400Z** may resist expulsion in multiple directions, e.g., forward flexion/extension and lateral bending.

Referring generally to FIGS. **44-48**, various views of a surgical tool **600** are disclosed. FIG. **44** illustrates the surgical tool **600** coupled to implant **400Z**. FIG. **44** is a perspective view of surgical tool **600**; FIG. **45** is a first side view of surgical tool **600**; and FIG. **46A** is a second side view of surgical tool **600**. FIG. **46B** is a third side view of surgical tool **600** in an operative position and FIG. **46C** is a perspective view of surgical tool **600** in the operative position of FIG. **46B**. FIG. **47A** is an exploded parts view of surgical tool **600** with select parts removed for ease of understanding and FIG. **47B** is an exploded parts view of surgical tool **600** showing additional detail and parts. FIG. **48** illustrates surgical tool **600** immediately prior to coupling with implant **400Z**. In the example embodiment, tool **600** may serve several purposes. For example, tool **600** may be used to insert implant **400Z**, expand implant **400Z**, and in some embodiments may also rotate breakoff screw **450** to cause it to separate into two portions along breakoff location **455** (see FIG. **34**).

With reference to the perspective view of FIG. **44** and the exploded parts views of FIGS. **47A** and **47B**, tool **600** may include a gripping handle **601** that is securely coupled to first body portion **607**. First body portion **607** may rotatably support an inner shaft **606** and an outer shaft **602** therein. Shafts **606**, **602** may extend through aperture **607A** of first body portion **607** in a longitudinal direction from a proximal end to a distal end. The inner shaft **606** may be securely connected to turn knob **603** at a proximal end **600P** and the outer shaft **602** may be securely connected to turn handle

604 at the proximal end **600P**. The inner shaft **606** may extend through outer shaft **602** and include a thread pattern **606T** at a distal end thereof for securely connecting to corresponding threads **456** of breakoff screw **450** (see FIG. **42B** for example). The inner shaft **606** may be independently rotatable relative to outer shaft **602** by rotating turn knob **603**. Additionally, the outer shaft **602** may be independently rotatable relative to inner shaft **606** by rotating turn handle **604**, for example. Generally, both the inner shaft **606** and outer shaft **602** will interact with and/or couple to breakoff screw **450** as will be explained in further detail below.

With reference to FIGS. **45** and **46A-46C**, tool **600** may include an actuator **605** in the form of a trigger that is connected to first body portion **607** at pin **605C**. Actuator **605** may be actionable to expand implant **400Z** by pivoting a linkage assembly **608** including first arm **610** and second arm **611** relative to a third fixed arm **620**, for example. Second arm **611** may be coupled to actuator **605** via pin **605B** such that pin **605B** may be linearly translatable forward and backward in a proximal-to-distal direction within slotted aperture **605A**. In turn, second arm **611** may be coupled to first arm **610** via pin **610B** such that second arm **611** may pivot up and down. Additionally, first arm **610** may be coupled to third arm **620** and in various embodiments third arm **620** may be fixed relative to first body portion **607**. Additionally, third arm **620** may include a protrusion **620B** which may be slidably seated within a curved vertical slot, for example slot **610A**, and thereby guide motion of arm **610** as it pivots up and down. As seen best in FIGS. **46B** and **46C**, depressing actuator **605** may cause linkage assembly **608** to pivot by causing second arm **611** to move distally such that first arm **610** pivots relative to third arm **620** and implant **400Z** may be expanded.

With reference back to FIG. **44**, third arm **620** may include an aperture extending therethrough in a proximal to distal direction for accommodating a linearly translatable shaft **621**. Shaft **621** may independently move forward and backward in a proximal to distal direction within the through aperture **624** of third fixed arm **620** (see FIGS. **47A** and **47B**), for example. Shaft **621** may include a charging handle at a proximal end thereof allowing an end user to manipulate shaft **621** in a forward and backward motion (proximal motion and distal motion in the proximal-to-distal direction). In some embodiments, an end user may translate charging handle **622** rotatably in a clockwise or counter-clockwise motion, for example to lock shaft **621** in a relative position. Additionally, in some embodiments, when shaft **621** is not in a locked position, depressing actuator **605** will not expand implant **400Z** for reasons explained in more detail below.

With reference to FIG. **48**, first arm **610** may include a gripping outdent **619** having a size and shape generally corresponding to a size and shape of gripping indentation **419Z**. Additionally, first arm **610** may include a counter torque surface **615** having a size and shape generally corresponding to a size and shape of the proximal surface of the superior endplate **410**, for example. Similarly, shaft **621**, extending through fixed arm **620**, may include a gripping outdent **629** having a size and shape generally corresponding to a size and shape of gripping indentation **429Z** (as shown in FIG. **42A**). Additionally, second arm **620** may include a counter torque surface **625** having a size and shape generally corresponding to a size and shape of the proximal surface of the inferior endplate **420**, for example. In this way, when shaft **621** is moved distally such that the outdent **629** is seated within indent **429Z** and outdent **619** is seated within indentation **419Z** (as shown in FIG. **42A**) an end user may

activate actuator **605** to expand implant **400Z** while surfaces **615**, **625** resist a twisting motion of implant **100**. For example, as shown in FIGS. **46B** and **46C** an end user may depress actuator **605** by squeezing the trigger which in turn may cause the linkage assembly **608** to pivot the first arm **610** relative to fixed arm **620** such that tool **600** may urge the superior endplate **410** and inferior endplate **420** away from one another.

With reference to the cross-section drawings of FIGS. **49**, **50**, and **51** additional features and functionality of tool **600** will be explained. In FIGS. **49** and **50**, it is shown that the threaded end of inner shaft **606** is threadably engaged with the threaded aperture **456** of breakoff screw **450**. Additionally, a proximal drive feature **453A** is mated within a corresponding female drive feature **602A** (see FIG. **48**) at the distal most end of outer shaft **602**. As explained previously, inner shaft **606** and outer shaft **602** are independently operable. In use, an end user may initially position inner shaft **606** adjacent to/within the front portion of threaded aperture **452** and rotate inner shaft **606** via turn knob **603**. As inner shaft **606** is rotated, implant **400Z** is pulled towards tool **600**. FIGS. **49-51** show implant **400** after inner shaft **606** has been sufficiently rotated such that implant **400Z** abuts the counter torque surfaces **615**, **625** and the gripping protrusion **619** may be seated within the corresponding gripping indentation **419Z**. With reference to FIG. **51**, once implant **400Z** is pulled sufficiently towards tool **600**, an end user may actuate shaft **621** by pushing on charging handle **622** to overcome the biasing force of spring **623** such that gripping protrusion **629** may be seated within the corresponding gripping indentation **429Z**, for example. In some embodiments, an end user may rotate charging handle **622** such that biasing force of spring **623** is prevented from urging shaft **621** away from implant **400Z**. At this point, the implant **400Z** may be in an operatively engaged position with tool **600**.

Once implant **400Z** is in the operatively engaged position, an end user may insert implant **400Z** into a disc space between a superior vertebra and inferior vertebra (cephalad vertebra and caudal vertebra), for example. After the implant **400Z** is positioned between the superior and inferior vertebrae, an end user may depress actuator **605** thereby causing the linkage assembly **608** to pivot and spread the superior endplate **410** apart from the inferior endplate **420**. Once the end user has expanded implant **400Z** to an appropriate position, the end user may rotate turn handle **604** which will rotate breakoff screw **450** via first drive feature **453A**. The end user may continue to rotate breakoff screw **450** such that it advances into implant **400Z** in a proximal to distal direction and locks implant **400Z** in the expanded configuration similarly as explained above. The end user may continue to rotate turn handle **604** until the torque applied to breakoff screw **450** is great enough that the breakoff screw **450** will shear into two pieces similarly as explained above. Notably, the broken off or sheared off portion of breakoff screw **450** may be retained by tool **600** on account of inner shaft **606** being threadably coupled to threaded aperture **452**.

Unless otherwise specifically defined herein, all terms are to be given their broadest possible interpretation including meanings implied from the specification as well as meanings understood by those skilled in the art and/or as defined in dictionaries, treatises, etc. It must also be noted that, as used in the specification and the appended claims, the singular forms "a," "an" and "the" include plural referents unless otherwise specified, and that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, elements, and/or components,

but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof

What is claimed is:

1. An expandable implant movable between a contracted position and an expanded position, comprising:

an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction and extending from a first lateral side to a second lateral side in a widthwise direction, the expandable body being defined by a superior endplate and an inferior endplate that are hingedly connected;

the superior endplate comprising a first core having a screw slot and a distal engagement surface;

the inferior endplate comprising a second core having a proximal engagement surface; and

a locking screw extending in a longitudinal direction from a proximal end to a distal end, being extendable through the first core and the second core, and being movable between a locked position and an unlocked position,

wherein, in the locked position, the locking screw urges the distal engagement surface of the first core against the proximal engagement surface of the second core, wherein the proximal end of the locking screw is defined by an exposed aperture having an internal threaded surface, and

wherein the locking screw comprises a proximal portion and a distal portion that are configured to separate along a recessed fracture surface, and, when separated, the recessed fracture surface is recessed relative to a sidewall of the distal portion and the fracture surface is recessed relative to a proximal most edge of the distal portion.

2. The expandable implant of claim 1, wherein:

the superior endplate comprises a first gripping indentation located at a proximal end thereof; and

the inferior endplate comprises a second gripping indentation located at a proximal end thereof.

3. The expandable implant of claim 2, wherein:

the first gripping indentation comprises a slot having a superior curved surface and an inferior curved surface; and

the second gripping indentation comprises a slot having a superior curved surface and an inferior curved surface.

4. The expandable implant of claim 1, wherein:

the superior endplate comprises a first plurality of engagement features that are angled 20 degrees to 40 degrees with respect to a proximal surface of the superior endplate; and

the inferior endplate comprises a second plurality of engagement features that are angled 20 degrees to 40 degrees with respect to a proximal surface of the inferior endplate.

5. The expandable implant of claim 1, wherein:

the superior endplate further comprises a channel located adjacent the distal end and extending in the widthwise direction; and

the inferior endplate further comprises a rail located adjacent the distal end and extending in the widthwise direction, the rail having a size and shape generally corresponding to a size and shape of the channel and being located within the channel.

6. The expandable implant of claim 5, wherein:

the distal engagement surface of the first core comprises a first curved surface;

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the proximal engagement surface of the second core comprises a second curved surface; and the first curved surface is defined by a radius of a circle having a center point that is offset from an axis of rotation of the rail.

7. The expandable implant of claim 5, wherein: the superior endplate comprises at least one hook portion adjacent the channel; the inferior endplate comprises at least one relief portion adjacent the rail; the at least one hook portion has a size and shape generally corresponding to a size and shape of the at least one relief portion; and the at least one hook portion is disposed within the at least one relief portion.

8. The expandable implant of claim 7, wherein, in a cross-section view, the channel comprises an arcuate shape and the rail comprises an arcuate shape.

9. The expandable implant of claim 1, wherein:

the locking screw comprises a threaded distal end and a drive feature;

the first core is disposed proximally with respect to the second core; and

the locking screw is threadably engaged with at least one of the first core and second core.

10. A system, comprising:

an expandable implant movable between a contracted position and an expanded position, the implant comprising:

an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction and extending from a first lateral side to a second lateral side in a widthwise direction, the expandable body being defined by a superior endplate and an inferior endplate that are hingedly connected;

the superior endplate comprising a first core having screw slot and a distal engagement surface;

the inferior endplate comprising a second core having a proximal engagement surface; and

a locking screw movable between a locked position and an unlocked position;

wherein, in the locked position, the locking screw urges the distal engagement surface of the first core against the proximal engagement surface of the second core; and

a surgical tool configured to move the implant from the contracted position to the expanded position and to move the locking screw between the locked position and the unlocked position, the surgical tool being capable of moving the locking screw into the locked position while supporting the implant in the expanded position,

wherein the surgical tool further comprises:

a body portion having an aperture extending there-through;

a pivotal linkage assembly including a first arm and a second arm;

a third arm that is fixed relative to the body portion;

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an outer shaft extending through the body portion and having a drive feature at a distal end thereof for rotating the locking screw from the unlocked position to the locked position; and

an inner shaft extending through the outer shaft and having a first thread pattern at an end thereof for drawing the implant towards the surgical tool.

11. The system of claim 10, wherein:

the superior endplate comprises a first gripping indentation located at a proximal end thereof;

the inferior endplate comprises a second gripping indentation located at a proximal end thereof;

the surgical tool comprises a first gripping protrusion having a size and shape generally corresponding to a size and shape of the first gripping indentation; and the surgical tool comprises a second gripping protrusion having a size and shape generally corresponding to a size and shape of the second gripping indentation.

12. The system of claim 11, wherein:

the first gripping indentation comprises a slot having a superior curved surface and an inferior curved surface; and

the second gripping indentation comprises a slot having a superior curved surface and an inferior curved surface.

13. The system of claim 10, wherein the surgical tool comprises a pivotal linkage assembly including a first arm and a second arm.

14. The system of claim 10, wherein:

the surgical tool is configured to separate the locking screw into the proximal portion and distal portion while the implant is in the expanded position and retain the proximal portion of the locking screw via the inner shaft.

15. The system of claim 10, wherein the surgical tool further comprises an actuator for causing the pivotal linkage assembly to pivot relative to the third arm and thereby expand the implant.

16. The system of claim 10, wherein:

the superior endplate further comprises a channel located adjacent the distal end and extending in the widthwise direction; and

the inferior endplate further comprises a rail located adjacent the distal end and extending in the widthwise direction, the rail having a size and shape generally corresponding to a size and shape of the channel and being located within the channel.

17. The system of claim 16, wherein:

the distal engagement surface of the first core comprises a first curved surface;

the proximal engagement surface of the second core comprises a second curved surface; and

the first curved surface is defined by a radius of a circle having a center point that is offset from an axis of rotation of the rail.

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